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DEPARTMENT OF PHYSICS

NUCLEAR, ATOMIC, MOLECULAR PHYSICS AND BIOPHYSICAL LABORATORY

TOPIC

Quantum Machine Learning Approach for Malaria Drugs Discovery

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General Introduction

Background of the study

Novel antimalarial therapies are required to combat drug-resistant *Plasmodium falciparum* (Pf) and help with malaria elimination. Malaria is one of the deadliest disease afflicting the mankind, with 282 million cases and over 610 000 reported deaths globally in 2024, concentrated in the WHO African Region (*World Malaria Report* 2025[1] [1]). The disease poses a risk to half of the world's population, with non-Mediterranean Africa and Southeast Asia being particularly affected [2]. Malaria continues to represent a significant public health challenge across the African continent, particularly in regions situated south of the Sahara Desert, which accounted for 94% of malaria cases and 95% of malaria deaths in 2023. Notable reductions in incidence and mortality were observed between 2000 and 2022; however, the disease continues to disproportionately affect children under five, who represent a significant malaria fatalities. Key countries, including Nigeria, the Democratic Republic of Congo, Tanzania, and Niger, contribute substantially to the global mortality burden. In Cameroon, the entire population is susceptible to infection, with the highest risk concentrated in the equatorial and coastal regions. As one of the 11 countries with the highest malaria burden, Cameroon accounts for 2.6% of all global malaria cases and 2.1% of malaria deaths. The transmission of malaria is particularly pronounced in forest, coastal, and humid savanna regions, with the rainy season (July to October) contributing to an escalation in risk due to increased rainfall and temperature. The nation faces a multitude of interconnected challenges, including drug and insecticide resistance, climate change, and disparities in healthcare access, which collectively impede efforts to control malaria transmission by infected Anopheles mosquitoes. Moreover, malaria imposes a substantial economic burden, manifesting in increased healthcare expenditures, diminished productivity, and stunted GDP[2] growth. This underscores the imperative for ongoing public health interventions and the development of novel control strategies.

In essence, a drug is any substance that has the capacity to cure or prevent diseases in humans and animals. For many centuries, drugs were derived from natural sources [3]. These substances were often discovered by chance, and their mechanisms of action were not always understood. Moreover, they were not always capable of being synthesized industrially. The process of discovering drugs was largely based on serendipity, or, in other words, fortunate accidents. The seminal contributions of Pasteur, who proposed the concept of a drug as a molecule with a biological mechanism of action, were informed by the preceding observations of van Leeuwenhoek. These observations, which included the identification of microorganisms and their association with diseases, laid the foundation for the field of microbiology. Indeed, at the beginning of the 20th century, Ehrlich articulated in *"Principle of Chemotherapy"* that an appropriate dose of a chemical could disrupt the growth of disease-causing microorganisms [4]. This "principle" became a reality in the 1930s when Fleming discovered penicillin. Subsequent to this breakthrough, considerable efforts were

[1] World Health Organization

[2] Gross Domestic Product (GDP) is defined as the total market value of all goods and services produced within a nation's borders over a specific period. The fluctuations in GDP over time are typically measured on an annual or quarterly basis.

made to synthesize or isolate compounds analogous to penicillin, which resulted in the development of various antibiotic families. Beginning in the 1950s, significant progress in the field of human genome research, complemented by substantial advancements in cell and molecular biology, paved the way for the development of rational drug design methodologies. In this novel approach, biologists formulate informed hypotheses concerning the mechanisms underlying diseases. Consequently, the process of drug discovery underwent a shift from a primarily chance-based approach to a more systematic one. This transformation occurred as researchers began to link diseases with specific proteins present within biological pathways, thereby facilitating the development of targeted organic compounds for therapeutic interventions. The objective of rational drug discovery is the identification of disease-associated targets and their ligands [5]. Consequently, in response to the AIDS epidemic that emerged in the 1980s, extensive research efforts culminated in the development of combinatorial chemistry, a field that facilitates the synthesis of a vast and diverse array of organic compounds. These compounds can then be tested for their efficacy in combating specific diseases. In the contemporary era, although chance still plays a role in the process of developing new drugs, recent technological advancements and discoveries have rendered the drug development process more rationalized [6].

The process of drug discovery can be defined as the identification of novel pharmaceutical compounds with the potential to treat diseases [7, 8]. In essence, this process entails the identification of novel chemical entities that possess specific chemical properties for the treatment of diseases. This process is often characterized by a significant investment of resources, both financial and temporal, and involves a rigorous testing and experimentation phase aimed at identifying drugs that are both safe and effective [9, 10]. The primary objective of a scientist in the realm of drug discovery is the identification of lead or hit molecules of the highest quality. However, it is imperative that these lead molecules not only exhibit pharmacological activity but also ensure safety for both humans and the environment. Subsequent to the identification of a promising lead molecule, a series of trials and assessments are conducted to predict its pharmacokinetic properties, including Absorption, Distribution, Metabolism, Excretion, and Toxicity (ADMET) characteristics [9, 11, 12]. The process of developing new drugs is costly and time-consuming, necessitating the testing of thousands of chemical compounds and conducting numerous experiments to identify drugs that are both safe and effective. In light of these challenges, there is an imperative to establish robust, standard, and reproducible computational methodologies to achieve the objectives set. These computational approaches are instrumental in determining the characteristics of receptors and compounds, thereby facilitating the binding process [13].

Despite progress in reducing malaria-related morbidity and mortality over the past two decades, the constant emergence of parasitic resistance to current antimalarial drugs has made the discovery of new drug candidates a major global health priority [14, 15]. Currently, artemisinin-combined therapy (ACT) since their implementation in 2006 remains the treatment of malaria, but Pf-resistance cases were recorded also. The regional picture has hardened: partial resistance to artemisinin derivatives, mediated by mutations in *Pf*kelch13, is now confirmed or suspected in at least eight African countries [16], spatial-temporal modelling indicates that the geographic footprint of *pfk13* markers on the continent is expanding [17], and genomic surveillance documents the co-occurrence of *Pfcrt* and *Pfdhfr* resistance variants across diverse transmission settings [18, 19, 20]. Malaria is known to have a large global burden, with strong poverty-promoting effects. It persists as chronic infections, in spite of the available effective medical treatments. While aggressive control measures have reduced prevalence rates in some regions, the emergence of drug resistance, particularly to artemisinin-based combination therapies, complicates treatment protocols and undermines elimination efforts. Therefore, the urgent need for innovative antimalarial compounds and therapeutic targets has become paramount [21]. In this context, understanding the sociocultural dynamics of health care partnerships in endemic regions can improve the effectiveness of interventions [22]. Taken together, these challenges underscore the importance of sustained investment in research aimed at overcoming these barriers and achieving long-term malaria control.

Natural products (NPs) and their structural analogues have historically made significant contributions to pharmacotherapy, particularly for infectious diseases such as malaria. A notable example is *artemisinin*, a breakthrough treatment derived from *Artemisia annua* (sweet wormwood), which was recognized with the Nobel Prize in Physiology or Medicine in 2015. However, natural products pose considerable challenges in the realm of drug discovery, primarily due to technical obstacles during the screening, isolation, characterization, and optimization processes. These challenges have contributed to a marked decline in the pharmaceutical industry's engagement in natural product research from the 1990s onwards [23]. NPs exhibit distinctive characteristics when compared to conventional synthetic molecules, thus presenting both benefits and difficulties in the context of drug discovery for malaria. These compounds exhibit a remarkable diversity in their scaffolds and structural intricacy. NPs characteristically exhibit higher molecular weights, a greater number of sp3 carbon and oxygen atoms, and a lower number of nitrogen and halogen atoms in comparison to synthetic libraries. They also feature increased numbers of hydrogen bond acceptors and donors, demonstrate lower calculated octanol-water partition coefficients (indicating greater hydrophilicity), and exhibit enhanced molecular rigidity compared to synthetic alternatives. These distinctive properties can prove advantageous in the development of antimalarial drugs. For instance, the increased rigidity of NPs is valuable when targeting protein-protein interactions in Plasmodium parasites. Notably, natural products represent a significant source of oral antimalarial medications that function beyond Lipinski's rule of five [24, 25, 26]. The increasing molecular mass of approved oral medications, including antimalarials, over the last two decades demonstrates the growing importance of compounds that do not conform to this conventional pharmaceutical rule [27].

Machine Learning (ML) have significantly advanced drug discovery. Pharmaceutical companies have greatly benefited from the utilization of various ML techniques to discover or re-purposing novel drugs candidate. The various ML algorithms have been used for predicting chemical, biological, and physical characteristics of compounds. Deep Learning (DL) techniques have become exemplar methods for drug activity prediction, target discovery and lead molecule discovery [28]. These algorithms have been extremely used alongside the vast data generated utilizing high throughput sequencing to enhance the target discovery process. High throughput screening (HTS) using either approach entails screening of a large library of compounds. This process is often inefficient and not cost-effective because a high failure rate at subsequent stages of drug discovery.

Biopharmaceutical industries are focusing on computational approaches in order to enhance the drug discovery processes as to reduce research and development expenses by diminishing failure rates in clinical trials and ultimately generate superior medicines. Drug discovery using ML techniques can be divided into four major steps: *(i)* identify and curate datasets (potential target), *(ii)* build and validate models, *(iii)* score and prioritize a drug library, *(iv)* validate predictions *in vitro* or *in vivo*. Different ML approaches help to identify drugs targets, find suitable molecules from data libraries, suggest chemical modifications, etc. The application of ML and Artificial Intelligence (AI) methods to drug discovery are reviewed in several recent papers. DL methods have made a massive impacts in science and technology generally, and in drug discovery particularly. Ashdown et *al.* [29] set out to develop a high-throughput cell-based, drug screening method that defines cell morphology and drug action from primary image data. From the outset, they sought to encompass asynchronous asexual cultures of the human malaria parasite Pf., by designing a semi-supervised DNN strategy that can analyze fluorescence microscopy images. The model can quantify the effects of antimalarial drugs both in terms of changes in life history distributions and morphological phenotype or drop points not observed during asexual development. Keshavarzi et *al.* [30] introduce DL based process (*DeepMalaria*), capable of predicting the anti-Pf inhibitory properties of compounds using a graph based model and their SMILES (Simplified Molecular Input Line Entry Specification) [31, 32]. Monika Samant et *al.* [33] have developed a protein-protein interaction network of Pf. and human host by integrating experimental data and computational prediction of interactions using the interlog method. To complement the medical professionals, Qazi Ammar et *al.* [34] have devised a DL based method to automatically detect/localize the Pf. parasites in the photograph of stained film. They

introduced a new large scale microscopic image malaria datasets, where cells are tagged from the microscopic images. Their model are able to classify Malaria life-cycle classification in thin blood smears images. Traditional machine learning approaches for drug discovery encounter computational limitations when processing complex molecular data, which is the stated motivation for quantum machine learning as an alternative. The claim that this alternative can achieve exponential speedups with relatively modest quantum resources is a hypothesis rather than an established property, and it is evaluated in this thesis against classical baselines on a fixed chemical space. The real question is, with all the modern technological advancements in drug discovery, how can we utilize innovative technologies to find new active compounds more efficiently, thus reducing the cost?

Quantum machine learning (QML) is an emerging field that integrates the principles of quantum computing with machine learning algorithms, and it is evaluated here as a hypothesis about computational capability rather than as an established one. This approach integrates the principles of quantum physics with the adaptability of machine learning, thereby offering new possibilities for solving complex problems across various domains [35, 36]. QML has been proposed for pharmaceutical development across enhanced molecular representation [37, 38], sophisticated feature extraction methods [39], accelerated screening processes [40], improved structure-activity relationship (SAR) analysis capabilities [41], optimized lead compound management [42], refined predictions of drug-target interactions [43], and streamlined handling of complex biological data [44, 45]; whether any of these proposals survives a strong classical baseline on antimalarial chemical space is an empirical question this thesis answers. The majority of QML applications in drug discovery are primarily focused on the initial stages of the drug discovery pipeline, particularly with regard to the identification of novel drug-like molecules. The complexity and intricacy of QML pose significant challenges in the interpretation of results, thereby hindering the identification of the molecular mechanisms that underpin drug-likeness [46]. The utilization of QML models necessitates high-performance computing resources and specialized software [47, 48].

Quantum computing is hypothesized to accelerate certain drug-discovery tasks, such as wavefunction simulation and combinatorial optimization, through exponential Hilbert-space scaling. These tasks sit in the initial phases of drug discovery, such as the identification of novel drug-like molecules and the prediction of molecular properties [49, 50, 51, 52]. Whether this hypothesis holds for molecular property prediction on finite-size antimalarial panels is an empirical question this thesis addresses. Quantum approaches to drug discovery are widely motivated by the prospect of accelerating the research and development phase and reducing costs, with some studies proposing that quantum simulations could shorten the R&D phase to mere months and decrease costs to a fraction of those incurred by traditional methods [53]; whether any such reduction is realized on real molecular prediction tasks remains an open empirical question, and no time-or-cost reduction is claimed by this work. QML frameworks, including those employing quantum support vector classifiers, have demonstrated superior performance in predicting the ADME-Tox properties of molecules compared to classical models [54]. This enhancement is critical for informed decision-making within the pharmaceutical industry [54]. A primary challenge in applying QML to drug discovery is the necessity of compressing extensive datasets of molecular descriptors to align with the capabilities of quantum computers, necessitating innovative data handling and algorithm design approaches [49, 54]. The amalgamation of quantum computing with machine learning is a complex endeavor requiring interdisciplinary collaboration. The integration process involves addressing technical challenges related to algorithm development and hardware limitations [51]. Despite the potential benefits, QML remains in its early stages, and its practical application in drug discovery is constrained by the current state of quantum computing technology. Hybrid solutions that combine classical and quantum methodologies are presently recommended to capitalize on the strengths of both [50]. QML techniques, such as generative adversarial networks and variational auto-encoders, are being investigated for the generation of small and large drug molecules and for virtual screening processes. These applications can improve the efficiency of identifying potential drug candidates [36, 55]. Claims of quantum advantage in drug discovery, particularly in ligand-based virtual screening and other computationally

intensive tasks, remain to be demonstrated against strong classical baselines, and this thesis treats them as hypotheses under test rather than as expectations.

Statement of the problem

The state of the art reviewed above converges on a single representational crisis with four interdependent facets, and together they define the gap this thesis addresses.

The crisis is this: classical molecular descriptors systematically fail to extrapolate across stereochemically complex African natural-product scaffolds because they encode two-dimensional topology, meaning atom-connectivity graphs, rather than three-dimensional geometry, meaning spatial coordinates, bond angles and stereocentres. The expansion developed in this thesis quantifies the challenge before any model is trained: 92.6 % of the expanded molecules are novel at the whole-molecule level while 69.3 % of the seed scaffolds are retained, so the novelty lies in peripheral substitution on conserved ring systems — precisely the variation that hashed two-dimensional encodings flatten. The crisis manifests in four empirical facets.

The scaffold paradox (chemical-space facet). African-natural-product-inspired libraries exhibit high whole-molecule novelty coexisting with high scaffold retention, on rigid polycyclic frameworks with multiple stereocentres. Classical fingerprints describe local atom environments with hashed bit collisions, fixed dictionaries, or bounded topological distances, and hash both the novel periphery and the conserved core into the same feature space, collapsing stereoisomers into identical bit vectors. None of these encodings was designed for the structures that dominate this chemical space, and the question of what that costs downstream is opened here.

The estimand-divergence problem (validation facet). Computational antimalarial prioritization rarely treats resistance explicitly: candidates are ranked on wild-type docking scores or on predicted activity against current strains, and the mutant states that drive treatment failure enter the analysis, if at all, as qualitative structural discussion. Static docking scores and short explicit-solvent molecular-dynamics trajectories disagree on the direction of resistance-mutant effects in 87.5 % of matched states, yet both are used interchangeably in prioritization pipelines. What each method actually measures has never been formalized on this chemical space, and neither the dynamics-based stress test nor the retrospective comparison against published genotype-resolved inhibition data that would bound the static score has been performed here.

The quantum-inspired null result (descriptor facet). Whether the descriptors proposed for exactly this purpose — topological invariants, tensor-compressed embeddings and quantum feature maps — actually improve activity prediction on such a panel is an empirical question that has not been answered for this chemical space; the literature that motivates them reports their geometric properties, not their benchmarked performance against a well-maintained classical baseline on the same data. This thesis answers it, and the answer is null: the simulated quantum kernel matches a bandwidth-tuned classical kernel (0.823 against 0.829, $p = 0.060$) despite a fourfold higher kernel–target alignment, and topological or tensor fusions add no predictive gain over the extended-connectivity fingerprint. The quantum-inspired designs inherit the information ceiling of the fingerprint they consume: garbage in, garbage out.

The learned-representation collapse (extrapolation facet). Graph neural networks and molecular transformers learn their own representations, yet under Bemis–Murcko scaffold separation the graph isomorphism network reaches 0.8047 against 0.8300 for the fingerprint baseline, and the deficit is largest exactly where structural novelty is greatest. Local message passing aggregates neighbourhoods on a two-dimensional graph; it cannot propagate stereochemical information beyond its receptive field, which is the failure mode the scaffold paradox predicts.

These four facets share a common root cause: no representation examined encodes stereochemistry, bond polarity or three-dimensional geometry directly. Hashed fingerprints project three-dimensional molecules onto fixed two-dimensional dictionaries. Local message passing cannot propagate stereochemical information beyond its receptive field. Quantum kernels fed by fingerprint-derived inputs cannot create three-dimensional information ex nihilo. Whether structure-direct quantum encoding — placing molecular structure into quantum feature space from the outset, so that the circuit itself is the representation — overcomes these blind spots is precisely the question that a genuine quantum machine learning implementation must answer on this chemical space, and it has not been answered.

A fifth, methodological consequence ties the four facets together: they cannot be adjudicated without an evaluation protocol that the literature does not provide. Published comparisons between molecular representations are difficult to compare with one another because each uses its own panel, label source, partition family and multiplicity handling, and several of the most-cited positive results were obtained under random splitting, which the representation-learning literature itself identifies as flattering to learned models. A benchmark that reports one split family, one label source and no power analysis cannot distinguish a genuine representational advantage from an artifact of panel composition. What is missing for antimalarial chemical space is a frozen, fully declared evaluation protocol under which classical and non-classical representations can be compared with the same data, the same splits and the same statistical machinery — which is what this thesis builds and uses for every result it reports.

Research questions

This thesis investigates a single overarching question through four empirical stages.

Central question: can structure-direct quantum molecular encoding overcome the stereochemical blind spots and scaffold-extrapolation failures that limit classical, quantum-inspired and learned molecular representations on African natural-product chemical space, and what must such an implementation demonstrate before any hardware-based advantage claim?

The six research questions follow the four-stage empirical arc of the thesis; each stage consumes the result of the one before it.

1. *Stage I (chemical space and validation).* Does an African-natural-product-inspired generative expansion produce a library that is genuinely novel at the whole-molecule level while retaining the scaffolds of its seed set, and can target-anchored docking on such a library support a resistance-aware prioritization layer?

2. *Stage II (estimand architecture).* Do static docking scores and short explicit-solvent molecular-dynamics trajectories estimate the same quantity across a resistance-mutant panel, and if not, how should the disagreement be used in a triage pipeline?

3. *Stage III (representation-ceiling diagnosis).* Do topological, tensor-compressed and quantum-inspired descriptors add predictive value over an extended-connectivity fingerprint baseline for antimalarial activity prediction on a frozen panel?

4. *Stage III (representation-ceiling diagnosis).* Do graph neural networks and molecular transformers close the gap to classical fingerprints when evaluation enforces structural extrapolation through scaffold-separated partitions, and what structural cause explains the failure if they do not?

5. *Stage IV (structure-direct quantum encoding).* Do the stereochemical blind spots and scaffold-extrapolation failures of hash-based fingerprint prioritization persist when molecular structure is encoded directly into quantum feature space?

6. *Stage IV (evaluation standard).* What evaluation standard must a genuine quantum machine learning study satisfy on this library before any hardware-based claim can be made?

Aim and objectives of the study

The aim of this thesis is to determine, under a single frozen evaluation protocol on African natural-product chemical space, whether quantum, quantum-inspired and learned molecular representations improve antimalarial candidate prioritization over strong classical baselines, and to establish what a genuine quantum machine learning implementation must demonstrate on this library.

The specific objectives, corresponding to the four empirical stages, are:

- *Stage I (chemical space and validation).* To construct an African-natural-product-inspired hybrid library and characterise its chemical space by target-anchored docking against four mechanistically distinct antimalarial proteins.
- *Stage I (chemical space and validation).* To validate the docking protocol by redocking and an external benchmark, declaring the estimand boundaries of static scoring upfront.
- *Stage II (estimand architecture).* To develop a resistance-resilience scoring layer on the docking campaign and stress-test it with explicit-solvent molecular dynamics and a retrospective genotype–phenotype confrontation.
- *Stage III (representation-ceiling diagnosis).* To benchmark topological, tensor-compressed and quantum-inspired descriptors against classical fingerprints under a frozen evaluation protocol with declared power boundaries.
- *Stage III (representation-ceiling diagnosis).* To benchmark graph neural networks and a molecular transformer against the same baselines under scaffold-separated evaluation, diagnosing the structural cause of any failure.
- *Stage IV (structure-direct quantum encoding).* To test whether structure-direct quantum molecular encoding overcomes the stereochemical blind spots and scaffold-extrapolation failures identified by the preceding benchmarks, against classical three-dimensional descriptors as well as fingerprint baselines.
- *Stage IV (evaluation standard).* To define the comparison standard, data, partitions and statistical machinery that the quantum machine learning programme must satisfy before hardware evaluation.

Significance of the study

The significance of this work is threefold.

Methodologically, the thesis delivers the first frozen, fully declared benchmark of classical, topological, tensor and quantum-inspired molecular representations on African natural-product chemical space, with scaffold-grouped partitions, declared label provenance and stated power boundaries. Its negative and null results are documented with the same machinery as its positive ones, which makes them usable: any future quantum implementation on this library inherits a comparison standard rather than a collection of incomparable one-off studies.

For antimalarial research, the resistance-aware layer contributes a concrete, reusable workflow: a corrected wild-type transporter receptor, a ratio-based retention score with a declared eligibility rule, a static-to-dynamic divergence diagnostic with its sampling limits quantified, and a retrospective phenotype check that bounds what the static score can claim. The workflow is deliberately protocol-local, but its components, particularly the WT-eligibility gate and the estimand-divergence framing, transfer to other resistance-relevant target panels.

For the wider drug-discovery community, the honest-negative record assembled here addresses a documented reporting gap: positive representational results are easy to publish and hard to reproduce, while careful null results, which are what actually bound a field, are rarely reported with the detail needed to build on them. This thesis reports its boundaries at the point of each result, so that the quantum machine learning programme that follows starts from measured limits rather than from aspiration.

Organization of the thesis

This thesis is organised into three chapters, followed by a general conclusion.

Chapter 1 presents the literature review. It begins with the biological and clinical background of malaria and its drug-discovery landscape, then covers natural products as a source of antimalarial chemistry, machine learning in drug discovery, the fundamentals of quantum computing and quantum machine learning, and the computational drug-discovery methods on which the thesis builds: molecular docking, molecular dynamics, chemical-space exploration, resistance-resilient scoring and polypharmacology. It closes with molecular representations for machine learning, generative models, integrated validation workflows, and a synthesis that positions the research questions of this thesis within that landscape.

Chapter 2 presents the tools and methods. It specifies the construction of the hybrid library, the four-target docking protocol and its validation, the resistance-resilience scoring framework with its estimand architecture, the molecular-dynamics protocol and its three cohorts, the classical, topological, tensor and quantum-inspired descriptors, the machine-learning architectures, and the statistical framework, including partition families, multiplicity handling, power boundaries, uncertainty taxonomy and the claim-ledger convention used throughout.

Chapter 3 reports the results and discussion as a single empirical arc, study by study. It covers the chemical-space construction and target-anchored docking results, the resistance-aware scoring results including the static-to-dynamic estimand divergence and the retrospective phenotype confrontation, the quantum-inspired and topological representation benchmark, and the learned-representation benchmark under chemical distribution shift; it then reports the structure-direct quantum encoding programme as work in progress — its archived exploratory phase, its frozen encoding substrate, its pipeline confirmation and its predeclared benchmark — before closing with a synthesis that states the comparison standard the quantum machine learning programme must meet.

The general conclusion summarises the findings, discusses their limitations, and suggests avenues for further studies, with the quantum machine learning programme as the principal forward direction.

Overview of the thesis argument

The empirical question at the centre of the thesis can now be stated precisely: under a single frozen evaluation protocol on African natural-product chemical space, do quantum, quantum-inspired and learned molecular representations improve antimalarial candidate prioritization over strong classical baselines, and what must a structure-direct quantum implementation demonstrate before any hardware-based advantage claim? The question is deliberately narrow and empirically answerable. It is addressed through five studies that share one library, one set of activity labels and one evaluation protocol. The chemical-space study constructs an African-natural-product-inspired hybrid library of 65 856 molecules and characterises it by target-anchored docking against four mechanistically distinct antimalarial proteins, together with a per-target resistance-resilience score. The resistance-scoring study converts that docking campaign into a resistance-aware triage framework and then stress-tests it with explicit-solvent molecular dynamics, which reveals that static docking scores and short trajectories are non-equivalent estimands rather than interchangeable measures. The representation study evaluates the descriptors that motivate the quantum framing, namely a persistent-homology topological fingerprint, a Tucker-compressed tensor network embedding and a simulated quantum kernel, against an extended-connectivity fingerprint baseline and a bandwidth-tuned classical kernel. The learned-representation study extends the comparison to graph neural networks and a molecular transformer under scaffold-separated evaluation, where the effect of chemical distribution shift on model ranking becomes measurable. The quantum machine learning programme, the only part of this thesis that uses genuine quantum machine learning, encodes molecular structure directly into parameterized quantum circuits through quantum molecular structure encoding, which places bond-order and Coulomb-adjacency information into one- and two-qubit rotation angles and trains a hybrid quantum-classical model end to end; its canonical benchmark is not yet computed, and no comparative result from it is reported here. Its reformulated benchmark question, stated in advance of the results, is whether this structure-direct encoding surpasses classical three-dimensional descriptors — extended three-dimensional fingerprints (E3FP), ultrafast shape recognition with pharmacophoric atom types (USRCAT) and three-dimensional autocorrelation functions (Autocorr3D) — on the high-complexity quartiles of the African-chemotype structural index under scaffold separation. The programme is presented as the comparison standard that the completed studies define rather than as a source of results, and its central question is whether the two blind spots this thesis documents, stereochemical collapse in hashed fingerprints and the scaffold-extrapolation deficit of learned models, are addressable by an encoding that never passes through a fingerprint at all.

The organising finding of the thesis is negative and is stated in advance so that it cannot be mistaken for a later qualification. Across every representation tested, ECFP4 with a random forest remained the strongest descriptor for activity prediction on the panels studied; no fusion with topological, tensor-compressed, graph-based or sequence-based representations improved on it; and the quantum kernel matched, but did not beat, a well-tuned classical kernel. The completed studies therefore establish three things that a quantum machine learning study should be expected to respect: a well-characterised chemical space in which resistance and breadth are separable properties, a set of classical baselines that are demonstrably hard to beat under scaffold-separated evaluation, and a set of diagnostic quantities, particularly from persistent homology, that describe molecular structure without predicting activity. The chapters that follow develop that argument from the biological and methodological background through to the results, and they report null and negative findings with the same detail as positive ones.

Chapter 1

LITERATURE REVIEW

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1.1 Introduction

This chapter establishes the scientific framework on which the thesis builds, moving from the biology of the disease to the computational methods that this work develops and evaluates. The approach is systematic. It begins with malaria itself: the biology of *Plasmodium falciparum*, the pharmaceutical process through which antimalarial drugs have been discovered, the treatments currently in use, and the resistance mechanisms that make new treatments urgent. These rudimentary elements lay the conceptual groundwork for everything that follows, because the target panel, the resistance mutations and the prioritization objectives of the later chapters are all inherited from this clinical reality.

The discussion then widens to the two source domains of the thesis. The first is natural-product chemistry, and specifically the underexplored chemical space of African medicinal plants, whose scaffold diversity and structural complexity motivate the generative expansion studied in the results. The second is machine learning, whose classical techniques, known limitations on high-dimensional molecular data, and quantum-inspired successors form the methodological spine of the thesis. Quantum computing fundamentals and recent quantum machine learning approaches are presented at the level needed to understand what the quantum kernel of the results chapters does and does not promise.

The chapter then reviews the computational drug-discovery methods employed in this work: molecular docking and structure-based design, molecular dynamics for drug–target interactions, chemical-space exploration and virtual screening, and resistance-resilient scoring with polypharmacology, each presented with the boundaries that later chapters test empirically. Molecular representations for machine learning, generative models for molecule design, and the integration of computational and experimental workflows complete the methodological landscape. The review serves three purposes: to provide the theoretical framework required to follow the thesis, to demonstrate that the research problem sits within a vibrant research domain, and to lay the groundwork for the critical evaluation of existing studies that positions the specific contributions of this work.

1.2 Malaria Drug Discovery Background

Malaria continues to represent a substantial global public health concern, exerting a profound impact on regions where it is endemic. The intricate biology of the malaria parasite, involving its lifecycle and interactions with Anopheles mosquitoes, underscores the necessity for a comprehensive understanding of the dynamics of the disease. Recent research underscores the influence of climatic factors on malaria distribution, particularly in the highlands of East Africa, where defying traditional boundaries signifies a potential threat to previously unaffected populations [56]. Advancements in computational biology have yielded novel insights into the intricacies of malaria transmission, facilitating targeted interventions and policy recommendations [57]. A foundational expertise in malaria biology is essential not only for academic purposes but also for devising effective strategies to combat the disease, thereby enhancing public health outcomes in vulnerable regions.

1.2.1 Overview of malaria biology

The persistent impact of malaria on global health is underscored by its historical prevalence and ongoing threat, particularly in tropical and subtropical regions. Malaria, a potentially lethal disease caused primarily by the Plasmodium falciparum parasite, imposes a substantial burden on public health systems, resulting in elevated mortality rates, particularly among vulnerable populations such as children and pregnant women. In response, concerted efforts have been made to combat the disease, propelling advancements in genomic research aimed at identifying novel biological targets

and drug candidates, which are essential for the development of effective treatment and prevention strategies [58]. Furthermore, shifts in climatic conditions have facilitated the resurgence of malaria in regions previously considered low-risk, such as the highlands of East Africa, thereby sparking debates regarding the role of climate change in its geographic expansion [56]. This multifaceted challenge underscores the necessity for a comprehensive understanding of the biological and environmental factors that influence malaria's persistence as a global health concern.

1.2.1.1 Different species of Plasmodium

Malaria is a parasitic disease instigated by protozoans of the *Plasmodium* genus, which are members of the *Apicomplexa* family. Transmission to humans occurs through the bite of an infected female Anopheles mosquito. Historically, four Plasmodium species were acknowledged as human pathogens. However, recent scientific investigations have uncovered a fifth species that had previously gone undetected [59, 60], thereby adding complexity to the epidemiological landscape of malaria. This finding signifies an increase in the number of Plasmodium species known to infect humans, now totaling five (see Figure 1.2.1). Thus we have :

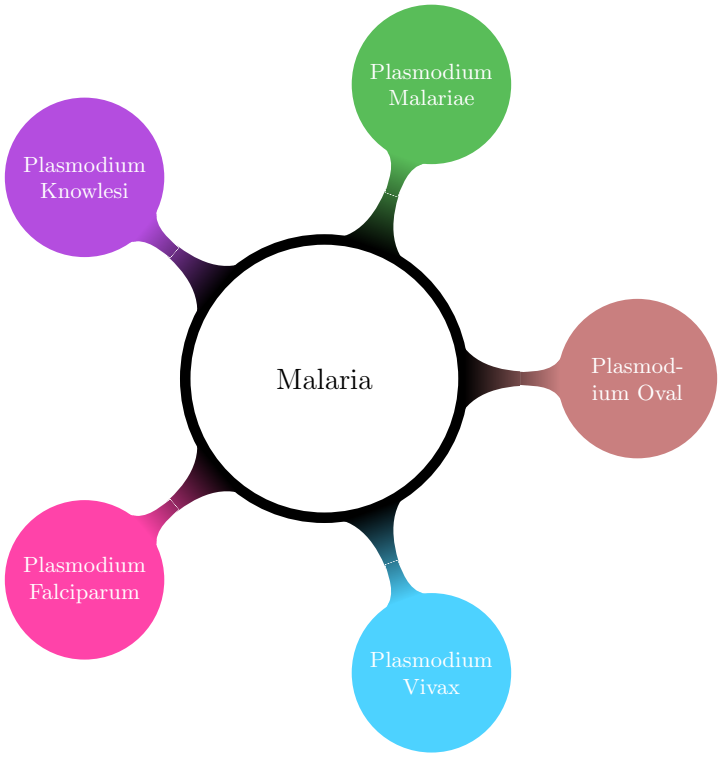


Figure 1.2.1: *Different species of plasmodium : (i) Plasmodium falciparum, (ii) Plasmodium vivax, (iii) Plasmodium oval, (iv) Plasmodium malariae, (v) Plasmodium knowlesi*

- ***Plasmodium Falciparum*** (Pf) *: This species is responsible for the most severe form of the disease, which frequently results in fatal outcomes. The species under consideration is present in the tropical zones of the African, Latin American, and Asian continents. This species has been observed to develop resistant forms, which have been documented in the context of existing antimalarial treatments.*
- ***Plasmodium Vivax*** (Pv) *: The phenomenon under consideration has been observed to co-exist with* Pf *in a variety of geographical regions worldwide, with a notable prevalence in South America and Asia. It has been identified as a causative agent of mild, benign fever. The prevalence of cases involving the consumption of* Pv*, which have been reported to*

result in severe, even fatal, outcomes, underscores the potential risks associated with this species [61, 62, 63].

- ***Plasmodium Oval*** *(*Po*) : The presence of the specie has been documented in the intertropical region of Africa, as well as in specific regions within the Pacific, with a notable prevalence in areas where the presence of the* Pv *is absent.* Po *does not result in immediate mortality; however, it has been observed to induce a state of reviviscence that persists for a duration ranging from four to five years.*
- ***Plasmodium Malariae*** *(*Pm*) : The species exhibits a scattered geographical distribution. Although* Pm *is not lethal, it has the potential to cause relapses twenty years after the initial infection.*
- ***Plasmodium Knowlesi*** *(*Pk*) : It is evident that infections in humans due to Pk are a recent phenomenon. The distribution of this species is limited to the Asian continent. This is explained because this species has long been confused with* Pm *[59, 60].*

1.2.1.2 Life cycle of malaria parasite

The complex interplay of malaria transmission is predominantly characterized by the interaction between the Anopheles mosquito and the human host. The various stages of parasite (see Figure 1.2.2) development play a pivotal role in determining the organism's pathogenic potential. Upon the bite of an infected mosquito, the parasite known as a sporocyst is injected into the victim. This sporocyst rapidly migrates to the liver, where it initiates asexual reproduction. Ultimately, the merozoites disperse into the bloodstream. The malaria parasite displays a multifaceted life cycle, which is initiated through the interaction with an Anopheles mosquito and subsequently continues within the body of a vertebrate host. Following entry into the bloodstream, these merozoites invade red blood cells, where they undergo further multiplication, leading to the symptomatic manifestations of malaria. Recent studies have emphasized the post-transcriptional regulation of gene expression in the parasite, with a focus on the over 1000-fold range. The presence of RNA-binding proteins has been identified, thereby suggesting the presence of a sophisticated mechanism of gene regulation that plays a crucial role in orchestrating the life cycle and survival of the parasite [64]. It is imperative to comprehend the intricacies of this phenomenon to formulate targeted interventions against malaria [65].

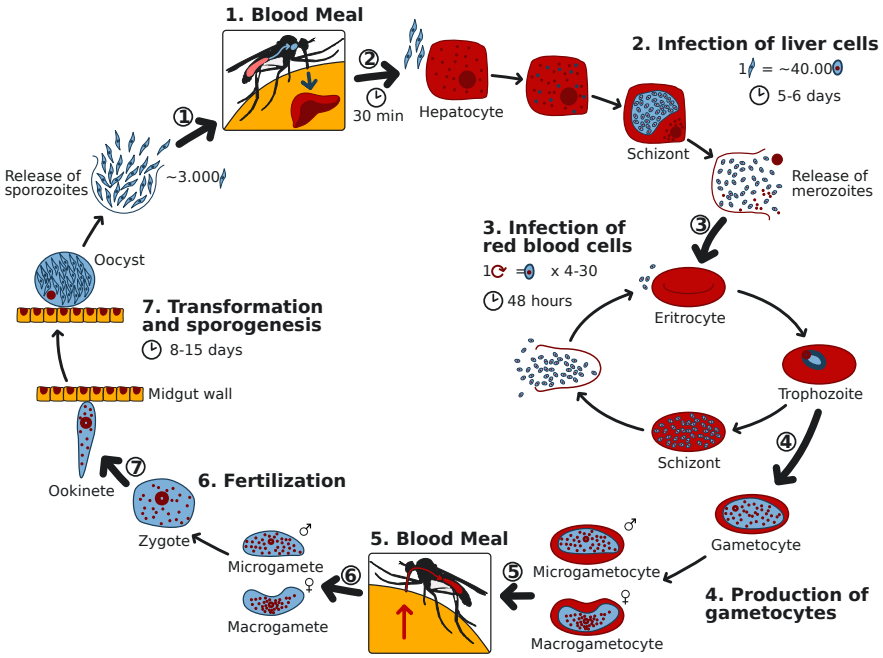


Figure 1.2.2: *Life cycle of the malaria parasite : In the human host, the parasite Plasmodium introduces itself into the bloodstream during a mosquito blood meal (**Step 1**) and rapidly targets hepatocytes (**Step 2**), where it undergoes an exoerythrocytic schizogony over approximately 5–6 days. In the context of a hepatocyte infected with a parasite, a single sporozoite will undergo differentiation into a schizont, yielding approximately 40 000 merozoites. These merozoites will then be released into the circulation. These merozoites invade erythrocytes (**Step 3**), thereby initiating the 48-hour asexual erythrocytic cycle. Each trophozoite matures into a schizont that liberates $4^{\sim}30$ new merozoites upon erythrocyte rupture. A subset of merozoites undergoes differentiation into sexual forms (**Step 4**), thereby giving rise to microgametocytes and macrogametocytes within circulating red cells. Subsequently, during a blood meal (**Step 5**), an Anopheles mosquito ingests these gametocytes. Within the midgut lumen, the gametocytes undergo exflagellation, fusion, and fertilization to form a diploid zygote (**Step 6**). The zygote subsequently transforms into a motileookinete, which traverses the midgut epithelium and establishes an oocyst on its basal surface (**Step 7**). It has been determined that during the 8–15 days of sporogony, each oocyst generates approximately 3000 sporozoites. These sporozoites subsequently migrate to the mosquito's salivary glands (**Step 8**), thereby completing the cycle and rendering the vector capable of transmitting infection to a new human host.*

1.2.1.3 Pathophysiology of malaria

The profound pathophysiological effects of malaria serve to underscore the disease's considerable impact on human health. As the disease progresses, particularly in cases of *Pf* infection, systemic inflammation and the sequestration of parasitized erythrocytes become pivotal. It has been established that these processes are responsible for severe manifestations, such as cerebral malaria, which is characterized by altered neurological functioning. Notably, studies indicate that these substances contribute to the onset of coma in the brain. In the placenta, they have been shown to contribute to low birthweight and preterm labor, and to increase the risk of abortion and stillbirth. In the context of *Pv*, recent advancements in genetics have facilitated a more nuanced comprehension of its mechanisms. Nevertheless, challenges persist due to the constrained capacity of laboratory studies for this species [66]. As demonstrated in the research by [61], both the inflammatory response and the genomics of pathogens play crucial roles in determining disease severity and the interindividual susceptibility observed among affected populations.

1.2.1.4 Mechanisms of disease and immune response in infected individuals

The intricate interplay between malaria pathogens and the host immune response unveils significant insights into disease mechanisms. As Plasmodium species invade erythrocytes, they employ antigenic variation to evade immune detection. This evasion is notably achieved through the expression of variant surface proteins, such as PfEMP1, which complicates the host's ability to mount

an effective immune response. This dynamic is further influenced by the splenic components that are essential for the processing of variant antigens. This phenomenon has been demonstrated in studies of *Pk* infections, where specific antibodies facilitate variant antigen switch events in vivo. As the immune system interacts with these constantly changing antigens, antibodies evolve, thereby enhancing the host's defensive capabilities over time. However, the critical challenge of the essential delay in antibody production underscores the necessity of immunization strategies in combatting malaria. Without prior exposure, the body struggles to respond swiftly to novel infections [67].

1.2.2 Pharmaceutical drug discovery process

In the medical, biotechnology, and pharmacology fields, the process of drug discovery entails the identification of novel potential medications. The contemporary process of drug discovery involves the identification of screening hits, the application of medicinal chemistry, and the refinement of these hits to enhance their affinity, selectivity (to minimize side effects), efficacy/potency, metabolic stability (to extend the half-life), and oral bioavailability. Once a compound that meets all these criteria has been identified, it progresses to drug development before entering clinical trials. This contemporary approach to drug discovery is ordinarily a capital-intensive endeavor, necessitating substantial investments from pharmaceutical companies and national governments. The culmination of the drug discovery process is the acquisition of a patent for the drug in question. The pharmaceutical company must conduct costly Phase I, II, and III clinical trials, and many drugs fail to succeed in these trials. It is imperative to acknowledge the pivotal role that small companies play in this context. Frequently, these enterprises find themselves in a position where they must relinquish their rights to larger firms that are better equipped to undertake the necessary trials. The identification of drugs that achieve commercial or public health success is characterized by a multifaceted interaction among investors, industry, academia, patent laws, regulatory exclusivity, marketing, and the necessity of balancing confidentiality with communication [68].

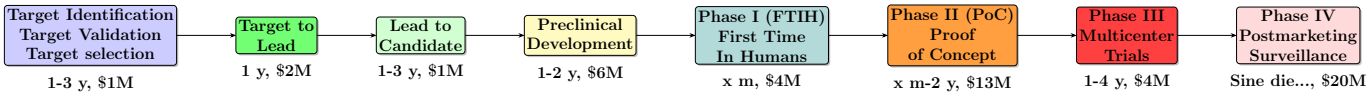


Figure 1.2.3: *Drug discovery process*

The main steps of the drug discovery process (see Figure 1.2.3) are listed below:

- **Target Validation** : The process of target validation in the pharmaceutical industry entails the evaluation of cellular or molecular structures associated with diseases. The term *target* remains a subject of debate; however, the industry distinguishes between established and new targets. The established targets have a substantial scientific literature documenting their role in physiology and disease, requiring less investment for therapeutic development and offering insights on chemical feasibility. New targets lack this established background but are currently pursued in discovery efforts, often representing newly identified proteins or those with recently elucidated functions. As of 2011, 435 human genome products had been identified as therapeutic drug targets for FDA-approved medications [69].

- **Target to lead** : The process of drug discovery is commonly initiated with high-throughput screening (HTS) of chemical libraries, with the objective of identifying compounds that exert an effect on the target. Substances that successfully complete the preliminary screening process are then subjected to cross-screening, a methodical evaluation that aims to substantiate their selectivity and mitigate potential risks of off-target toxicity. The elimination of *pan-assay interference compounds* is initiated at this stage. In instances where a plurality of active compounds exhibit shared chemical characteristics, practitioners of medicinal chemistry may

develop pharmacophores [70]. Subsequently, they enhance lead compounds through iterative structure-activity relationship (SAR) studies to improve potency, selectivity, and ADMET properties.

- **Clinical trials** : The design of new pharmaceuticals clinical trials is generally divided into four phases, although individual trials may span multiple phases, such as combined phase I/II or phase II/III trials. Consequently, a more systematic approach might involve the classification of these studies as either early phase or late phase [71]. The process of developing a new pharmaceutical compound typically traverses all four phases over the course of several years. The successful navigation of Phases I, II, and III by a pharmaceutical agent typically results in its approval for public use by the national regulatory authority. Phase IV encompasses studies conducted subsequent to approval.

  1. *Phase I* trials were formerly referred to as "first-in-man studies," but the field generally transitioned to the gender-neutral language phrase "first-in-humans" in the 1990s. These trials represent the initial stage of testing in human subjects. These studies are conducted with the objective of ascertaining the maximum tolerable dosage of the pharmaceutical agent in question. The primary endpoint of these investigations is to identify the highest dosage that can be administered to a subject without incurring adverse effects that would compromise the patient's well-being or safety. Typically, a cohort of 2–100 healthy volunteers is recruited for this purpose.
  2. *Phase II* : Subsequent to determining a dose or range of doses, the subsequent objective is to evaluate whether the pharmaceutical agent exhibits any biological activity or effect. These trials are conducted on larger groups (100–300) and are designed to assess the efficacy of the drug, as well as to continue Phase I safety assessments in a larger group of volunteers and patients.
  3. *Phase III* : The objective of this phase is to assess the effectiveness of the new intervention and, consequently, its value in clinical practice. Phase III studies are randomized controlled multicenter trials on large patient groups (300–3000 or more, depending on the disease or medical condition studied) and are aimed at providing a definitive assessment of the efficacy of the drug in comparison with the current **gold standard** treatment.
  4. *Phase IV* is also referred to as a postmarketing surveillance trial or, more colloquially, a confirmatory trial. These trials entail the safety surveillance (pharmacovigilance) and ongoing technical support of a drug following its authorization for commercialization (e.g., subsequent to approval under the FDA Accelerated Approval Program) [72].

1.2.3 Current treatments

Notwithstanding substantial progress in the domain of medical science, infectious diseases continue to present considerable challenges to global health, particularly in tropical and subtropical regions, where access to healthcare resources exhibits significant variability. The repercussions of malaria extend beyond the realm of public health, encompassing socioeconomic development and impeding the capacity of healthcare systems to function effectively. The aforementioned complexity is further compounded by cultural factors that influence treatment preferences. This phenomenon is evident in regions where traditional medicine is often preferred due to its perceived efficacy or limited access to conventional biomedical treatments [73]. Moreover, the emergence of drug resistance against conventional antimalarial therapies is indicative of broader concerns in the management of infectious diseases, akin to the challenges observed with antibiotics, where stagnation in new drug discovery and increasing resistance jeopardize treatment efficacy [74].

1.2.3.1 Traditional treatments for malaria

Historical methodologies employed in the treatment of malaria have placed significant emphasis on traditional remedies derived from plant sources [75], as well as on indigenous knowledge, predating the emergence of modern pharmacology by considerable periods of time. Quinine, a natural substance derived from the bark of the cinchona tree, was among the earliest and most significant treatments for malaria. It laid the foundation for the development of modern antimalarial drugs. Despite the advancements in the field of synthetic drug development, many communities persist in their utilization of herbal concoctions, driven by factors such as accessibility and entrenched cultural beliefs. However, the efficacy and safety of such traditional treatments vary widely and often lack rigorous scientific validation. While certain traditional practices may offer potential therapeutic benefits, the emergence of drug-resistant malaria strains highlights the urgent need for treatments that are subjected to rigorous scientific evaluation. Complementary and alternative medical practices, despite their prevalence, often exhibit outcomes that are no more efficacious than placebo when subjected to rigorous study methods. This observation underscores the significant challenges associated with the integration of these remedies into the realm of evidence-based medicine. Consequently, the reliance on traditional treatments must be balanced with advancements in clinically proven therapies to effectively combat malaria.

1.2.3.2 Current advances in malaria treatment

Recent advancements in malaria treatment have expanded beyond conventional indoor interventions to address the challenge posed by outdoor-biting mosquitoes, which maintain transmission even in areas with widespread use of bed nets and indoor spraying. Innovative tools such as the *Mosquito Landing Box*, baited with human odors and treated with mosquitocidal agents, have demonstrated efficacy in attracting and killing outdoor host-seeking vectors, particularly Anopheles species responsible for malaria transmission. As demonstrated by field studies, the deployment of these devices has been shown to enhance the capture of mosquitoes when baited with natural human odors. Furthermore, the optimization of contact with effective insecticides, such as pirimiphos-methyl, has been observed to result in an increase in vector mortality. However, challenges persist in ensuring the rapid and sustained insecticidal action, given the brief visits of mosquitoes [76]. Moreover, the global fight against malaria is intricately linked to a broader array of global concerns regarding antimicrobial resistance. This underscores the necessity for a coordinated development and distribution of new diagnostics, vaccines, and treatments that are adapted to local conditions. As Bhagwandin asserts [77], effective governance and innovative incentive mechanisms are crucial to sustaining research and overcoming resistance hurdles.

1.2.3.3 Antimalarial medecines

In recent decades, there has been a shift in focus towards artemisinin-based combination therapies (ACTs), which have demonstrated efficacy against strains of Plasmodium falciparum that are resistant to other treatments. However, the emergence of resistance to these newer treatments underscores the urgent need for innovative solutions and ongoing research in the field of anti-malarial drug development [78, 79]. The exploration of new candidates is underway, with a particular focus on the identification of compounds that target novel mechanisms of action to combat resistance and enhance treatment outcomes. This persistent endeavor is of paramount importance as malaria continues to represent a substantial public health concern on a global scale. The global burden of malaria remains substantial, with millions of cases reported annually, particularly in sub-Saharan Africa, where the disease is most prevalent [80]. Addressing this challenge necessitates not only the development of new drugs but also a comprehensive approach that includes vector control, effective surveillance, and community engagement in prevention efforts. The integration of these strategies is imperative to enhance the effectiveness of anti-malarial interventions and ensure sustainable disease

control. Moreover, the integration of traditional medicinal knowledge may offer novel insights into the potential identification of natural products that could serve as novel anti-malarial agents [81].

1.2.4 Emerging resistance issues

The efficacy of antimalarial medications is increasingly undermined by the complex strategies employed by malaria parasites to evade drug treatments. Genetic alterations in Plasmodium species have the potential to modify the targets of pharmaceutical drugs or enhance the parasites' capacity to expel therapeutic substances, thereby reducing the drugs' effectiveness. For instance, the resistance exhibited by parasites to artemisinin [82, 83], the prevailing treatment, has been linked to mutations in the Kelch13 gene. These mutations have been shown to influence the parasite's protein balance and its ability to recover from the adverse effects of the drug. The concern is no longer confined to Southeast Asia: partial artemisinin resistance is now confirmed or suspected in at least eight African countries [16], the geographic footprint of *pfk13* resistance markers on the continent is expanding according to spatial-temporal modelling [17], and genomic surveillance documents the co-occurrence of *Pfcrt* and *Pfdhfr* resistance variants across diverse transmission settings [18, 84, 20, 19]. For a computational prioritization workflow this regional epidemiology has a concrete consequence: a candidate ranked only against wild-type receptors is ranked against a parasite population that increasingly carries the very substitutions the ranking ignores, which is the motivation for the resistance-resilient scoring layer reviewed in Section 1.1 and developed in the following chapters. Furthermore, the development of resistance to other antimalarials, including quinine derivatives and antifolates, is attributable to alterations in parasite transport proteins and enzymes that are essential for metabolic processes. This complicates the formulation of effective treatment plans. Recently, sophisticated techniques, including photoaffinity labeling, have played a pivotal role in elucidating these molecular changes by ascertaining the precise biological targets and resistance pathways of antimalarial drugs in live parasites. It is imperative to comprehend these mechanisms to develop novel therapeutic approaches that can counteract resistance and maintain malaria control efforts. The increasing resistance of malaria parasites and vectors to current interventions poses a significant challenge to existing treatment and control strategies [85]. Artemisinin-based combination therapies (ACTs), which were once the foundation of effective treatment, are now threatened by the rise of artemisinin resistance, especially in Southeast Asia. This development risks a reversal of progress made in recent decades. This situation necessitates the implementation of immediate modifications to therapeutic protocols, encompassing the exploration of triple artemisinin combination therapies (TACTs) and the evaluation of alternative drug candidates to ensure the maintenance of efficacy. Concurrently, the emergence of insecticide resistance in vector populations has been observed, thereby diminishing the efficacy of established interventions such as insecticide-treated nets and indoor residual spraying. Innovations such as gene drive technologies and next-generation insecticides hold great promise; however, they must be subjected to a thorough evaluation process to ensure their safety and long-term sustainability. The effectiveness of malaria control strategies is contingent upon the continuous implementation of genomic surveillance, a process that facilitates the expeditious identification of resistance patterns and the subsequent development of adaptive interventions. In the absence of integrated, dynamic strategies, attaining the WHO's ambitious 2030 reduction goals remains improbable. Moreover, recent advancements in genome-based technologies are facilitating the development of innovative approaches to address drug resistance, enabling the identification of new therapeutic targets and the creation of more effective treatment regimens. This collaborative endeavor among various stakeholders is imperative to ensure a comprehensive strategy to combat malaria and achieve long-term control and eradication goals [86].

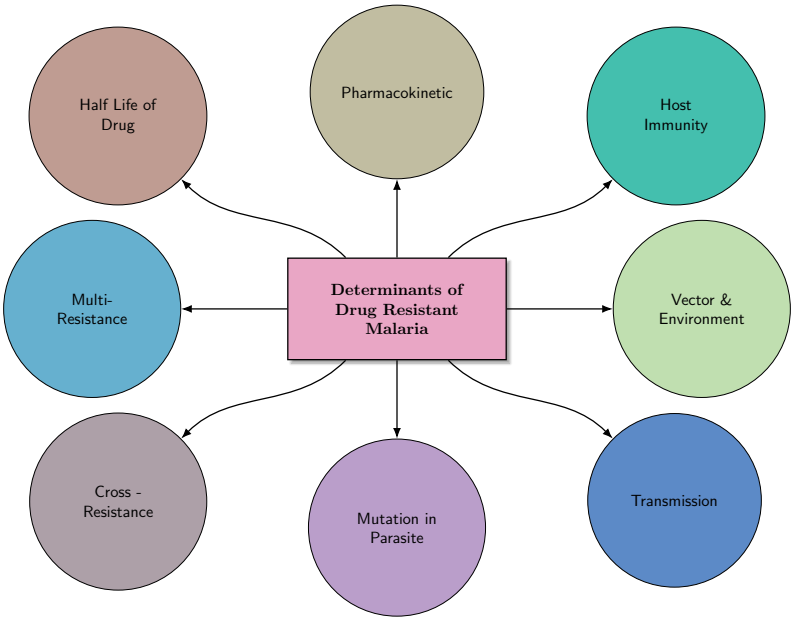


Figure 1.2.4: *Different parameters that contribute to antimalarial resistance [85].*

1.3 Natural Products in Drug Discovery

Natural products (NPs) are substances derived from natural sources such as plants, animals or minerals. They have been used in traditional medicine for centuries and have gained popularity in recent years as people seek natural alternatives to conventional medicines and other synthetic products [87]. There are many different types of NPs, including herbal remedies, essential oils, dietary supplements and organic foods. Some NPs have been scientifically proven to have health benefits, while others have not been thoroughly studied or may be harmful.

1.3.1 Natural products as drug

Many pharmaceutical drugs are derived from NPs or have been inspired by natural compounds. For example, aspirin is derived from salicylic acid, which is found in the bark of willow trees, and the antimalarial drug artemether is derived from artemisinin, which is extracted from the plant *Artimisia annua*, sweet wormwood, a herb used in traditional Chinese medicine. NPs can have a wide range of therapeutic benefits, including antimicrobial, anti-inflammatory, analgesic and immunomodulatory properties. They may also have fewer side effects than synthetic drugs, making them an attractive alternative for some people. However, developing NPs into drugs can be challenging due to issues such as low yields, difficulty in synthesising the compounds and variability in the composition of natural sources. The drug discovery process for NPs is given in Figure 1.3.1. This process fits into the traditional drug discovery process of target identification and validation, hit identification, hit to lead, lead to candidate and development. In addition, NPs can interact with other drugs and have potential side effects, so it's important to use them under the guidance of a healthcare professional. Overall, NPs have the potential to be valuable sources of new medicines, but rigorous scientific research is needed to fully understand their efficacy, safety and mechanisms of action.



Figure 1.3.1: *Natural products drug discovery and development from plant*

1.3.2 Successes of natural products and the untapped potential of African medicinal plants

The exploration of the remarkable successes of natural products, especially within the captivating and fascinating context of African medicinal plants, reveals a vibrant and rich tapestry that is intricately woven with a wealth of traditional knowledge, immense biodiversity, and immense potential for groundbreaking, modern pharmacological applications. An extensive body of literature exists that highlights the considerable significance of these plants as vital and indispensable resources for healthcare, particularly in regions where access to conventional medicine is severely limited or even nonexistent. This underscores the critical importance of meticulously preserving, studying, and understanding these invaluable plants. They hold the extraordinary promise of unlocking new therapeutic options and significantly enhancing our understanding of medicinal chemistry and the intricate relationships between natural compounds and human health.

Obembe et *al.* [88], underscores the effectiveness of Nigerian medicinal plants, which are often preferred over commercially available drugs. The chemical diversity inherent in Nigeria's flora presents a promising avenue for the discovery of novel lead compounds. However, the unsustainable harvesting practices pose a threat to the survival of certain species, underscoring the necessity for conservation efforts and the acknowledgement of traditional medicine's significance in healthcare. Ntie-Kang et *al.* build upon this foundation by underscoring the substantial reservoir of phytochemicals present in African medicinal plants, particularly within the context of traditional medicine [89]. The study illustrates how local populations have historically relied on these plants for treating various medical conditions. The authors advocate for a collaborative approach involving traditional healers and scientists to validate the medicinal uses of these plants, thus facilitating the development of modern therapeutic agents. Additionally, Onguéné et *al.* explore the anti-malarial properties of alkaloids and terpenoids derived from African medicinal plants [90]. The authors posit that the integration of traditional knowledge into the process of drug discovery is of the utmost impor-

tance. However, they also acknowledge the challenges associated with translating this knowledge into practical applications. The necessity for a transition from exclusively academic research to the development of phytomedicines is emphasized, along with the establishment of African centers of excellence in drug discovery. This ongoing investigation into the pharmacological properties of these compounds reflects a growing recognition of the therapeutic capabilities embedded in African flora. Lifongo et *al.* [91], continue this discourse by exploring the bioactivity of Nigerian medicinal plants. The assertion is made that the quality control and standardization of these plant preparations are essential for their acceptance in modern medicine. The authors advocate for a scientific validation of traditional uses to support the development of new drugs, emphasizing the importance of local support for these initiatives. Salmerón-Manzano et *al.* [92], present a comprehensive overview of global research trends concerning medicinal plants, emphasizing the historical context of their therapeutic properties. The authors underscore the persistent reliance on traditional medicine in underdeveloped regions, where access to synthetic drugs is constrained. However, there is a caveat regarding the unregulated use of these plants, as they have the potential to pose health risks. As posited by Ungogo et *al.* [93], the challenges of increasing the production of natural compounds are particularly salient in the context of climate change and environmental degradation. The review under consideration emphasizes the necessity of a multidisciplinary approach to drug discovery that fosters collaboration between academia and the pharmaceutical industry, especially in the context of Nigerian bioresources. Adase et *al.* [94], provides a comprehensive analysis of Parquetina nigrescens, emphasizing its phytochemical and pharmacological characteristics. Their work emphasizes the necessity of rigorous scientific investigation to ensure the efficacy and safety of herbal medicines, thereby supporting the development of novel drugs for widespread health issues. Finally, Fajinmi et *al.* [95], discuss the propagation of medicinal plants in South Africa, highlighting the cultural significance and economic potential of traditional medicine. They advocate for sustainable practices to ensure the availability of these essential resources, which are integral to the healthcare systems in many African communities. This extensive review of the extant literature showcases the complex and varied role of numerous African medicinal plants within the broad field of healthcare, highlighting their immense potential as valuable sources for new and groundbreaking pharmaceuticals. As illustrated in Table 1.3.1, certain aspects indicate that African NPs have the potential to serve as a viable source of pharmaceutical drugs. The text delves into the challenges associated with their use, including critical sustainability issues, the pressing need for rigorous quality control, and the paramount importance of integrating traditional knowledge with modern scientific methods to ensure optimal outcomes. This analysis underscores the necessity for collaborative efforts among researchers, traditional healers, and policymakers to enhance the effectiveness and availability of these extraordinary natural resources.

Aspect	Successes of Natural Products	Untapped Potential	Socioeconomic and Conservation Aspects
Drug Discovery and Biological Activity	African medicinal plants have yielded diverse bioactive compounds (e.g., terpenoids, flavonoids, alkaloids, quinones) with antimicrobial, antimalarial, and anti-ulcer properties.	Many African plant compounds show favorable drug-like properties and are strong candidates for virtual screening and lead development.	Greater focus on preclinical and clinical studies is needed to validate traditional knowledge and accelerate therapeutic discovery.
Commercialization	Plants like *Harpagophytum procumbens* (devil's claw), *Pelargonium sidoides*, and *Aloe vera* have found global commercial success.	Less than 10% of over 5400 known medicinal species in Africa have been fully explored or commercialized.	Sustainable commercialization benefits rural communities and encourages biodiversity through regulated cultivation.
Antimicrobial and Antimalarial Successes	Compounds from African plants have demonstrated effectiveness against pathogens such as *Helicobacter pylori* and malaria parasites, especially where conventional drug resistance is an issue.	Chemical diversity in unexplored species offers new opportunities for antimicrobial innovation.	Traditional medicinal knowledge is at risk due to poor documentation; preservation and integration into research is essential.

Table 1.3.1: *Successes and Potential of African Medicinal Plants*

1.3.3 Computational methods for natural product analysis

The integration of computational methodologies has significantly enhanced the study of NPs, facilitating more precise characterization and identification of bioactive substances. Recent advancements in this field include the application of cheminformatics to assess chemical diversity and the utilization of virtual screening techniques to identify potential therapeutic agents from natural sources [96]. These advancements facilitate the systematic investigation of extensive natural product databases, ultimately leading to the discovery of new drugs that can more effectively target various diseases. Furthermore, the integration of bioinformatics with conventional approaches has led to a paradigm shift in the realm of drug discovery, enhancing its efficiency and cost-effectiveness. Consequently, researchers can now utilize these tools to identify molecular targets and evaluate the druggability of natural compounds with increased precision. This collaboration between computational techniques and NP research establishes the foundation for innovative therapeutic developments in the future. This methodological evolution not only streamlines the drug discovery process but also increases the likelihood of finding novel compounds that might have been overlooked using traditional methods. By leveraging computational techniques, scientists can more effectively navigate the complexities of natural product chemistry and enhance the development of new therapeutic agents. This transition towards computational methodologies signifies a substantial shift in the manner in which researchers approach natural product drug discovery, thereby facilitating a more informed and strategic approach to identifying promising candidates [95, 96]. The future of this field appears promising, with ongoing advancements expected to further refine and improve the effectiveness of natural product-based therapies.

1.4 Machine Learning in Drug Discovey

Machine learning (ML) and deep learning (DL) have rapidly transformed the process of drug discovery. These technologies have enabled the analysis of large, complex datasets and the automation of key steps in the drug development pipeline. These technologies have become indispensable at various stages, from target identification to clinical trials, offering the potential to accelerate timelines, reduce costs, and improve the accuracy of predictions. In the pharmaceutical industry, the processes of drug screening and optimization are of critical importance. The purpose of these procedures is twofold: first, to identify compounds that have therapeutic potential, and second, to ensure the safety and efficacy of these compounds. Recent advancements in ML and DL have significantly enhanced the efficiency and accuracy of these processes, allowing for better candidate selection and optimization strategies. These methodologies employ extensive datasets to predict bioactivity and ADMET properties, thereby facilitating the streamlining of the drug development pipeline [97, 98, 99, 100]. The integration of classical machine learning techniques with deep learning applications has been demonstrated to yield more robust and reliable outcomes in the domain of drug discovery.

In this context, significant applications encompass supervised learning methods, which forecast the biological activity and toxicity of compounds, alongside unsupervised learning techniques that identify patterns within extensive datasets. These approaches refine the accuracy of selecting drug candidates and enhance the predictive modeling of drug efficacy and safety outcomes, thereby supporting more informed decision-making throughout the development process. Despite their promise, the use of traditional ML techniques in drug discovery faces several obstacles, such as challenges related to data quality and availability, model interpretability, and computational requirements [101, 102]. The complex nature of many ML models can lead to a *"black box"* phenomenon, raising concerns about the transparency and reliability of their predictions. Additionally, issues like class imbalance in datasets can hinder the creation of robust predictive models. Overcoming these challenges is crucial to fully harness ML's potential in optimizing drug development and ensuring dependable outcomes in clinical settings. As machine learning continues to advance, it offers both exciting opportunities and significant ethical considerations for the future of drug discovery. The integration of

advanced techniques, along with a focus on personalized medicine, holds the promise of transforming therapeutic approaches and improving patient outcomes. It is essential to recognize the importance of ongoing efforts to enhance data integration, transparency, and collaboration within the field. These initiatives are vital for successfully navigating the complex intricacies of this transformative landscape.

1.4.1 Type of Machine Learning Techniques

Traditional ligand-based and docking-based drug discovery approaches have seen their limitations addressed by the advent of ML. ML methods learn from known data points to predict various properties, such as the binding strength between a potential drug candidate and its target protein [103]. Here is an overview of the type of machine learning used in computational drug discovery (See Figure 1.4.1).

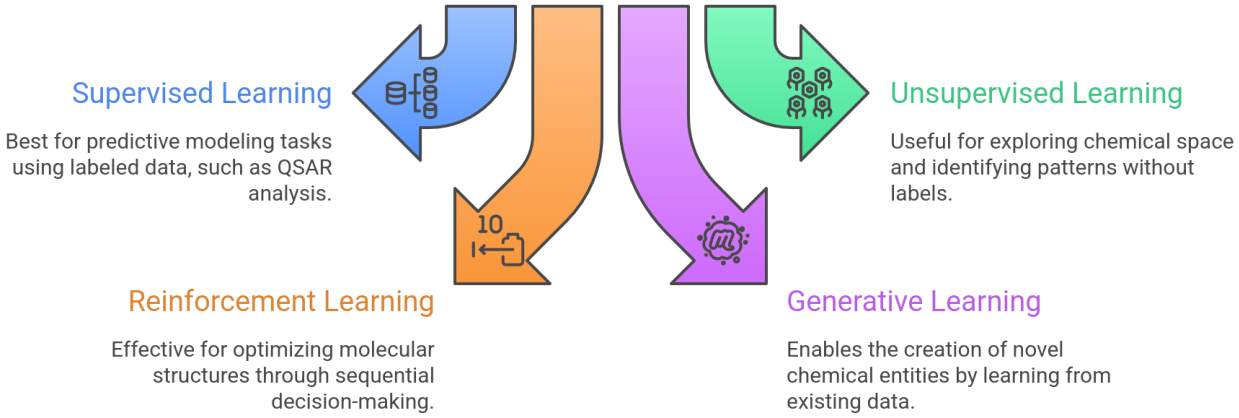


Figure 1.4.1: *Type of machine learning with drug discovery applications*

ML techniques employed in drug discovery can be broadly categorized into supervised, unsupervised, and generative learning paradigms, each addressing distinct stages of the drug development pipeline. Supervised learning methods, including classical algorithms such as support vector machines, random forests, and deep neural networks, are widely used for quantitative structure–activity relationship (QSAR) modeling, enabling the prediction of biological activity, toxicity, and pharmacokinetic properties from molecular descriptors and learned representations. Unsupervised learning techniques, such as clustering, dimensionality reduction, and latent space modeling, play a crucial role in chemical space exploration by identifying structural similarities, discovering hidden patterns among compounds, and reducing the complexity of high-dimensional molecular data. Generative learning approaches, including variational autoencoders, generative adversarial networks, and reinforcement learning–based molecular optimization frameworks, facilitate the de novo design of novel antimalarial candidates by learning the underlying distribution of bioactive molecules and proposing chemically valid and synthetically feasible structures.

1.4.2 Limitations in handling high-dimensional data and non-linearity

1.5 Quantum Computing Fundamentals

The fundamental principles and concepts of quantum computing, which are distinct from those of classical computing, are predicated on the phenomena of quantum mechanics. Quantum computing utilizes these phenomena to process information in unprecedented ways. The foundation of

this technology is rooted in quantum bits, or qubits, which exhibit a fundamental difference from classical bits in that they exist in multiple states simultaneously through a phenomenon known as superposition. This unique property enables quantum computers to perform complex computations with a level of efficiency that far surpasses that of classical systems. Consequently, this technological advancement has led to significant advancements in various fields, including cryptography, optimization problems, and drug discovery. Notwithstanding its potential, the domain of quantum computing confronts substantial challenges, including concerns regarding decoherence, scalability, and error correction, which impede the practical implementation of quantum systems. It is imperative to acknowledge the significance of ongoing research in addressing the prevailing obstacles and realizing the full potential of quantum computing in addressing problems that have historically been considered intractable for classical computers.

1.5.1 Quantum computing concepts

Quantum computing operations generally entail the manipulation of a quantum register through the application of a sequence of pulses that execute unitary transformations based on a designated algorithm. Subsequent to these operations, the quantum state undergoes measurement, resulting in the collapse of the superposition and the production of a single outcome. To ensure the accuracy and reliability of the results, this procedure is typically repeated thousands of times. The outcomes from all measurements are averaged to mitigate quantum uncertainty and achieve statistical precision. It is imperative to reiterate that repetition is of the essence, given that each measurement yields only a probabilistic representation of the quantum system. The following are some fundamental concepts in quantum computing.

- ***Quantum bits (qubits) and quantum gates*** *: In contradistinction to classical computers, which rely on bits, quantum computers utilize qubits. A qubit is defined as the fundamental unit of information capable of existing simultaneously in a superposition of both 0 and 1. This property of being in multiple states at once is crucial for conducting parallel computations and examining various possibilities concurrently. In contrast to classical bits, which are limited to a binary representation of either 0 or 1, qubits possess the capacity to simultaneously represent both values due to the superposition principle. Quantum gates are operations applied to qubits to manipulate and process quantum information, thereby enabling transformations that alter the probabilities of measuring different qubit states.*

- ***Quantum superposition*** *: A single qubit, when considered in isolation, offers limited utility. However, it has been demonstrated that this system is capable of performing a significant feat of quantum physic computation by placing the quantum information it holds into a superposition, thereby merging all possible qubit configurations. In the context of quantum physics, when qubits are in a state of superposition, they generate complex, multidimensional computational environments. These environments facilitate the representation of complicated problems in innovative ways. The principle of quantum superposition enables qubits to exist in multiple states simultaneously. This characteristic is employed in quantum algorithms to facilitate large-scale computations by concurrently examining numerous potential solutions.*

- ***Quantum entanglement****: This phenomenon occurs when two or more qubits become interconnected in such a manner that the state of one qubit is dependent on the state of the others. In the context of quantum physics, when qubits are said to be* "entangled"*, this implies that the observation of one immediately influences the state of the other, regardless of their spatial separation. This entanglement facilitates the formation of highly correlated*

quantum systems and serves as a vital resource in quantum computing algorithms. In the domain of drug discovery, quantum computing employs the principles of superposition and entanglement to conduct intricate simulations, refine molecular structures, and address computational challenges encountered in areas such as molecular dynamics simulations, drug target identification, and drug molecule optimization. By leveraging the capabilities of quantum physics, quantum computing has the potential to transform the drug discovery process and expedite the advancement of personalized medicine.

1.6 Quantum Machine Learning

Quantum machine learning (QML) represents a promising frontier at the intersection of quantum computing and artificial intelligence. QML algorithms leverage quantum mechanical phenomena—such as superposition and entanglement—to enhance certain computational tasks within machine learning workflows. The fundamental premise is that quantum systems can process information in ways that classical architectures cannot, potentially offering computational advantages for specific problems in chemistry and drug discovery. Nevertheless, the practical realization of these advantages remains an area of active investigation, with current hardware limitations and algorithm maturity presenting significant challenges.

1.6.1 Recent QML algorithms and their potential for accelerating drug discovery

Several QML algorithmic frameworks have been proposed for applications in computational chemistry and drug discovery. Quantum kernel methods, which leverage the Hilbert space of quantum states to compute similarity measures between molecular representations, have been investigated as alternatives to classical kernel approaches. Variational quantum eigensolvers (VQE) offer a hybrid quantum-classical approach to solving electronic structure problems, potentially enabling more accurate ground-state energy calculations for molecular systems of pharmaceutical interest. Quantum neural networks, composed of parameterized quantum circuits, have been explored for supervised learning tasks such as molecular property prediction.

Despite theoretical promise, empirical evaluations on near-term quantum hardware have yielded mixed results. Recent benchmark studies suggest that for current problem sizes and noise levels, quantum kernel methods do not consistently outperform well-optimized classical baselines such as radial basis function kernels or tree-based ensemble methods. For instance, comparative assessments on molecular activity prediction panels have observed no significant quantum advantage when evaluated under rigorous statistical testing and appropriate multiple comparison corrections. This observation underscores the importance of honest-negative reporting in the field: claims of quantum superiority must be substantiated through controlled experiments with clearly defined metrics, appropriate baselines, and statistical validation.

A more recent development addresses a structural weakness of the fingerprint-fed designs. In kernel and variational schemes built on top of hashed fingerprints, the quantum circuit consumes a compressed classical descriptor, so the quantum component carries no structural information beyond what the fingerprint already captured; whatever stereochemistry, bond polarity or ring geometry the fingerprint discarded is invisible to the quantum layer. Quantum molecular structure encoding (QMSE) proposes the opposite route: molecular bond-order and Coulomb-adjacency matrices are encoded directly as one- and two-qubit rotation angles, with bond orders, Z/E geometry and stereocentre configuration represented by signed rotation parameters, so the circuit itself becomes the molecular representation rather than a downstream consumer of one [104]. Whether this

structure-direct route can overcome the documented blind spots of hashed fingerprints, which collapse stereoisomers, and of local message passing, which cannot distinguish ring geometries across unseen scaffolds, is an empirical question rather than a property of the scheme, and it defines the design of the quantum machine learning programme of this thesis.

1.6.2 Quantum applications for drug discovery and opportunities for malaria research

The potential application of quantum computing to malaria drug discovery is predicated on several computational bottlenecks that quantum algorithms may address. Accurate simulation of molecular electronic structure, particularly for transition metal complexes or systems with strong electron correlation, remains computationally expensive for classical methods. Quantum simulation algorithms could, in principle, provide more tractable pathways to these calculations. Similarly, optimization tasks in chemical space exploration—such as identifying molecular candidates that simultaneously satisfy multiple pharmacological constraints—may benefit from quantum annealing or variational optimization approaches.

However, practical deployment of quantum methods for malaria-specific drug discovery faces substantial challenges. Current quantum hardware exhibits limited qubit counts, short coherence times, and high error rates, restricting the size and complexity of molecular systems that can be reliably simulated. Furthermore, the translation of quantum computational results into actionable drug discovery decisions requires integration with experimental validation pipelines, including *in vitro* assays and *in vivo* models, which remain outside the quantum computing paradigm. Consequently, the most promising near-term applications may involve hybrid quantum-classical workflows, where quantum subroutines are invoked for specific subtasks within a predominantly classical computational framework.

1.7 Computational Drug Discovery Methods

The integration of computational techniques into drug discovery has fundamentally transformed the efficiency and scope of therapeutic candidate identification. Computational approaches enable the systematic exploration of vast chemical spaces, the prediction of molecular properties prior to synthesis, and the prioritization of candidates based on multi-objective criteria. These methods complement experimental workflows by reducing the number of compounds requiring physical synthesis and biological testing, thereby accelerating the overall discovery timeline and reducing associated costs.

1.7.1 Molecular docking and structure-based design

Molecular docking is a computational technique that predicts the preferred orientation and binding affinity of a small molecule ligand when bound to a macromolecular target, typically a protein. Docking algorithms employ scoring functions to estimate the free energy of binding, enabling the ranking of candidate compounds based on their predicted affinity for a given target. Structure-based drug design leverages three-dimensional structural information, often derived from X-ray crystallography or cryo-electron microscopy, to guide the identification and optimization of lead compounds.

In antimalarial drug discovery, molecular docking has been extensively applied to key targets within the *Plasmodium falciparum* proteome. These targets include dihydrofolate reductase (PfDHFR), which is essential for folate metabolism; the chloroquine resistance transporter (PfCRT), implicated in drug efflux mechanisms; the ATPase PfATP4, involved in sodium homeostasis; and

the protease PfClpP, which plays a role in protein quality control. Computational screening campaigns targeting these proteins have identified numerous candidate compounds with predicted binding affinities in the low micromolar to nanomolar range.

Nevertheless, docking scores must be interpreted with appropriate caution. Scoring functions are approximations of the true binding free energy and exhibit systematic biases depending on the target and chemical series. Docking-predicted affinities do not directly correspond to experimentally measured IC_{50} or EC_{50} values, and false positives are common. Validation through orthogonal methods, including surface plasmon resonance, isothermal titration calorimetry, or cellular assays, remains essential. Furthermore, docking studies typically assume a static receptor structure, neglecting conformational flexibility and solvent effects that can significantly influence binding thermodynamics.

1.7.2 Molecular dynamics simulations for drug-target interactions

Molecular dynamics (MD) simulations provide a time-resolved, atomistic description of molecular systems by numerically integrating Newton's equations of motion. In drug discovery, MD simulations are employed to model the dynamic behavior of drug-target complexes, assess binding stability, and estimate free energy differences between bound and unbound states. Force fields, which parameterize atomic interactions, are central to the accuracy of MD simulations. Common choices for biomolecular systems include CHARMM, AMBER, and OPLS, while small molecule parameterization may employ CGenFF, GAFF, or more recent methods based on quantum chemical calculations.

MD simulations can complement docking studies by providing insights into the conformational dynamics of ligand-receptor interactions over nanosecond to microsecond timescales. Free energy perturbation (FEP) and molecular mechanics/Poisson-Boltzmann surface area (MM-PBSA or MM-GBSA) methods enable the estimation of relative or absolute binding free energies, which correlate more closely with experimental affinity measurements than static docking scores. However, MD-based free energy calculations are computationally intensive and require careful protocol design, including adequate sampling, convergence checks, and consideration of protonation states and solvent models.

In the context of antimalarial research, MD simulations have been applied to model drug resistance mechanisms. For instance, substitutions in PfDHFR (N51I, C59R, S108N and I164L) and in PfCRT (K76T) are associated with resistance to antifolate and chloroquine-based therapies, respectively; the K76A state is used as a mechanistic comparator for perturbation of Lys76 rather than as a clinical resistance marker. MD simulations can elucidate how these mutations alter the binding mode or stability of drug molecules, providing a molecular-level understanding of resistance emergence. Such insights inform the design of next-generation compounds that retain activity against resistant parasite strains.

1.7.3 Chemical space exploration and virtual screening

Chemical space refers to the multidimensional descriptor space in which molecular structures can be represented. Exploration of chemical space aims to identify regions enriched in bioactive compounds while avoiding areas associated with toxicity or poor pharmacokinetics. Virtual screening techniques systematically evaluate large libraries of compounds—often numbering in the millions—to identify candidates with desirable properties. Virtual screening workflows typically integrate molecular fingerprints, such as extended-connectivity fingerprints (ECFP) or molecular access system (MACCS) keys, with similarity-based or machine learning–based models to predict activity.

A critical challenge in virtual screening is achieving chemical diversity while maintaining drug-like properties. Lipinski's rule of five and related heuristics provide guidelines for oral bioavailability, but these rules are not absolute and may exclude biologically relevant chemical scaffolds. Modern approaches employ multi-objective optimization to balance conflicting criteria, such as predicted

potency, selectivity, synthetic accessibility, and predicted resistance resilience. Pareto-optimal solutions, which represent trade-offs among these objectives, offer a transparent framework for candidate prioritization.

African natural product libraries represent an underexplored source of chemical diversity for antimalarial drug discovery. These libraries contain scaffolds distinct from synthetic compound collections, often featuring complex polycyclic structures and stereochemical diversity. Computational screening of African medicinal plant–derived compounds has identified numerous candidates with predicted activity against malaria targets. However, translating these computational predictions into validated hits requires addressing challenges related to compound availability, purity, and structural confirmation.

1.7.4 Resistance-resilient scoring and polypharmacology

Drug resistance remains a persistent challenge in malaria treatment, driven by genetic mutations in parasite targets and transporters. Computational methods can integrate resistance-related information into candidate prioritization by evaluating predicted binding not only against wild-type targets but also against clinically relevant mutant forms. A resistance resilience score (RRS) is defined per target and per mutation as the ratio of the mutant to the wild-type docking-score magnitude, expressed as a percentage; candidates whose wild-type score magnitude falls below a declared binding threshold are excluded from that target's denominator rather than scored against a weak baseline. The quantity is a score-retention statistic rather than a free-energy difference, and the classes derived from it are operational categories rather than biological resistance phenotypes. Its interpretation therefore depends on the structural confidence of the receptor and on explicit null controls, as the analysis in Chapter 3 shows for the corrected PfCRT channel[105]. A further development treats the static-to-dynamic contrast itself as an observable: when rigid-grid scores and short explicit-solvent trajectories are applied to the same mutant panel, the two procedures can disagree in a consistent direction because they estimate different quantities, a phenomenon framed as estimand divergence. Reported disagreement of this kind is used as a calibration flag that prioritizes states for experimental follow-up rather than as evidence against either method, since docking scores and nanosecond trajectories carry complementary structural information.

Polypharmacology, the concept that a single compound can modulate multiple targets, offers an additional strategy to combat resistance. Multi-target compounds may reduce the selective pressure for resistance emergence and provide synergistic therapeutic effects. Computational polypharmacology workflows evaluate candidate molecules against panels of targets, assessing both on-target activity and potential off-target liabilities. Network-based approaches, such as polypharmacology network scoring (PNS), quantify the connectivity of a compound to multiple targets within a biological network, providing a systems-level perspective on drug action.

Empirical studies have explored correlations between computational polypharmacology metrics and resistance resilience scores, and the observed associations depend strongly on cohort size. On a 17-candidate cohort, network similarity correlated negatively with mean RRS ($\rho = -0.714$, $p = 0.0013$), as did the number of targets meeting a within-target favourability criterion ($\rho = +0.714$, $p = 0.0013$), whereas a chemical-space descriptor of African-natural-product character did not ($\rho = -0.190$, $p = 0.465$); on the narrower 12-candidate two-target subset none of the three associations reached significance after correction. The two dimensions therefore measure different properties, but at these sample sizes the tests are descriptive rather than confirmatory: a two-sided test at $n = 17$ reaches 80 % power only for $|\rho| \geq 0.62$. Computational predictions of polypharmacology do not constitute evidence of biological efficacy, and experimental validation through multi-target assays and phenotypic screens remains necessary. The same boundary applies to the resistance scores themselves: when static docking score differences are confronted retrospectively with published genotype-resolved inhibition shifts for dihydrofolate reductase, the two quantities fail

to track each other at any genotype, so rigid-receptor score differences must not be read as resistance predictions even when a computational panel shows favourable retention ratios.

1.8 Molecular Representations for Machine Learning

The performance of machine learning models in drug discovery critically depends on the choice of molecular representation. Representations encode chemical structures into numerical vectors suitable for input to predictive models. Classical representations, such as molecular fingerprints and physicochemical descriptors, have been widely adopted due to their computational efficiency and interpretability. More recent approaches explore learned representations derived from graph neural networks or sequence-based models, as well as representations informed by quantum chemical or topological considerations.

1.8.1 Classical molecular fingerprints and descriptors

Extended-connectivity fingerprints (ECFP), also known as Morgan fingerprints or circular fingerprints, are among the most widely used molecular representations in cheminformatics. ECFP encodes the local chemical environment of each atom within a specified radius, producing a fixed- or variable-length binary vector. ECFP4 (radius 2) is a common choice, balancing structural resolution with computational efficiency. These fingerprints are particularly effective for substructure searching and similarity-based virtual screening.

Physicochemical descriptors, such as molecular weight, LogP, topological polar surface area (TPSA), and hydrogen bond donor/acceptor counts, provide interpretable features related to drug-like properties. Descriptor-based models are transparent and readily interpretable, facilitating mechanistic insights. However, fixed descriptor sets may fail to capture complex structure-activity relationships, particularly for structurally diverse or novel chemical scaffolds.

1.8.2 Graph neural networks and learned representations

Graph neural networks (GNNs) operate directly on molecular graphs, where atoms are nodes and bonds are edges. Message-passing neural networks (MPNNs), graph isomorphism networks (GINs), and graph attention networks (GATs) iteratively aggregate information from neighboring atoms to produce node-level or graph-level embeddings. These learned representations can capture higher-order structural patterns not explicitly encoded in classical fingerprints.

Despite their architectural sophistication, GNNs do not universally outperform classical fingerprint-based models. Under a stratified random split, a graph isomorphism network and a SMILES transformer both approach an ECFP4 random-forest baseline; under Bemis–Murcko scaffold separation the ranking is preserved but the deficits become resolvable, with ECFP4–RF reaching a mean test ROC-AUC of 0.8300 against 0.8047 for a graph isomorphism network, 0.8138 for the same network with topological fusion, and 0.7867 for the transformer across 25 fold–seed replicates. The deficit is not uniform: it is largest for test compounds whose nearest training neighbour is least similar, and it is not closed by increasing model capacity. This pattern suggests that for moderately sized datasets (on the order of 10^4 compounds), the inductive bias provided by ECFP's substructure-aware design remains highly effective, and that the expressive ceiling of local message passing matters most in the low-similarity tail rather than uniformly. GNNs may offer advantages in larger-scale datasets or when incorporating auxiliary information, such as protein–ligand interaction data, but these scenarios remain less common in early-stage drug discovery.

1.8.3 Topological and quantum-inspired descriptors

Topological data analysis (TDA) provides a framework for extracting global structural features from molecular graphs through persistent homology. Persistent homology quantifies the multi-scale connectivity and void structures within a molecular graph, producing topological fingerprints (TFP) that encode information orthogonal to classical descriptors. Similarly, quantum-inspired descriptors, such as those derived from quantum kernel methods, embed molecules into Hilbert spaces defined by quantum feature maps.

Empirical evaluations of topological and quantum-inspired descriptors on antimalarial activity prediction tasks have yielded consistent negative results when the comparison is made against a well-optimized baseline. Fusing multi-scale persistent-homology features into a graph network improves on the base graph architecture in every structural cohort examined, yet the same descriptors stacked onto ECFP4 change the area under the ROC curve by -0.0002 ($p = 0.084$) and every feature-level fusion is worse than ECFP4 alone. Quantum kernel methods behave similarly: a six-qubit kernel is indistinguishable from a bandwidth-tuned radial basis function kernel internally ($p = 0.060$) and performs lower than it under a separately labelled transfer panel, despite achieving a substantially higher kernel–target alignment. Persistent homology and tensor compression remain informative about molecular structure, but that information does not translate into predictive advantage on the panels studied. These findings underscore the importance of honest-negative reporting: novel representations should be assessed not only for their theoretical novelty but also for their empirical utility relative to well-optimized baselines.

The clearest practical role for these representations is diagnostic rather than predictive. Persistent homology decomposes the apparent conflict between whole-molecule novelty and scaffold conservation in generative expansion into conserved ring topology and diverging connectivity[106], and it therefore provides an interpretable check on what a molecular generator has actually produced. Where topological fusion does help a learned model, the benefit is small and does not scale with structural complexity: the gain over the base graph architecture was largest on flat aromatic scaffolds and smallest, with an interval including zero, on the small polycyclic cohort.

1.9 Generative Models for Molecule Design

Generative machine learning models offer the capability to propose novel molecular structures that satisfy specified property constraints. These models learn the distribution of known bioactive molecules and generate new candidates by sampling from the learned distribution. Generative approaches are particularly valuable for de novo design, where the goal is to explore uncharted regions of chemical space rather than screen existing libraries.

1.9.1 Variational autoencoders and generative adversarial networks

Variational autoencoders (VAEs) and generative adversarial networks (GANs) are two prominent generative modeling paradigms. VAEs compress molecular representations into a continuous latent space and reconstruct molecules through a decoder network. By sampling from the latent space, novel molecular structures can be generated. GANs employ a generator-discriminator framework, where the generator learns to produce molecules indistinguishable from real training examples.

Scaffold-based VAEs, such as junction tree VAEs or graph-based VAEs, have been developed to ensure that generated molecules adhere to chemical validity constraints. These models decompose molecules into substructures (e.g., rings, functional groups) and reassemble them during generation, reducing the frequency of chemically infeasible outputs. However, generative models face challenges related to mode collapse, where the generator produces a limited diversity of outputs, and the difficulty of incorporating multiple property constraints simultaneously.

1.9.2 Reinforcement learning and multi-objective optimization

Reinforcement learning (RL) frameworks enable the optimization of molecular properties through iterative design cycles. In RL-based molecular design, an agent proposes molecular modifications (e.g., adding or removing atoms, changing bond types) and receives rewards based on the predicted properties of the resulting molecule. Policy gradient methods, such as proximal policy optimization (PPO), guide the agent toward regions of chemical space enriched in desirable compounds.

Monte Carlo tree search (MCTS), combined with RL, provides a systematic approach to exploring the exponentially large space of possible molecular modifications. MCTS balances exploration (trying new modifications) and exploitation (refining promising candidates) through a tree-structured search process. Multi-objective MCTS extends this framework to optimize multiple conflicting objectives—such as predicted potency, synthetic accessibility, and resistance resilience—simultaneously, and Pareto-guided variants have been applied specifically to expose multi-objective trade-offs that scalar antimalarial generation objectives miss[107].

1.10 Integration of Computational and Experimental Workflows

The ultimate success of computational drug discovery depends on the seamless integration of computational predictions with experimental validation. Computational models provide prioritization hypotheses, but experimental assays remain the definitive arbiter of biological activity, safety, and efficacy. Effective integration requires bidirectional feedback: computational models inform experimental design, and experimental results refine computational models.

1.10.1 Validation cascades and hit-to-lead progression

A typical validation cascade begins with computational screening to prioritize a manageable subset of candidates for experimental evaluation. Initial experimental assays may include biochemical binding assays (e.g., surface plasmon resonance, fluorescence polarization) to confirm target engagement. Candidates demonstrating acceptable binding affinities progress to cellular assays, assessing activity against parasite cultures. Compounds exhibiting potent cellular activity undergo counter-screens to evaluate selectivity and cytotoxicity against human cell lines.

Hit-to-lead optimization involves iterative cycles of structural modification, computational re-evaluation, and experimental testing. Structure-activity relationship (SAR) studies guide modifications aimed at improving potency, selectivity, and pharmacokinetic properties. Computational methods, including free energy calculations and ADMET predictions, inform which modifications are most likely to yield improved candidates. This iterative process continues until lead compounds meet predefined criteria for advancement to preclinical development.

1.10.2 Prospective validation and retrospective analysis

Prospective validation, where computational predictions are tested in experiments designed after the predictions were made, provides the strongest evidence of model utility. Retrospective validation, where models are evaluated on historical data, is less definitive due to potential overfitting and lack of temporal separation between training and test sets. Rigorous prospective validation studies are resource-intensive but essential for establishing the predictive value of computational workflows.

In the context of antimalarial research, prospective validation campaigns have evaluated computational predictions against experimental outcomes for compound panels not used in model training. These studies assess metrics such as hit rate (the fraction of predicted actives that are experimentally

confirmed), enrichment factor (the improvement over random selection), and area under the receiver operating characteristic curve (ROC-AUC). Results vary depending on the target, chemical series, and computational methodology, underscoring the importance of context-specific model evaluation.

1.11 Conclusions and Future Perspectives

This chapter has provided a comprehensive review of the multifaceted landscape of malaria drug discovery, spanning from the biological foundations of the disease to cutting-edge computational methodologies. The persistent global burden of malaria, particularly in sub-Saharan Africa, underscores the urgency of developing novel therapeutic interventions that can overcome emerging resistance mechanisms and provide durable protection for vulnerable populations. The integration of traditional knowledge regarding natural products, particularly African medicinal plants, with modern computational drug discovery approaches offers a promising pathway toward identifying structurally diverse antimalarial candidates.

The emergence of drug resistance represents one of the most formidable challenges in malaria chemotherapy. Substitutions in established resistance determinants, notably PfDHFR and PfCRT, have compromised the efficacy of antifolate and chloroquine-based therapies, while additional parasite vulnerabilities such as PfATP4 and PfClpP remain active areas of target discovery, necessitating the development of compounds that retain activity against resistant strains. Computational methods, including resistance-resilient scoring frameworks and molecular dynamics simulations of mutant protein-ligand complexes, provide valuable tools for understanding resistance mechanisms at the molecular level and guiding the design of next-generation antimalarials. However, computational predictions must be validated through rigorous experimental testing, including phenotypic assays against resistant parasite lines and pharmacokinetic profiling in relevant animal models.

The exploration of chemical space through computational screening has revealed vast regions of potentially bioactive molecules that remain experimentally untested. Large-scale virtual screening campaigns, leveraging molecular docking, machine learning models, and multi-objective optimization, have identified numerous candidate compounds with predicted activity against malaria targets. Notably, the integration of African natural product libraries into these screening efforts has uncovered scaffolds distinct from conventional synthetic compound collections, offering opportunities for discovering novel modes of action. Nevertheless, the translation of computational hits into validated leads requires addressing challenges related to compound availability, structural confirmation, and synthetic feasibility.

Machine learning and deep learning methodologies have demonstrated considerable utility in predicting molecular properties, prioritizing candidates, and accelerating hit-to-lead optimization. Classical representations, such as extended-connectivity fingerprints, have proven remarkably effective for antimalarial activity prediction, often outperforming more complex learned representations from graph neural networks or transformer models. This observation highlights a critical lesson: model complexity does not guarantee superior performance, and well-optimized classical baselines should serve as benchmarks for evaluating novel methodologies. Topological and quantum-inspired representations offer diagnostic information that classical fingerprints do not expose, but in the benchmarks reviewed here they provided no detectable predictive improvement when integrated into ensemble models, and their contribution must therefore be judged by what they reveal about molecular structure rather than by an expected gain in accuracy.

Generative machine learning models, including variational autoencoders, generative adversarial networks, and reinforcement learning–based molecular design frameworks, offer the potential to explore uncharted regions of chemical space and propose de novo candidates, provided that their evaluation is guided by rigorous experimental design, appropriate baseline comparisons, and transparent reporting of both successes and limitations.

Quantum machine learning represents a frontier area with theoretical promise but limited empirical evidence of practical advantage for current drug discovery applications. Quantum kernel methods, variational quantum eigensolvers, and quantum neural networks have been proposed for molecular property prediction and electronic structure calculations. Benchmark evaluations on antimalarial activity prediction tasks have not demonstrated superiority over classical methods such as radial basis function kernels or tree-based ensembles: on a 19 849-molecule antimalarial panel, a simulated quantum kernel reached an area under the ROC curve of 0.823 against 0.829 for a bandwidth-tuned radial basis function kernel, and 0.817 against 0.847 on a separately labelled transfer panel. The discrepancy between a kernel's geometric properties and its classification performance is itself an empirical finding, because the same quantum kernel achieved a kernel–target alignment more than four times that of the classical kernel without translating that alignment into predictive accuracy. Structure-direct encodings such as QMSE change what the quantum component receives rather than how it is compared: they remove the fingerprint layer whose information ceiling the benchmarks above expose, and they therefore shift the empirical question from whether a quantum kernel matches a tuned classical kernel to whether molecular structure encoded as circuit parameters transfers across unseen scaffolds [104]. Near-term quantum computing hardware faces substantial limitations, including restricted qubit counts, short coherence times, and high error rates, which constrain the size and complexity of molecular systems that can be reliably simulated. Hybrid quantum-classical workflows may offer the most pragmatic pathway for integrating quantum computational methods into drug discovery pipelines.

The successful integration of computational and experimental workflows requires bidirectional feedback, where computational models inform experimental design and experimental results refine computational predictions. Validation cascades, beginning with biochemical assays and progressing through cellular phenotypic screens to in vivo efficacy studies, provide the necessary evidence for advancing candidates toward clinical development. Prospective validation studies, where computational predictions are tested in experiments designed after the predictions were made, offer the strongest evidence of model utility. Retrospective analyses, while informative, are susceptible to overfitting and lack the temporal separation required for definitive validation.

Looking forward, several key opportunities and challenges merit emphasis. First, the systematic integration of resistance-related information into computational screening workflows will enhance the identification of compounds with broad-spectrum activity against wild-type and mutant parasite strains. Second, the expansion of chemical diversity through the exploration of African natural product libraries and the application of generative modeling approaches will enrich the pool of candidates available for optimization. Third, the development of more accurate force fields and free energy calculation methods will improve the predictive power of molecular dynamics simulations, enabling more reliable assessments of binding affinity and resistance resilience. Fourth, the maturation of quantum computing hardware and algorithms may eventually enable quantum methods to address computational bottlenecks that remain intractable for classical approaches, although this prospect remains speculative at present.

Finally, the translation of computational discoveries into clinical therapies requires sustained collaboration among computational chemists, medicinal chemists, biologists, pharmacologists, and clinicians. No single methodology, whether classical or quantum, can independently solve the multifaceted challenges of antimalarial drug discovery. Rather, the integration of complementary approaches, guided by rigorous validation and honest reporting of both positive and negative results, will provide the most robust pathway toward developing the next generation of life-saving therapies. The work presented in this thesis contributes to this integrative vision by applying quantum machine learning and classical computational methods to the identification and optimization of antimalarial candidates, with a focus on rigorous evaluation, transparent reporting, and practical utility for advancing global health outcomes.

Chapter 2

MATERIAL AND METHODOLOGY

2.1 Introduction

This chapter presents the computational methodologies, algorithms, and protocols employed throughout this thesis to investigate quantum machine learning approaches for antimalarial drug discovery. The work encompasses five interconnected studies, each addressing a distinct aspect of the computational drug discovery pipeline: a chemical-space construction and multi-target docking study on an African natural-product-inspired library; a resistance-aware scoring study that validates its predictions through explicit-solvent molecular dynamics; an evaluation of quantum-inspired molecular descriptors against classical representations; a benchmark of graph neural networks and transformer models against classical fingerprint-based approaches; and a quantum machine learning programme, refocused on quantum molecular structure encoding, in which bond-order and Coulomb-adjacency matrices are encoded directly into parameterized quantum circuits and trained end to end with a classical TensorFlow head; the protocol is specified in this chapter, and its canonical benchmark is pending completion, so no comparative quantum result appears in this thesis.

Each study adheres to rigorous provenance standards, ensuring that all computational predictions are clearly distinguished from experimental validations. Null and negative results are reported with the same transparency as positive findings, reflecting a commitment to scientific integrity.

The chapter is organised as follows. Section 2.2 specifies the construction of the hybrid library, its two complementary expansion routes, the scoring terms of the prioritisation pipeline, and the selection of the four protein targets with their structural preparation and the correction of the transporter receptor. Section 2.3 details the docking protocol, its three-part validation by redocking, external benchmarking and positive-control enrichment, the consensus arm, and the multi-seed and perturbation audits. Section 2.4 establishes the resistance-resilience scoring framework: the estimand architecture, the score definition with its eligibility rule and class table, the network and chemical-space descriptors, and the retrospective genotype–phenotype confrontation. Section 2.5 describes the molecular-dynamics protocol, its three cohorts, the trajectory analysis, the endpoint estimation and the interaction-fingerprint complement. Section 2.6 presents the classical fingerprints and the topological, tensor and quantum-inspired descriptors, including the conformer-generation governance and the kernel diagnostic hierarchy. Section 2.7 covers the random-forest baselines, the graph and transformer architectures, and the quantum molecular structure encoding protocol of the hybrid quantum-classical programme. Section 2.8 closes with the statistical framework: partition families, activity-label provenance, performance metrics with their prevalence-imposed ceilings, conformal calibration and the distance-aware triage rule, significance testing with its power boundaries, the uncertainty taxonomy, the reproducibility infrastructure and the claim-ledger convention used throughout the thesis.

2.2 Chemical Space Construction and Virtual Screening

2.2.1 Hybrid library construction

The chemical space exploration aims to identify structurally diverse antimalarial candidates by combining African natural products with synthetic antimalarial chemistry and expanding around both seed sets. The seed collection comprises 396 African natural products and 454 synthetic antimalarial compounds. Canonical SMILES strings are used as the molecular identity, and duplicates are removed after canonicalisation so that every entry in the library corresponds to a unique structure.

Expansion proceeds through two complementary approaches. First, the Cheese API[108] performs a similarity search with Morgan fingerprints (radius 2, 2048 bits) at a Tanimoto threshold of 0.7 against the ZINC15 and ENAMINE-REAL collections, retrieving 100 analogues per seed. Second, STONED-SELFIES[109] generates 100 variants per seed through single-character SELFIES

mutations drawn from an alphabet of size 50, which explores substitutions that a similarity search cannot reach. Generated structures are filtered for pan-assay interference compounds, validated with RDKit and Open Babel, and deduplicated by canonical SMILES, yielding a hybrid library of 65 856 unique molecules of which 94.1 % are Lipinski-compliant. Chemical novelty relative to the African natural-product seed set is quantified by the maximum whole-molecule Tanimoto similarity computed with the same Morgan parameters; molecules whose maximum similarity falls below 0.4 are treated as whole-molecule-novel. Because the two expansion routes act on different parts of the molecule, the library was additionally characterised by scaffold recovery, computed as the fraction of annotated seed Bemis–Murcko scaffolds detected in the expanded set.

Prioritisation of the expanded library proceeds in two stages. A SMILES variational autoencoder[110] provides a continuous latent representation of the library, and k-means clustering of the selected 64-dimensional representation yields 484 centroids. This centroid-based sampling reduces the cost of the subsequent docking campaign from exhaustive docking of every molecule to docking of the cluster representatives, while preserving structural diversity through the representatives themselves. Multi-parameter optimization then combines docking-derived scores, DiffDock confidence, the quantitative estimate of drug-likeness, an ADMET composite, and a Lipinski penalty. The 76 centroids above the optimization threshold map to 53 clusters and expand to 40 481 molecules, of which 19 913 remain after synthetic-accessibility and multi-parameter filtering. The 19 913 molecules are computational hypotheses rather than confirmed synthetic targets.

The latent representation itself is produced by a SMILES variational autoencoder of the continuous-representation architecture[110], trained on the full 65 856-molecule library before any candidate ranking. The encoder uses three convolutional layers with kernel sizes 9, 9 and 10 and filter counts 9, 9 and 11; training runs for 100 epochs with the Adam optimiser at learning rate 0.001 and batch size 128 under a KL-divergence weight of 0.5. Hyperparameters were not hand-set: an Optuna search of 150 trials per latent dimensionality[111] optimised the learning rate, batch size, dropout rate, kernel size and encoder architecture, and convergence was assessed by validation-loss stability over the final ten epochs. Two latent dimensionalities, 32 and 64, were trained and evaluated side by side before the production clustering; the comparison is a measured trade-off rather than a dominance claim, and the model-selection evidence with the clustering-quality diagnostics is reported with the library characterisation in Chapter 3.

The generative stage has declared failure modes that the filtering stage is designed to absorb, and naming them clarifies what the library is. SELFIES-based mutation guarantees chemical valence validity by construction, but validity is not plausibility: single-character mutations can produce strained ring systems, reactive functionality or fragmentated structures that a chemist would not recognise as analogues. Similarity-based retrieval produces plausible analogues but inherits the scope of the retrieved catalogue, so structures absent from the commercial collections are unreachable by that route. The two routes therefore cover complementary failure profiles, and the PAINS filter plus valence and standardisation checks remove the portion of both outputs that would corrupt downstream scoring. The scaffold-recovery statistic closes the loop: because it measures how many annotated seed ring systems survive expansion, it detects the failure mode in which expansion drifts away from the medicinally relevant framework entirely, a failure that whole-molecule novelty statistics cannot see because they reward exactly that drift.

The individual scoring terms of the prioritisation pipeline are declared explicitly, because each carries its own estimation assumption. The quantitative estimate of drug-likeness (QED) weights eight physicochemical desirability functions (molecular weight, octanol–water partition coefficient, hydrogen-bond donor and acceptor counts, polar surface area, rotatable bonds, aromatic ring count and structural alerts) into a desirability score in $[0, 1]$[112]. Synthetic accessibility is estimated with the SYBA Bayesian classifier, which scores a molecule from its fragment contributions without requiring a retrosynthetic search[113]; the natural-product-likeness score, used wherever African-chemotype character enters a composite, follows the Ertl fragmentation scheme calibrated on natural-product and flat synthetic reference collections[114]. The ADMET composite is a weighted

mean of predicted absorption, distribution, metabolism, excretion and toxicity endpoints from a single prediction service and is used only for ranking, never as a substitute for experimental pharmacokinetics. Because the composite terms are heterogeneous in origin and uncertainty, the multi-parameter score is treated as a prioritisation device whose components are reported individually wherever a claim depends on one of them.

2.2.2 Target selection and structural preparation

Four protein targets are selected to span mechanistically distinct and resistance-relevant vulnerabilities rather than to provide four equivalent binding sites. Dihydrofolate reductase (*Pf*DHFR, PDB 7F3Y, methotrexate copy A702) anchors the antifolate axis and a canonical resistance pathway. The chloroquine resistance transporter (*Pf*CRT) is derived from the 6UKJ cryo-electron microscopy structure of the 7G8 isoform and is corrected to a 3D7-like wild-type receptor by restoring lysine at position 76, since the deposited structure carries the K76T chloroquine-resistance substitution at that position. The caseinolytic protease *Pf*ClpP (PDB 2F6I) represents parasite proteostasis through its chain-A Ser252/His223/Asp219 catalytic triad, and the P-type ATPase *Pf*ATP4 (PDB 9N10) represents ion homeostasis through the D451 phosphorylation-loop region and the DPPR hinge. The paralog structure 4GM2 is not used for *Pf*ClpP interpretation.

Structures are prepared by removing crystallographic waters, assigning protonation states consistent with physiological pH, and adding hydrogen atoms. Binding-site definition combines crystallographic ligand positions with computational pocket detection. The corrected *Pf*CRT receptor was validated independently with P2Rank, whose top predicted pocket scored 162.7 at a probability of 0.999 with its centre 5.7 Å from the docking grid centre, confirming that the restored wild-type cavity is the pocket targeted by the grid. Grid boxes are target-specific: 25 Å edge length for *Pf*DHFR and *Pf*CRT, 20 Å for *Pf*ClpP, and 24 Å for *Pf*ATP4. Table 2.2.1 summarises the docking-stage specification per target. The structural evidence behind the four entries is deliberately heterogeneous. *Pf*DHFR carries a co-crystallized inhibitor; the *Pf*CRT site is a cavity proxy rather than a validated small-molecule pocket; and the *Pf*ClpP and *Pf*ATP4 regions are mechanism-motivated because no small-molecule inhibitor is co-crystallized. The panel is therefore treated as a gradient of structural confidence throughout the analysis.

The correction of the *Pf*CRT receptor is described in detail because it is the largest single structural intervention in the panel and because every downstream *Pf*CRT number depends on it. The deposited 6UKJ structure resolves the 7G8 isoform, which carries the K76T substitution associated with chloroquine resistance; docking against the deposited receptor would therefore score chloroquine-like chemotypes against a resistance-state transporter while reporting them as wild-type comparisons. The correction restores lysine at position 76 by side-chain replacement on the frozen backbone, and the restored receptor is audited by an independent pocket-detection method rather than assumed correct: the top predicted pocket must localise to the docking box, which it does with a high score and near-coincident centre. The corrected receptor is then used uniformly, so that all wild-type and mutant *Pf*CRT comparisons are computed against the same restored reference. The intervention is recorded as a declared deviation from the deposited structure in the provenance manifest, with the deposited entry retained for reproducibility, and the alternative of using the uncorrected structure is not analysed, because a resistance-state receptor in the wild-type position would contaminate the retention ratios that the analysis exists to compute.

Target	PDB entry	Role in the panel	Box edge	Structural confidence
*Pf*DHFR	7F3Y	Antifolate axis; resistance panel	25 Å	Co-crystallized inhibitor
*Pf*CRT	6UKJ (corrected)	Resistance transporter; K76 restored	25 Å	Proxy cavity, P2Rank-audited
*Pf*ClpP	2F6I	Proteostasis	20 Å	Mechanism-motivated
*Pf*ATP4	9N10	Ion homeostasis	24 Å	Mechanism-motivated

Table 2.2.1: *Docking-stage specification per target. Box edges are cubic and centred on the corrected binding-site coordinates; the pocket audit compares the fixed box centre with the top P2Rank prediction. Structural confidence grades from a co-crystallized inhibitor (*Pf*DHFR) through a corrected proxy cavity (*Pf*CRT) to mechanism-motivated sites without a co-crystallized ligand (*Pf*ClpP,* Pf*ATP4).*

2.3 Molecular Docking Protocols

2.3.1 AutoDock Vina docking

Molecular docking employs AutoDock Vina 1.2.7, a widely validated docking engine that balances computational efficiency with predictive accuracy. The Vina scoring function approximates the binding free energy through a knowledge-based combination of steric, electrostatic, and hydrophobic terms. Ligands are standardised with Open Babel and converted to the PDBQT format with Meeko before docking. Docking is run against fixed target-specific boxes centred on the coordinates described in Section 2.2.2, with the exhaustiveness parameter set to 64 for all reported scores. Each run returns poses ranked by predicted binding affinity, and the top-ranked pose is retained for downstream analysis.

Docking score interpretation requires careful consideration of the scoring function's limitations. Vina scores are reported in kilocalories per mole and represent an approximation of the binding free energy under idealized conditions. They do not correspond to experimentally measured IC_{50} or K_d values, because the scoring function carries systematic biases, neglects entropy contributions, and treats the receptor as a static object. Docking scores are therefore used for relative ranking within a target and never as quantitative predictions of affinity, and no cross-target arithmetic mean is computed because Vina scores are not calibrated across heterogeneous pocket architectures. Every pose is additionally subjected to an automated geometric gate that requires the rank-1 centroid to lie inside the box, at least one ligand atom within 10 Å of the target anchor, and at least 90 % of ligand atoms inside the padded box. Passing the gate is a pose-quality criterion, not evidence of binding.

2.3.2 Docking validation and benchmarking

Validation of the docking protocol proceeds through three complementary assessments: redocking to the native ligand pose, external benchmarking against the DEKOIS 2.0 dataset, and positive-control enrichment using known actives from the MMV collection. Redocking evaluates the ability of the protocol to reproduce experimentally determined binding modes: for each target with an alignable co-crystallized ligand, the ligand is extracted, redocked into the prepared receptor, and the heavy-atom root-mean-square deviation (RMSD) between the docked and crystallographic poses is computed, with 2.0 Å used as the success threshold. Redocking reproduced the crystal pose for four of the five alignable ligands (80 % success), with at least one successful reproduction for each of the three targets used in the primary analysis. The single failure was the *Pf*DHFR reference ligand methotrexate, whose RMSD was approximately 30 Å because the *Pf*DHFR grid is centred on the NADPH-adjacent site rather than the catalytic methotrexate pocket. This failure is retained in the record rather than repaired, and it illustrates that enrichment outcomes can reflect grid positioning as well as scoring-function behaviour.

The DEKOIS 2.0 benchmark pairs each active compound with property-matched decoys that are topologically distinct, so that enrichment performance reflects molecular recognition rather than trivial separation on molecular weight or lipophilicity. The *Pf* DHFR set comprises 40 actives and 1200 decoys docked with the same protocol applied to the hybrid library, and performance is summarised by the area under the receiver operating characteristic curve (ROC-AUC) and the enrichment factor at 5 percent ($\mathrm{EF}_{5\%}$). Because the near-chance outcome of this benchmark conditions the interpretation of every absolute Vina score in the docking campaign, a controlled two-arm comparison was added on the identical DEKOIS panel: the deposited receptor with the retained methotrexate was compared with the same receptor after methotrexate removal, using the same grid, frame and exhaustiveness.

Positive-control enrichment uses the MMV Malaria Box under the dual-filter consensus protocol combining AutoDock Vina with DiffDock. A subset of experimentally confirmed antimalarials is docked against the three targets with docking data, and the ROC-AUC of the ranked predictions measures retrodictive consistency with known activity. Enrichment labels for *Pf* CRT and *Pf* ATP4 are score-stratified (top and bottom 25 percent) and therefore measure internal consistency rather than independent experimental discrimination. The complete validation results are reported alongside the corresponding docking campaign in Chapter 3.

The consensus arm of this protocol deserves its own description, because it is a distinct estimator rather than a refinement of the Vina score. DiffDock treats docking as a generative task: a diffusion model over the rotation, translation and torsion-angle space of the ligand produces sampled poses, and the model's confidence output replaces the physical scoring function as the ranking signal[115]. The dual filter retains a candidate only when both estimators place it favourably, which trades recall for precision: a physically parameterised score and a learned generative confidence must agree before a compound advances. The consequence for interpretation is that consensus rankings inherit the failure modes of neither tool alone but also the calibration of neither alone, so consensus output is used for prioritisation and never quoted as an affinity estimate.

Multi-seed reproducibility is quantified for the docking protocol itself. Selected candidate–receptor pairs are redocked under five random seeds with identical ligand preparation and grid boxes, and the dispersion of the lowest-energy modes is reported as the seed sensitivity of the protocol. This bound is protocol-local: it characterises the stochasticity of the search algorithm under fixed inputs and says nothing about pose accuracy, force-field adequacy or binding. It is nevertheless the quantity that any ratio-based score computed from docking outputs inherits, because numerator and denominator fluctuate under the same seed distribution. A bounded score-perturbation audit completes the robustness picture: the resistance-score classes are recomputed after adding deterministic perturbations of up to $\pm 1\,\mathrm{kcal\,mol^{-1}}$ to every score, and the fraction of class assignments that survives the perturbation is reported with the results.

2.3.3 Independent and CNN-based cross-scoring

Two further cross-checks extend the validation of the docking protocol beyond the primary campaign. The first is a secondary cross-docking calibration on the QuickVina engine, a Vina derivative, applied to the primary-lead panel against three independent receptor structures; the resulting composite MPO–score association is null (Spearman $\rho = 0.0129$, $p = 0.0913$), which bounds any claim that the multi-parameter optimization and the docking scores measure the same quantity.

The second is a deep-learning-based rescoring of a 39-compound sensitivity panel across the same eight *Pf* DHFR and *Pf* CRT states as the primary cohort, producing 312 finite docking complexes scored by both AutoDock Vina and the GNINA convolutional-neural-network affinity model. The panel serves two declared purposes: it measures the sensitivity of the ratio-based score to a second scoring function (bootstrap mean docking RRS 100.45, 99.59 to 101.32; Class-A fraction 0.974,

0.923 to 1.000), and it tests whether class assignments survive independent rescoring (100 % class-level agreement with Vina, with no per-mutant rank transfer claimed; per-mutant rank correlations are near zero for C59R and I164L). Because almost the whole panel falls in Class A, the agreement is close to degenerate and constrains gross class assignment only. This is a computational sensitivity check, not validation of the candidate ranking or of experimental resistance.

2.3.4 Retrospective genotype–phenotype confrontation

The static docking estimand is additionally confronted, retrospectively and outside the candidate cohort, with published genotype-resolved inhibition data. Inhibition records for *Pf* DHFR (ChEMBL target CHEMBL1939) are curated by assay description, which encodes the enzyme genotype, yielding 28 compounds with paired wild-type and quadruple-mutant (N51I/C59R/S108N/I164L) IC_{50} measurements, 24 of which also carry paired double-mutant (C59R/S108N) values, and triple-mutant (C59R/S108N/I164L) pairs complete for all 28. The compounds are docked against wild-type, double, triple and quadruple *Pf* DHFR receptors under the declared protocol of Section 2.3.1 (exhaustiveness 64, fixed *Pf* DHFR box, single seed 42; 112/112 jobs completed); mutant receptors are generated from the wild-type coordinates by in-silico side-chain replacement with backbone coordinates unchanged, then prepared identically. For each genotype the score difference $\Delta\Delta G = s_{\mathrm{mutant}} - s_{\mathrm{WT}}$ is compared with the published fold-shift $\log_2(\mathrm{IC}_{50,\mathrm{mut}}/\mathrm{IC}_{50,\mathrm{WT}})$ by Spearman rank correlation, direction agreement, and an exact binomial probability of the observed departure from chance agreement.

The comparison carries two structural limitations that are declared with the protocol. The reference IC_{50} values are whole-parasite growth measurements in DHFR-genotyped but non-isogenic strains, so they form a phenotype-level reference rather than a binding-affinity ground truth, and permeability, efflux and background mutations all contribute to the endpoint; and the curated compounds are disjoint from the candidate cohort, so the result audits the estimand itself rather than validating or invalidating any candidate class. Assays identifiers, raw records and provenance are frozen with the analysis.

2.3.5 Candidate selection cascade

The transition from the expanded library to the resistance-analysis cohort is a filter cascade whose every stage is declared, because each stage changes the population that later statistics describe. Starting from the 19 913 molecules that survive synthetic-accessibility and multi-parameter filtering, the cascade applies, in order: a multi-parameter optimization threshold (score at least 0.70, retaining the primary leads), a predicted synthetic-accessibility requirement (positive SYBA score), a predicted selectivity requirement (selectivity index above 10), a target-coverage requirement (docking evidence against at least two of the four receptors), and a scaffold-diversity cap (at most three compounds per Bemis–Murcko scaffold). The cascade yields the seventeen-member Set-C cohort used throughout the resistance analyses. Every filter is a computational prediction, so the cohort is a chain of conditional selections and its composition is conditional on all of them; class frequencies within the cohort are reported with this conditioning attached and are never extrapolated to the parent library. The cascade also fixes which comparisons are legitimate: candidates excluded at different stages or evaluated against different target subsets are not pooled into a single statistic without the coverage difference being declared, which is why the two-target primary set and the single-target remainder are always reported separately.

2.4 Resistance-Resilient Scoring and Polypharmacology

2.4.1 Estimand architecture of the resistance analysis

The resistance analysis distinguishes, by construction, four layers of analysis that answer different questions and are never merged. The primary layer is the docking-score-retention analysis: for the candidates with eligible wild-type docking scores on both the dihydrofolate-reductase and chloroquine-resistance-transporter panels, ratio-based retention classes are computed under the declared protocol. The second layer is the static-to-dynamic divergence comparison, which asks whether the primary retention metric and a trajectory-based geometric retention ratio point in the same direction on the matched systems for which both exist; this layer is a comparison of estimands, not a validation of either. The third layer comprises secondary structural-retention diagnostics, including the non-overlapping parent-study wild-type complexes and the single-ligand feasibility probe, which address pose retention and endpoint feasibility on different coverage and sampling questions and are therefore reported separately from the primary classes. The fourth layer is technical validation: protocol reproducibility, seed sensitivity, rescoring by an independent scoring function, and bounded perturbation audits. Each layer carries its own scope statement, and no quantity from one layer is presented as evidence within another. This architecture exists because the most damaging interpretive error in this class of study is not a wrong number but a right number quoted under the wrong estimand: a protocol-local retention class, a geometric pose-retention ratio, and a phenotype-level resistance measurement are three different quantities, and the analysis never allows one to substitute silently for another.

The reference controls for the docking and dynamics layers are five approved antimalarial drugs spanning distinct mechanisms of action: artemisinin, pyrimethamine, chloroquine, lumefantrine and cipargamin. They serve as calibration anchors for the docking protocol and, where trajectories exist, as familiar behavioural reference points for the dynamics; they are not part of the candidate cohort and contribute no retention classes.

2.4.2 Resistance-resilience score computation

Resistance-resilient scoring (RRS) quantifies a compound's predicted ability to retain binding across clinically relevant resistance mutations. The score is defined per target and per mutation from the ratio of Vina score magnitudes rather than from a difference of free energies, because absolute Vina values are not comparable across receptor states. For candidate i, mutation m and target t, with S_{Vina} denoting the signed Vina output,

$$\mathrm{RRS}_{i,m,t} = 100 \times \frac{|S_{\mathrm{Vina},i,m,t}|}{|S_{\mathrm{Vina},i,\mathrm{WT},t}|}. \tag{2.4.1}$$

Candidates whose wild-type score magnitude falls below 5.0 kcal mol^{-1} are excluded from that target's score-retention analysis. This refers to the separate wild-type/mutant panel for the target in question, not to the four-target wild-type matrix, and non-binding targets are excluded rather than assigned a zero score or used as a denominator. The available-target summary averages the defined target-specific values across eligible targets and mutation states. RRS is a relative docking-score-retention statistic: it is not a free-energy difference and not a measurement of resistance.

Classes are assigned from retention thresholds and are mutually exclusive. Class A* requires all available mutant RRS values to be at least 80 % and the minimum eligible wild-type score magnitude to be at least 7.0 kcal mol^{-1}. Class A requires all available values to be at least 80 % without the wild-type anchor criterion. Class B requires all available values to be at least 70 % with at least one below 80 %. Class C requires at least one value to be at least 80 % without meeting the A, A* or B criteria, and Class D is assigned when no available mutant RRS value reaches 80 %. The cut-offs

follow the Vina score distribution in this panel and are not calibrated to biochemical resistance or clinical outcome, so sensitivity to alternative thresholds is reported as part of the resistance-scoring analysis. Table 2.4.1 states the class definitions in one place, because the mutually exclusive partition is easier to audit as a table than as running text.

Class	Retention condition	Anchor condition
A*	all available values $\geq 80\,\%$	anchor $\geq 7.0\,\mathrm{kcal\,mol^{-1}}$
A	all available values $\geq 80\,\%$	anchor $< 7.0\,\mathrm{kcal\,mol^{-1}}$
B	all $\geq 70\,\%$, at least one $< 80\,\%$	—
C	at least one $\geq 80\,\%$; not all $\geq 70\,\%$	—
D	no value $\geq 80\,\%$	—

Table 2.4.1: *Resistance-score class definitions. The classes form a mutually exclusive partition computed from the available (eligible) mutant retention values;* anchor *denotes the minimum eligible wild-type score magnitude for the candidate. The cut-offs are operational choices calibrated to the Vina score distribution of the panel, not to biochemical resistance.*

For *Pf* DHFR, four resistance-associated substitutions are modelled: N51I, C59R, S108N and I164L. For *Pf* CRT, two states are modelled: K76T, which is the chloroquine-resistance marker, and K76A, which serves as a mechanistic comparator for perturbation of Lys76 and carries no clinical prevalence claim. No mutant crystal structures were available, so homology models were generated with SWISS-MODEL from the wild-type templates 7F3Y and 6UKJ[116]. A model was retained only when the QMEAN score exceeded −4.0, the GMQE exceeded 0.7, Ramachandran-favoured residues exceeded 95 %, and the backbone RMSD relative to the template was below 1.5 Å; all six models met these thresholds. Model scores are not experimental structures, and this constraint is carried into the interpretation of every mutant comparison.

The eligibility rule that excludes weak wild-type denominators is stated as part of the score definition rather than as a post-processing choice, because it changes the estimand. A ratio of score magnitudes is unstable when its denominator approaches zero: a wild-type score of $-1\,\mathrm{kcal\,mol^{-1}}$ against a mutant score of $-4\,\mathrm{kcal\,mol^{-1}}$ would produce a retention ratio of 400 % that reflects score noise rather than mutation response. Excluding targets whose wild-type magnitude falls below $5.0\,\mathrm{kcal\,mol^{-1}}$ removes this failure mode at the cost of coverage, since candidates without an eligible denominator on a target are reported as single-target rather than dropped silently. The eligibility threshold itself was subjected to a bounded sensitivity check (moving it from $4.0\,\mathrm{kcal\,mol^{-1}}$ to $5.0\,\mathrm{kcal\,mol^{-1}}$), and the class assignment is invariant under that move. This property is reported with the results rather than assumed, because invariance under one alternative threshold does not establish invariance under all of them.

2.4.3 Polypharmacology network similarity

The resistance-aware workflow does not compute a cross-target arithmetic composite, because Vina scores are not calibrated across different pocket architectures. Target breadth is read instead from the pattern of target-specific evidence, using a within-target favourability criterion: a target counts as favourable for a candidate when the candidate's Vina estimate falls at or below that target's median across the cohort. On the 17-member cohort the per-target medians were -5.92 $\mathrm{kcal\,mol^{-1}}$ for *Pf* DHFR, -8.06 $\mathrm{kcal\,mol^{-1}}$ for *Pf* CRT, -6.02 $\mathrm{kcal\,mol^{-1}}$ for *Pf* ClpP and -6.38 $\mathrm{kcal\,mol^{-1}}$ for *Pf* ATP4. This within-target definition avoids a single cross-target cutoff and is a prioritisation device rather than an affinity threshold.

Alongside this per-target profile, a polypharmacology network similarity (PNS) is computed as a network-weighted ranking score from a STRING protein–protein interaction network at a confidence

threshold of 700[117]. The score formalises the network layer as follows. The association network assigns every protein pair a confidence weight, and a candidate's target set induces a subnetwork of the proteins it engages; the candidate's proximity to the antimalarial reference targets is computed as a weighted combination of shortest-path distances through the association graph, in which each edge is discounted by the confidence weight of the association so that speculative links contribute less than experimentally supported ones. Formally, for candidate i with target set T_i, reference target set R, and network distance $d_{\mathrm{net}}(u, v)$ defined as the minimum over paths of the sum of inverse-confidence edge costs,

$$\mathrm{PNS}_i = \frac{1}{|R|} \sum_{r \in R} \min_{u \in T_i} d_{\mathrm{net}}(u, r), \tag{2.4.2}$$

so that a candidate engaging a protein directly linked to a reference target receives a lower, more favourable value than one whose nearest connection is indirect. The score measures how closely a candidate's target profile sits to established antimalarial network hubs. *Pf* CRT is absent from the queried network, so its centrality is imputed as the network mean (0.151) and the sensitivity of the ranking to that imputation is reported explicitly. Two reference scores are computed for comparison: a network position that tracks established targets, and a measure of chemical-space position described in the next subsection. Associations between RRS, PNS and this chemical-space descriptor are evaluated with Spearman rank correlation, with permutation testing over 100 000 permutations, Bonferroni adjustment for three tests, and percentile bootstrap intervals from 10 000 resamples. At the cohort sizes involved these correlations are descriptive rather than powered tests of independence.

The network layer carries declared robustness checks of two kinds. The first is imputation sensitivity: the absent *Pf* CRT centrality is replaced not only by the network mean but also by zero, one-half and twice the reference value, and the rank correlation of the resulting PNS ordering against the reference ordering is reported, so that the reader can see how much of the ranking depends on a value that no data source provides. The second is network-construction sensitivity: the STRING confidence cutoff itself (400, 700 and 900) is varied, and the stability of the ranking across the three networks is quantified by pairwise rank correlations. Both checks bound the same quantity from different sides: the first asks how the score responds to a missing input, the second asks how it responds to an arbitrary threshold on a complete one. Neither check speaks to biological target importance, and neither is presented as doing so.

2.4.4 African-chemotype structural index

The African-chemotype structural index (ACSI) places a candidate relative to approved-drug and African natural-product reference spaces. It is a weighted composite of four min–max normalised components,

$$\mathrm{ACSI} = 0.40\, D_{\mathrm{DrugBank}} + 0.25\, D_{\mathrm{ANPDB}} + 0.20\, f_{sp^3} + 0.15\, \mathrm{NPL}, \tag{2.4.3}$$

where D_{DrugBank} and D_{ANPDB} are the minimum Tanimoto distances to the approved-drug and African natural-product reference sets respectively, f_{sp^3} is the fraction of sp^3-hybridized carbons, and NPL is a natural-product-likeness score. All four components are normalised over the cohort under analysis, so ACSI is a relative rather than an absolute coordinate and its weights carry no mechanistic claim; weight-sensitivity checks are reported with the resistance-scoring results. ACSI describes chemical-space position and is not used as a substitute for target-specific mutation analysis.

2.5 Molecular Dynamics Simulation Methods

2.5.1 System preparation and parameterization

Molecular dynamics (MD) simulations provide time-resolved, atomistic descriptions of drug–target interactions and allow the geometric and energetic consequences of a docking pose to be examined. All simulations are run with GROMACS[118], with proteins parameterised under the CHARMM36m force field[119]. System preparation begins from the selected docking pose for each compound–target pair. The protein is parameterised with CHARMM36m, which has been extensively validated for proteins. Two ligand parameterisation protocols were used and are reported separately. For the four parent-study wild-type complexes, ligands were parameterised with GAFF2. For the 16-system Set-C pilot, ligands were parameterised with the OpenFF 2.2.0 force field using AM1-BCC partial charges. The substitution of OpenFF for GAFF2 in the Set-C pilot, and its earlier substitution for the originally planned CGF force field, are recorded in the provenance manifests as declared, approved policy deviations; because the two cohorts use different ligand force fields, their absolute MM-GBSA values are never compared with one another.

The protein–ligand complex is placed in a rectangular simulation box with periodic boundary conditions, ensuring a minimum distance of 12 Å between the solute and the box edges, and solvated with the TIP3P water model. Counterions are added to neutralise the system and to reach a physiological ionic strength of 0.15 M. Energy minimisation proceeds in two stages: steepest descent with heavy-atom restraints, followed by unrestrained conjugate-gradient minimisation. Convergence is assessed by monitoring the maximum force against a fixed threshold.

2.5.2 Equilibration and production protocols

Equilibration proceeds through a three-stage protocol designed to relax the system gradually while avoiding structural artifacts. The first stage runs under the NVT ensemble at 310.15 K with position restraints on heavy atoms to allow solvent equilibration, using a velocity-rescaling thermostat. The second stage runs under the NPT ensemble at the same temperature and 1.0 bar with reduced position restraints to equilibrate the system density, using a Parrinello–Rahman barostat. The third stage continues under NPT without restraints to reach the final equilibrated state. Equilibration quality is assessed by monitoring temperature, pressure, density, potential energy, and the RMSD of protein backbone atoms; a system is considered equilibrated when these properties fluctuate stably around their mean values for a sustained interval. Table 2.5.1 summarises the ensemble, thermostat, barostat and restraint schedule per stage.

Stage	Ensemble	Thermostat / barostat	Restraints	Quantity monitored
Minimisation	—	—	heavy atoms	maximum force
NVT equilibration	NVT	velocity rescale	decreasing	temperature, density
NPT equilibration	NPT	Parrinello–Rahman	reduced	pressure, energy, RMSD
Final equilibration	NPT	Parrinello–Rahman	none	stability of all monitors
Production	NPT	Parrinello–Rahman	none	trajectory output

Table 2.5.1: *Minimisation, equilibration and production schedule for the molecular-dynamics protocol. Restraint force constants decrease to zero across the equilibration stages; production runs unrestrained under NPT at 310.15 K and 1.0 bar.*

Production simulations run for 10 ns under the NPT ensemble at 310.15 K and 1.0 bar. The integration timestep is 2 fs, with hydrogen-involving bond lengths constrained by the LINCS algorithm. Long-range electrostatics are treated with the particle-mesh Ewald method, and van der

Waals interactions use a force-switch smoothing function. Coordinates are saved at a fixed interval for downstream analysis. The trajectory length is chosen as a pose-retention and local-geometry check; it is not intended as a measurement of residence time or as a converged free energy, and this limitation is stated wherever an endpoint is reported. GPU acceleration is used for production runs, with the non-bonded and PME calculations offloaded to the device while bonded interactions and coordinate updates remain on the central processor.

The molecular dynamics programme comprises three distinct cohorts that are never merged. The primary Set-C pilot covers the 16 matched systems formed by PP-01 and PP-02 against the wild-type and mutant *Pf*DHFR and *Pf*CRT states, and provides the matched-row comparison for the static-to-dynamic divergence analysis. A secondary parent-study cohort of four wild-type complexes from earlier work addresses pose retention and endpoint feasibility on a different compound set. A single-ligand feasibility probe, PP-15, is reported outside the Set-C estimand and carries no resistance interpretation. Table 2.5.2 summarises the cohorts and their roles. All 16 Set-C trajectories completed production and satisfied the geometric contact-stability criterion of a minimum protein–ligand distance below 3.5 Å.

Cohort	Systems	Ligand force field	Purpose	Interpretive scope
Set-C pilot	16 matched	OpenFF 2.2.0 (AM1-BCC)	static-to-dynamic divergence	protocol-local contrast
Parent study	4 wild type	GAFF2	pose retention; endpoint feasibility	feasibility check
PP-15 probe	2 wild type	as parent study	single-ligand feasibility	no resistance reading

Table 2.5.2: *The three molecular-dynamics cohorts. Cohort membership, force field and analytical role are fixed per cohort; absolute endpoint energies are never compared across cohorts because the ligand parameterisation differs. QC denotes the geometric contact-stability criterion.*

2.5.3 Trajectory analysis and binding free energy estimation

Trajectory analysis employs standard GROMACS utilities together with MDAnalysis[120] for Python-based workflows. Structural stability is assessed through the RMSD of protein backbone atoms and ligand heavy atoms relative to their initial conformations, and per-residue flexibility is assessed through the root-mean-square fluctuation (RMSF). RMSD is computed as

$$\mathrm{RMSD}(t) = \sqrt{\frac{1}{N}\sum_{i} \left|\mathbf{r}_i(t) - \mathbf{r}_i(\mathrm{ref})\right|^2}, \tag{2.5.1}$$

where N is the number of atoms, $\mathbf{r}_i(t)$ is the position of atom i at time t, and $\mathbf{r}_i(\mathrm{ref})$ is the reference position; RMSF is computed as

$$\mathrm{RMSF}(i) = \sqrt{\frac{1}{T}\sum_{t} \left|\mathbf{r}_i(t) - \langle \mathbf{r}_i \rangle\right|^2}, \tag{2.5.2}$$

where T is the number of frames and $\langle \mathbf{r}_i \rangle$ is the time-averaged position of residue i. Contacts, hydrogen bonds and the radius of gyration are computed on a decimated subset of frames, while RMSD is evaluated on every frame after periodic-boundary discontinuities are removed. Ligand retention is summarised by the minimum protein–ligand heavy-atom distance computed per frame, from which a bound fraction is derived. Ligand–protein interaction fingerprints are computed over stored snapshots to measure the persistence of individual contacts.

Binding free energy estimation uses the molecular mechanics/generalised Born surface area (MM-GBSA) approach implemented in `gmx_MMPBSA`[121], with the GB$^{\mathrm{OBC2}}$ model and a salt concentration of 0.15 M. The endpoint is computed as

$$\Delta G_{\mathrm{bind}} = \langle G_{\mathrm{complex}} \rangle - \langle G_{\mathrm{protein}} \rangle - \langle G_{\mathrm{ligand}} \rangle, \tag{2.5.3}$$

where each term is decomposed as

$$G = E_{\mathrm{MM}} + G_{\mathrm{solv}} - TS, \tag{2.5.4}$$

with E_{MM} the molecular mechanics energy, G_{solv} the solvation free energy, and TS the entropic contribution. Beyond the total endpoint, the per-term decomposition (E_{vdW}, E_{elec}, polar and non-polar solvation) and the per-residue contribution to binding are computed from the same snapshots. The per-residue decomposition serves a diagnostic purpose: it identifies which pocket residues dominate the enthalpic endpoint, which can be compared with the interaction-fingerprint anchors as a consistency check between the energetic and the geometric view. It is not used for lead optimisation, because per-residue endpoint contributions at this sampling carry the same within-protocol scope as the total endpoint. Configurational entropy is not included in the reported endpoints, and the MMPBSA conversion from CHARMM36 to AMBER parameters is documented as part of the protocol. Endpoints are reported with the frame-level standard error for the interpretable complexes and with the propagated within-trajectory standard deviation for the Set-C table; neither quantity is a replicate-level uncertainty estimate. Endpoint energies at this trajectory length are within-protocol contrasts and are not used to rank dissimilar compounds.

Where independent wild-type replicates exist, their endpoint difference defines the empirical noise floor of the endpoint estimator: the magnitude of the difference between two nominally identical systems bounds the variation that the protocol itself produces without any mutation. This floor is reported with the results and used as the reference scale against which an inter-replicate sensitivity is judged material. It is an estimator-scale quantity, not a physical energy: it characterises the protocol, not the binding.

Trajectory quality control comprises the absence of fatal errors in the GROMACS log files, the absence of LINCS warnings indicating unstable bond constraints, the absence of not-a-number values in energy or coordinate files, and the absence of gaps in the trajectory files. A geometric contact-stability criterion is applied in addition: a trajectory enters endpoint analysis only if the minimum protein–ligand heavy-atom distance remains below 3.5 Å. Systems failing any check are flagged, and the failure is reported rather than corrected.

The quality-control contract is deliberately asymmetric between acceptance and interpretation. Acceptance is binary: a trajectory either satisfies every check or it is excluded from the corresponding analysis class, and the excluded record keeps its identifier and failure mode in the output ledger. Interpretation is scope-bound: a passing trajectory supports statements about pose retention and local geometry at the simulated timescale, and nothing beyond that. The contract also specifies what is not checked. No convergence test on free-energy estimates is applied, because endpoint values at this trajectory length are not converged quantities and pretending otherwise would manufacture a precision that the sampling cannot support; no equivalence bound between replicates is assumed, because replicate variance is itself an output of the analysis where replicates exist rather than a known constant; and no visual inspection enters the acceptance decision, so that the contract is auditable from the recorded files alone.

Interaction-fingerprint analysis provides the contact-level complement to these global geometric summaries. The ProLIF library converts each stored snapshot into a binary interaction fingerprint whose bits encode hydrophobic contacts, π–π stacking, hydrogen bonds and cation–π interactions between specific protein–ligand atom pairs[122]. Averaging the fingerprints over snapshots yields a per-residue occupancy between 0 and 1 for each interaction type, which is the quantity reported as contact persistence. Occupancy answers a different question from the minimum-distance or endpoint-energy summaries: it identifies which specific residues anchor the ligand and how consistently each anchor persists, whereas the geometric summaries measure whether the ligand stays in

the pocket as a whole. The two views can disagree, and the disagreement is informative, because a ligand can retain a bounded position while exchanging one anchor for another, a pattern that distance statistics smooth over and occupancy statistics expose.

Beyond endpoint energies, trajectory-based retention is summarised by a distance ratio defined on the matched Set-C systems,

$$\mathrm{MD\text{-}RRS}_d = 100 \times \frac{\langle d_{\min} \rangle_{\mathrm{mut}}}{\langle d_{\min} \rangle_{\mathrm{WT}}}, \tag{2.5.5}$$

where $\langle d_{\min} \rangle$ is the mean minimum protein–ligand heavy-atom distance over the trajectory of the mutant or wild-type system. Values at or below 100 % indicate that the mutant pose is retained at least as tightly as the wild-type endpoint. The quantity is deliberately named as a distance ratio and never as an affinity: it is a geometric retention statistic of the same ratio form as the docking RRS of Section 2.4.2 but estimated from an entirely different procedure, which is precisely what makes the two quantities non-interchangeable estimands in the analysis of Chapter 3.

The comparison that the ratio enables is directional by design, and the direction rule is fixed before the trajectories are analysed. For each matched mutant state with an eligible wild-type docking baseline, the comparison records whether the static retention ratio falls below 100 %, indicating that the docking scores penalise the mutant, while the distance ratio lies at or below 100 %, indicating that the trajectory retains the pose at least as tightly as the wild type. The fraction of matched states exhibiting this disagreement is reported with an interval that accounts for the small number of matched comparisons, so that the reader can see both the observed direction and the sampling limit on it. Three interpretive boundaries are attached. The comparison requires no assumption that either quantity estimates affinity, only that each is internally consistent under its own protocol; the matched-state set is fixed by eligibility rather than chosen after inspection; and the fraction is a property of this panel under this protocol, so it is a calibration observation rather than a general docking-failure rate.

2.6 Classical and Quantum-Inspired Molecular Descriptors

2.6.1 Extended-connectivity fingerprints

Extended-connectivity fingerprints (ECFP), also known as Morgan or circular fingerprints, encode the local chemical environment of each atom within a specified radius. The algorithm proceeds iteratively: at iteration zero each atom is assigned an initial identifier derived from its atomic properties, and at each subsequent iteration the identifier is updated by hashing the current identifier together with the identifiers of neighbouring atoms. After a fixed number of iterations the set of all identifiers encountered constitutes the fingerprint. ECFP4 (radius 2) is the standard choice, balancing structural resolution against computational efficiency.

ECFP4 is computed with RDKit 2025.03.6 as a 2048-bit Morgan fingerprint with radius 2. Fingerprint similarity between two molecules is quantified by the Tanimoto coefficient,

$$T(A, B) = \frac{|A \cap B|}{|A \cup B|}, \tag{2.6.1}$$

where A and B are the sets of substructures encoded in the fingerprints. Values range from 0 (no shared substructures) to 1 (identical fingerprints). Across the representation analyses the following classical fingerprints are computed as baselines with the RDKit toolkit[123]: ECFP4, MACCS 167-bit structural keys, atom-pair, binding-pattern, functional-class ECFP4, 2D-pharmacophore, and

Morgan atom-pair counts. The seven descriptors span three distinct encoding principles, and the spread is deliberate. Hashed circular fingerprints (ECFP4 and its functional-class variant) encode local atom environments without a dictionary; keyed structural benchmarks (MACCS) and pharmacophore fingerprints map onto predefined dictionaries of substructures or feature triples; and the atom-pair family encodes pairs of atoms at a given topological distance rather than connected neighbourhoods. Because a representation benchmark can only be interpreted relative to the family its baseline belongs to, the primary baseline is ECFP4, and the remaining six serve as family representatives whose relative ordering is reported descriptively. ECFP4 serves as the primary baseline against which every quantum-inspired and learned representation is evaluated.

2.6.2 Topological fingerprints via persistent homology

Topological data analysis extracts global structural features from molecular graphs through persistent homology, which tracks the birth and death of topological features as a filtration parameter increases. For a molecule, a three-dimensional conformer is generated with the RDKit ETKDG distance-geometry generator[124] at a fixed random seed, and a weighted undirected graph is constructed with atoms as vertices and edge weights

$$w_{ij} = d_{ij}\left(1 + \frac{Z_i Z_j}{100}\right), \tag{2.6.2}$$

where d_{ij} is the Euclidean distance between atoms i and j and Z_i, Z_j are their atomic numbers, so that interactions involving heavier atoms are weighted more strongly. Vietoris–Rips persistence diagrams are computed up to homology dimension two using Ripser.py. Persistence in dimension zero describes connected components, dimension one describes cycles, and dimension two describes voids, so the representation captures ring topology and cavity structure that local substructure hashing does not resolve.

Each diagram is vectorised into scalar summaries per homology dimension, including the persistence entropy

$$H_d^{\mathrm{ent}} = -\sum_i p_i \log p_i, \qquad p_i = \frac{\mathrm{pers}_i}{\sum_j \mathrm{pers}_j}, \tag{2.6.3}$$

the count of features whose persistence exceeds a fixed threshold, the maximum persistence and the mean persistence. The primary activity analysis uses a 78-feature representation obtained by combining these homology summaries with persistence-image and Betti-curve components, while the 12-feature core is evaluated separately as a descriptive topological comparison. Because the descriptors are derived from three-dimensional conformers, they encode geometric persistence rather than graph-theoretic indices.

2.6.3 Tensor network embeddings

Tensor networks compress high-dimensional data by exploiting low-rank structure, and provide a fixed-length, physically interpretable embedding of a molecule. For each molecule a three-dimensional tensor $T \in \mathbb{R}^{N\times F\times 3}$ is constructed with $N = 100$ maximum atoms (zero-padded), $F = 10$ per-atom features (atomic number, degree, total hydrogens, formal charge, aromaticity, hybridization, mass, electronegativity, bond count and ring membership), and the three Cartesian coordinates, so that $T_{i,f,c} = x_{i,f}\, r_{i,c}$.

Each tensor is compressed by Tucker decomposition with mode ranks $(d, d, 3)$ at bond dimension $d = 8$,

$$T \approx \mathcal{G} \times_1 U^{(1)} \times_2 U^{(2)} \times_3 U^{(3)}, \tag{2.6.4}$$

where $\mathcal{G} \in \mathbb{R}^{d \times d \times 3}$ is the core tensor, and the descriptor is the flattened core of $d^2 \times 3 = 192$ elements. For a molecule with N actual atoms the real compression ratio is $\mathrm{CR}_{\mathrm{real}}(N) = 10N/d^2$, which at the mean panel size of approximately 39 atoms corresponds to roughly 6.1-fold compression relative to the real-atom representation. Compression quality is quantified by the relative Frobenius reconstruction error. Because mode rank and bond dimension are the free parameters of the representation, the descriptor is evaluated at several settings in the dedicated sensitivity analysis.

Every three-dimensional descriptor in this thesis inherits a conformer-generation stage, and the stage is governed rather than left implicit. A single low-energy conformer per molecule is generated with fixed random seed and standard distance-geometry bounds, without subsequent geometry optimisation, so that the mapping from graph to geometry is deterministic given the software version. This choice has a declared cost: conformational flexibility is collapsed to one representative geometry, so descriptors that read ring topology and connectivity are robust to the choice while descriptors sensitive to substituent orientation are not. Where a descriptor family consumes three-dimensional input, the same conformer set is shared across all consumers in a given analysis, so that differences between descriptors reflect their encoding rather than different geometry samples. The dependency is recorded in the provenance manifest together with the generator version and seed, because a conformer regeneration with a different seed or generator can shift a descriptor value without any change to the molecular graph, which would otherwise look like an unexplained reproducibility failure.

2.6.4 Quantum kernel similarity

Quantum kernel methods embed molecular representations into a Hilbert space defined by a quantum feature map, allowing similarity to be computed from quantum state overlaps. The quantum kernel similarity between molecules A and B is

$$K_{\mathrm{quantum}}(A, B) = \left| \langle \phi(A) \mid \phi(B) \rangle \right|^2, \tag{2.6.5}$$

where $|\phi(A)\rangle$ and $|\phi(B)\rangle$ are quantum states prepared from the corresponding molecular feature vectors. The feature map is implemented with the IQP embedding template of PennyLane 0.45.1 on a six-qubit statevector simulator, so that the state is prepared as

$$|\phi(\mathbf{x})\rangle = U_\phi(\mathbf{x}) \, |0\rangle^{\otimes n}, \tag{2.6.6}$$

where $U_\phi(\mathbf{x})$ applies single-qubit rotations and two-qubit entangling gates whose angles are functions of products of feature values. The input vector for each molecule is the Morgan atom-pair count vector (radius 2, 2048 features) reduced to six dimensions by uniform manifold approximation and projection under the Jaccard metric at a fixed random state, then scaled to the interval $[-1, 1]$.

The quantum kernel score (QKS) is the area under the ROC curve obtained by a support vector machine trained on the precomputed quantum kernel matrix. The corresponding classical comparator is a radial basis function kernel,

$$K_{\mathrm{RBF}}(A, B) = \exp\left(-\gamma \left\| \mathbf{x}_A - \mathbf{x}_B \right\|^2\right), \tag{2.6.7}$$

whose bandwidth γ is selected by inner cross-validation over a fixed candidate grid, and a linear kernel provides a second reference point. Descriptor representations are evaluated with random forests and kernel representations with support vector machines, because a support vector machine accepts a precomputed kernel directly; the two families are therefore never presented as a like-for-like model comparison. All quantum kernel values reported here are computed on a classical statevector

simulator. No hardware-level quantum advantage is claimed from this implementation, and evaluation on near-term quantum hardware is deferred to the quantum machine learning programme, whose protocol is established in the part of this thesis that follows.

Because two kernels can be compared at several levels that are easy to conflate, the diagnostic hierarchy is fixed in advance. The first level is kernel–target alignment, which measures how strongly the kernel matrix concentrates on same-label pairs and is reported as a descriptive geometric quantity. The second level is classification performance under cross-validation, which is the quantity that matters for any claim about predictive utility. The hierarchy exists because the two levels can move in opposite directions: a feature map whose induced geometry aligns well with the labels can still yield a classifier indistinguishable from a tuned classical kernel, because alignment is a property of the average geometry while classification is decided by the decision boundary in the region where classes overlap. The alignment statistic is therefore reported alongside performance, never in its place, and the interpretation rule is declared before the results: a difference at the alignment level without a difference at the performance level is recorded as a geometric observation, not as an advantage.

2.7 Machine Learning and Deep Learning Architectures

2.7.1 Random forest baselines

Random forests provide a robust and interpretable baseline for molecular property prediction. A random forest is an ensemble of decision trees in which each tree is trained on a bootstrap sample of the training data and considers only a random subset of features at each split point, with predictions aggregated across trees by majority voting or averaging. Random forests are well suited to high-dimensional descriptor spaces, resist overfitting through ensemble averaging, and expose feature importance scores that support interpretability, which makes them the appropriate baseline when the object of study is a molecular representation rather than a model class.

Two random-forest configurations appear in this thesis and are reported separately wherever they are compared. For the descriptor benchmark, descriptor representations are evaluated with random forests of 200 trees under stratified cross-validation. For the learned-representation benchmark, the ECFP4–RF baseline uses a random forest of 500 trees trained on 2048-bit ECFP4 fingerprints. Class imbalance, where present, is addressed through class-weight adjustment rather than resampling, so that the training distribution is not altered. In both benchmarks the baseline configuration is fixed before the comparative experiments are run and is held constant, so that differences between representations are not attributable to baseline tuning.

2.7.2 Graph isomorphism networks

Graph isomorphism networks (GINs) are a class of graph neural networks designed to achieve maximal expressive power within the message-passing framework. The GIN layer updates node representations according to

$$h_v^{(k+1)} = \mathrm{MLP}^{(k)}\left((1+\varepsilon^{(k)})\cdot h_v^{(k)} + \sum_{u\in N(v)} h_u^{(k)}\right), \tag{2.7.1}$$

where $h_v^{(k)}$ is the representation of node v at layer k, $N(v)$ denotes the neighbours of v, $\mathrm{MLP}^{(k)}$ is a multi-layer perceptron, and $\varepsilon^{(k)}$ is a learnable parameter. This formulation is provably as powerful as the Weisfeiler–Lehman graph isomorphism test, which means that a GIN can distinguish any pair of non-isomorphic graphs that the test can distinguish. It also fixes the expressive ceiling of

the architecture: structural motifs that the test cannot resolve, including certain bridged polycyclic cages and macrocyclic systems, are not separated by local message aggregation regardless of depth or width. This bound motivates the fusion architectures evaluated in the learned-representation benchmark.

For molecular property prediction, atoms serve as nodes with initial features encoding atomic number, degree, formal charge, hybridization, aromaticity and ring membership, and bonds serve as edges with features encoding bond type and stereochemistry. The GIN implementation comprises three message-passing layers with 128 hidden units, dropout of 0.1, rectified linear activations and batch normalisation after each message-passing layer, followed by global sum pooling. Two fusion variants are evaluated alongside the base architecture. GIN–TFP concatenates the graph embedding with the 78-dimensional persistent-homology vector described in Section 2.6.2, and GIN–TNE replaces that block with the 192-dimensional tensor network embedding described in Section 2.6.3. Training uses the Adam optimiser with a fixed learning rate and weight decay, and the message-passing configuration is held constant across the fusion variants so that the comparison isolates the contribution of the fused descriptor block. A separate capacity sweep over hidden width and dropout bounds the alternative explanation that any deficit of the graph architecture is a tuning artifact.

2.7.3 Transformer-based molecular representations

Transformer models, originally developed for natural language processing, have been adapted for molecular property prediction by treating SMILES strings as sequences of tokens. ChemBERTa is a transformer pre-trained on a large corpus of unlabeled chemical structures by masked language modelling, so that the model learns contextualised representations of molecular substructures by predicting masked tokens within SMILES strings.

For antimalarial activity prediction, ChemBERTa is fine-tuned with the Hugging Face Transformers library. Canonical SMILES strings are tokenised, the pre-trained model is loaded, and a classification head is placed on top of the pooled sequence representation before fine-tuning on the activity labels. A methodological safeguard is applied specifically to this architecture during cross-validation: pretrained weights are restored from the original checkpoint before each fold, so that no fold's fine-tuned weights seed another fold. Enforcing per-fold weight restoration reduces the risk of cross-fold information transfer that can inflate performance estimates of a sequence model during cross-validation, and it is reported as a reproducibility safeguard rather than as an optimisation. Fine-tuning and evaluation follow the same partition families applied to the fingerprint and graph baselines.

2.7.4 Quantum molecular structure encoding for the hybrid quantum-classical programme

The final programme of the thesis implements genuine quantum machine learning through quantum molecular structure encoding (QMSE)[104], which places molecular structure directly into the parameters of a quantum circuit rather than into an intermediate classical fingerprint. Two complementary matrices are derived from the molecular graph. The bond-order matrix assigns the bond order between atoms i and j (single, double, triple and aromatic bonds mapped to 1, 2, 3 and 1.5 respectively) to the off-diagonal entry weighted by the product of nuclear charges, $Z_i Z_j / b_{ij}$; stereocentre configuration enters as the sign of the diagonal entry and double-bond geometry as the sign of the bond order, so that cis/trans and stereochemical distinctions survive the encoding. The Coulomb-matrix alternative places $0.5\, Z_i^{2.4}$ on the diagonal and $Z_i Z_j / d_{ij}$, with average bond lengths as distances, off the diagonal. Matrix entries are flattened into rotation angles: diagonal elements parameterize single-qubit rotations (R_X, R_Y, R_Z) on each qubit, and upper-triangle off-diagonal

elements parameterize two-qubit entangling rotations (R_{XX}, R_{YY}, R_{ZZ}), so that the pairwise correlation structure of the molecule is carried by the entangling layer itself. A shallow variational ansatz with trainable parameters follows, the Pauli-Z expectation values of the qubits form the quantum feature vector, and the circuit is wrapped as a `qml.qnn.KerasLayer` on the PennyLane default-qubit simulator. The classical head is a TensorFlow dense block (rectified-linear activation, batch normalisation, dropout and a final sigmoid unit), and the model is trained end to end, with gradients propagated through the circuit simulation and the classical head by automatic differentiation. The distinction from the simulated quantum kernel of Section 2.6 is therefore twofold: the encoding bypasses ECFP4 entirely, and the circuit parameters are trained rather than fixed by the feature map.

The programme is organised in four declared phases with frozen inputs at each boundary. The encoding phase preprocesses the complete 19 849-molecule representation panel; all 19 849 molecules encoded successfully with zero failures, and the angle datasets are frozen with recorded hashes in the provenance manifest, so the encoding substrate is fixed before any model is trained. The pipeline-confirmation phase runs the full hybrid stack end to end and executes a leave-one-out smoke-test on a 17-molecule Set-A panel (17/17 folds, zero encoding failures, thirty epochs, untuned); its outcome confirms that the circuit–head interface trains and predicts, and it is explicitly labelled as a pipeline confirmation rather than a benchmark, because seventeen molecules at untuned settings support no performance claim. The hyperparameter-search phase (a joint search over encoding type, circuit depth, entangling gate and classical-head configuration) and the canonical benchmark phase, defined as evaluation on the frozen panel under the partition families, label provenance and multiplicity machinery of Section 2.8 against the ECFP4 baseline of Table 3.4.1 and against declared classical three-dimensional comparators (extended three-dimensional fingerprints, ultrafast shape recognition with pharmacophoric atom types, and three-dimensional autocorrelation functions), are pending completion; the comparative status of the programme is therefore NOT COMPUTED, and no comparative quantum result appears in this thesis. Three outcome scenarios are predeclared before the benchmark: an advantage of the structure-direct encoding under scaffold separation, equivalence within the sampling limits, or a deficit attributable to simulator depth, and each is treated as interpretable under the claim-ledger convention; the decision between them is quantified by effect sizes (Cohen's d) with paired tests under the multiplicity corrections of Section 2.8, and performance is stratified by the African-chemotype structural index so that any effect is localized with respect to chemotype complexity. An earlier implementation of this programme, which compressed ECFP4 vectors into shallow variational circuits scored by a support-vector oracle, is archived as exploratory under a different estimand: its quantum component received no structural information beyond what the fingerprint already carried, so it cannot answer the structure-direct question and it informs none of the claims of this thesis.

2.8 Statistical Analysis and Validation Frameworks

2.8.1 Activity-label provenance and panel freezing

Every supervised comparison in the representation analyses consumes the same label layer, and its provenance is declared here once. The primary panel is frozen at the molecules with complete descriptor coverage; binary activity labels are produced by a single external prediction model at a fixed decision threshold, giving a cohort in which roughly three quarters of the molecules are labelled active. These labels are model outputs, not assay measurements, so every classification metric in the results chapters measures agreement with one model's opinion of activity, and the phrase *activity prediction* is used in exactly that sense throughout. Three structural consequences

follow and are respected by construction. First, the prevalence determines the ceiling of any early-recognition metric, so enrichment-style metrics are reported only together with their prevalence-imposed ceilings. Second, the label model has its own inductive bias, most likely favouring scaffold families over-represented in its training data, so the scaffold-grouped analyses double as a partial control on label-model bias. Third, the transfer analysis uses a separately sourced label set built from experimentally reported inhibition values under a declared potency cutoff, which changes the label semantics between panels; internal and transfer results are therefore never pooled, and the transfer result is interpreted as a label-source-shift assessment rather than as a hold-out validation, because molecule-level disjointness between the panels was not established.

2.8.2 Chemical-space characterisation of the candidate cohort

Before any resistance or representation analysis, the candidate cohort is characterised on a fixed descriptor set so that its chemical-space position is known independently of any performance result. The characterisation reports the mean and dispersion of molecular weight, octanol–water partition coefficient, quantitative drug-likeness estimate, fraction of sp^3-hybridised carbons and stereocentre count, together with the composite multi-parameter optimization score and the synthetic-accessibility estimate. For the Set-C cohort these summaries locate the panel in a compact, moderately polar region dominated by natural-product cores, and the values are reported with the results. Two boundaries attach to this characterisation. The descriptors describe the sources from which the panel was drawn and carry no ethnobotanical validation of any candidate; and the summary statistics are cohort-conditional, so they should not be extrapolated to the parent library from which the cohort was filtered.

2.8.3 Computational environment

All analyses were executed on a single workstation-class server with forty-eight central-processing cores, 128 GB of memory and a consumer-grade graphics accelerator used for molecular-dynamics offloading. The software stack is pinned per analysis and recorded in each provenance manifest; Table 2.8.1 lists the principal tools and their roles. Where a tool exists in one pinned version per analysis, the version is recorded in the manifest rather than quoted here; the environment as a whole is treated as part of the frozen experimental setup.

2.8.4 Partition families and cross-validation

Rigorous evaluation of predictive models requires careful partitioning of the data, and the choice of partition determines what the resulting performance estimate means. Four partition families are used across the two representation benchmarks. A stratified random split provides the in-domain baseline. A canonical scaffold split based on Bemis–Murcko ring-core separation enforces structural extrapolation by assigning molecules that share a scaffold to the same side of the split. A Butina cluster split based on ECFP4 Tanimoto distance at a similarity cutoff of $C = 0.55$ enforces a coarser notion of chemical separation. Finally, a nearest-neighbour Tanimoto decile stratification orders test molecules by the similarity of their nearest training neighbour, so that performance can be resolved as a function of structural distance from the training distribution rather than reported as a single aggregate.

The learned-representation benchmark uses a repeated 5×5 cross-validation design, giving 25 fold–seed execution records per model architecture and partition family, and paired hypothesis testing on the five per-seed performance means. The descriptor benchmark uses 5-fold stratified cross-validation with fixed random seeds. Scaffold separation is confirmed independently by the decile analysis, because a model can be accurate within analogue series and still fail on unseen

Tool	Pipeline role	Consumed by
RDKit	fingerprints, descriptors, conformers	all descriptor analyses
Open Babel / Meeko	ligand standardisation, PDBQT conversion	docking stage
AutoDock Vina / QuickVina	docking scores	docking, sensitivity checks
DiffDock	generative pose sampling and confidence	consensus prioritisation
GNINA	CNN affinity rescoring	independent cross-scoring
SWISS-MODEL	homology modelling of mutant receptors	resistance panel
P2Rank	computational pocket detection	grid audit
GROMACS	MD integration	all trajectories
MDAnalysis	trajectory analysis	geometric summaries
ProLIF	interaction fingerprints	contact persistence
`gmx_MMPBSA`	endpoint energies and decomposition	MM-GBSA endpoints
scikit-learn	classical models, cross-validation	representation benchmarks
PyTorch stack	graph and transformer models	learned-representation benchmark
PennyLane	quantum kernel simulation	quantum kernel analysis
STRING database	protein–association network	network scoring

Table 2.8.1: *Principal software components of the computational environment. Versions are pinned per analysis in the provenance manifests; the table records the tool, its pipeline role and the analyses that consume its output.*

chemotypes; reporting both is necessary for a claim about prospective use. An exploratory partition at the level of scaffold clusters is used in the chemical-space analysis to verify that candidate sets remain disjoint.

2.8.5 Performance metrics

Performance metrics for binary classification include the area under the receiver operating characteristic curve (ROC-AUC), which quantifies discrimination across all decision thresholds; the area under the precision–recall curve, which is more informative under class imbalance; the enrichment factor at a fixed early fraction of the ranked list; and the Matthews correlation coefficient, which is robust to imbalance. Accuracy and macro F1 are reported alongside ROC-AUC where they aid interpretation. For regression against docking scores, the coefficient of determination R^2 is reported under cross-validation. One metric requires explicit qualification: because the activity panels used in this thesis are highly enriched (approximately 74 % active), the enrichment factor at 1 percent saturates at a prevalence-imposed ceiling and therefore cannot support a virtual-screening precision claim. Where that saturation occurs it is reported alongside the metric rather than omitted.

The saturation argument is stated formally because it recurs across the results. The enrichment factor at fraction f of the ranked list is $\mathrm{EF}_f = p_f/\pi$, where p_f is the active fraction within the top f of the list and π is the overall active prevalence. Its maximum value is bounded by $(1-(1-f)(1-\pi)^{|L|})/(f\pi)$ only for an ideal ranking; in the limit of a large ranked list the practical ceiling approaches $1/\pi$ times the top-fraction active rate, which cannot exceed 1. At $\pi = 0.742$ and $f = 0.01$, a perfect ranking would place almost only actives in the top one percent, so $\mathrm{EF}_{1\%}$ collapses toward $1/\pi \approx 1.35$ regardless of model quality, and observed values of 1.32 to 1.35 reflect the panel composition rather than discriminative performance. The same reasoning bounds precision at the top of the list near the prevalence-imposed maximum. Enrichment-style metrics are therefore reported only for panels whose prevalence leaves them identifiable, and on the enriched panels they are reported with their ceilings attached.

2.8.6 Conformal calibration and the distance-aware triage rule

Two calibration tools accompany the learned-representation benchmark, and both are declared as operational hypotheses rather than as validated procedures. The first is split-conformal prediction: a calibration split of the training data is scored, the nonconformity scores of the calibration split are ranked, and the quantile at level $\lceil (n_{\mathrm{cal}}+1)(1-\alpha) \rceil / n_{\mathrm{cal}}$ defines a threshold that yields prediction sets whose empirical coverage is guaranteed marginally at the target level $1-\alpha$ under exchangeability. The calibration is applied per model arm, and the reported diagnostics are empirical coverage and mean set size, which trade off against each other: a rule that covers more must predict more ambiguity. The second tool is post-hoc isotonic recalibration: the model's predicted scores are mapped to calibrated probabilities through an isotonic regression fitted on the calibration split, and calibration error is re-evaluated after the mapping. Isotonic regression is chosen over parametric calibration because it imposes no shape assumption on the score-to-probability map, at the cost of requiring sufficient calibration mass in every score region; regions without calibration support retain their uncalibrated values and the support boundary is reported.

The distance-aware triage rule assembles these pieces into a single operational proposal that the results chapters evaluate descriptively. The rule routes each test molecule to a representation arm according to the Tanimoto similarity of its nearest training neighbour: molecules in the interpolation regime, where classical fingerprints retain their advantage, are scored by the fingerprint baseline, and molecules in the structural-extrapolation tail, where the fingerprint advantage narrows, are routed to the arm with the best tail performance. Its coverage and calibration diagnostics are computed on the same frozen panel as the benchmark, and its evaluation inherits every boundary declared for that benchmark; in particular, because the rule is proposed and evaluated on the same panel, it is a testable operational hypothesis for prospective screening and not a demonstrated screening procedure. Prospective evaluation on an independent, low-prevalence library is the experiment that would convert the hypothesis into a validated rule, and that experiment is reserved for future work.

2.8.7 Statistical significance testing and multiplicity

Comparisons between models require appropriate statistical testing. For paired comparisons on identical cross-validation folds, paired t-tests or Wilcoxon signed-rank tests are used, with the paired t-test applied to per-seed means where a repeated design is available. Because the number of comparisons is large, multiplicity is handled deliberately rather than opportunistically. The descriptor benchmark declares a single confirmatory family comprising the 12 tests that support the comparative claims on the primary panel, including the classical descriptors against ECFP4, the hybrid descriptor against ECFP4, the quantum kernel against a linear kernel, and the ECFP4-inclusive fusion arms, and applies Holm–Bonferroni adjustment across that family at a family-wise error rate of 0.05. Analyses outside that family, including the external label-source-shift transfer and the secondary correlation analyses, are reported as exploratory with uncorrected p-values and effect sizes, and for the transfer analysis a variance-corrected resampled paired test is reported in addition to the naive fold-level test because the training sets overlap. The learned-representation benchmark applies Benjamini–Hochberg false-discovery-rate adjustment across the scaffold-split contrasts, reporting exact adjusted p-values rather than a single tail bound. Where a comparison fails to detect a difference, the result is stated as a failure to detect rather than as evidence of equivalence, since fold-level tests that share training data have limited degrees of freedom.

A parallel limitation governs every correlation reported in this thesis, and it is quantitative rather than rhetorical. With a cohort of n observations, a two-sided Spearman test at significance level α reaches a given power only above a minimum detectable association; for the cohorts analysed here the boundary is material. At $n = 17$, a two-sided test at $\alpha = 0.05$ reaches 80 % power only for $|\rho| \geq 0.62$, and at the Bonferroni-corrected threshold the power to detect a moderate association of $\rho = 0.5$ falls below 0.4. The consequence is asymmetrical: significant findings in these cohorts

are interpretable, but the absence of significance is not evidence of independence, and every null association in the results chapters is reported with this power boundary attached rather than as a demonstration of independence. Power considerations are computed before interpretation for every correlation family, and the detectable-effect boundary is stated wherever a null result carries interpretive weight.

2.8.8 Uncertainty taxonomy and formal test mechanics

The quantities reported in this thesis carry three kinds of uncertainty that are never interchanged. Sampling uncertainty is the frame-level or fold-level dispersion of an estimate within one execution of a declared protocol, and it is what a standard error or a cross-validated interval describes. Protocol sensitivity is the response of a derived quantity to a declared arbitrary choice, such as a threshold, a seed, a weight or a scoring function, and it is what seed-dispersion bounds, weight-sensitivity checks and cross-scoring agreements quantify. Replicate variance is the dispersion across independent executions of the same protocol on the same system, and it is the only one of the three that bears on the reproducibility of an endpoint estimate; it is available only where replicates exist. The distinction matters because the three are routinely conflated into a single error bar: a frame-level standard error says nothing about protocol sensitivity, and a protocol-sensitivity bound says nothing about replicate variance. Where replicate variance is unavailable, which is the case for the single-replicate pilot trajectories, the reported intervals are explicitly labelled as within-execution dispersions and no reproducibility claim is attached to them.

The formal machinery of each reported test is fixed rather than chosen per analysis. Spearman rank correlations are computed on paired observations with permutation-based null distributions, the permutation count fixed at 100 000 with a fixed random seed, and multiplicity handled by the declared family structure of the analysis in question. Paired cross-validated comparisons use per-seed means as the unit of analysis, so that the test operates on the appropriate degrees of freedom rather than on pseudo-replicated fold records. Exact binomial tests are used for direction-agreement statistics against a 50 % chance level. For a proportion estimated from a small number of matched comparisons, such as a direction-agreement fraction or a divergence fraction, the interval reported is the Wilson score interval rather than the Wald interval, because the Wald interval is known to under-cover near the boundaries of the parameter space, which is exactly where small-sample proportions typically fall; the Wilson interval maintains nominal coverage at small counts and never produces bounds outside $[0, 1]$. The count and the denominator are always reported alongside the interval, since an interval without its sample size invites over-reading. Bootstrap intervals are percentile intervals from a fixed resample count with a fixed seed, reported for descriptive effect sizes rather than for hypothesis tests. The seed of every stochastic procedure is part of the frozen manifest, so that every reported interval is exactly reproducible.

A final governance rule concerns hyperparameters and the order in which choices are frozen. Baseline configurations, cross-validation designs, threshold values and the candidate grid of every tuned comparator are fixed in the analysis plan before the corresponding comparative experiment is run, and the frozen values are recorded in the provenance manifest. Where a sensitivity analysis is run after the fact, it is labelled as such and its role is bounded: it tests the robustness of a declared choice, it does not replace the choice retroactively. This ordering matters because the alternative, selecting hyperparameters after observing cross-validated performance, converts the reported performance estimate into an optimistic bias whose magnitude cannot be quantified after the fact. Nested designs, in which inner cross-validation selects a hyperparameter and outer cross-validation evaluates it, are used wherever the tuned comparator is part of a confirmatory claim; where a design is not nested, the claim it supports is correspondingly limited and reported as exploratory.

2.8.9 Honest-negative reporting and provenance documentation

This thesis adopts a policy of honest-negative reporting, in which null results, negative findings and limitations are documented with the same rigour as positive results. Honest-negative reporting prevents publication bias, provides realistic expectations for future work, and supports scientific integrity. Null results, meaning cases where a method does not significantly outperform a well-optimised baseline, are reported alongside positive findings with the statistical test that supports the absence of a detectable difference. Two consequences of this policy appear throughout the thesis: quantum-inspired representations are not presented as advantageous when a matched classical comparator performs equivalently, and learned representations are not presented as superior when scaffold-separated evaluation reverses the ranking observed under random splitting.

The policy is implemented through three documentation practices that make the negative results auditable rather than merely stated. First, failed and excluded work is retained in the record with an explicit status, so that the denominator of every success statistic is visible; a trajectory that failed quality control and an analysis that detected nothing are both first-class entries. Second, every negative claim is accompanied by the analysis that would have detected the opposite, including its power boundary, so the reader can distinguish *we found nothing* from *nothing was detectable*. Third, the scope of every negative result is fixed to the panel, protocol and label source under which it was established, so that a negative result is never converted by phrasing into a universal one. These practices cost space and repetition, and they are retained deliberately: in a thesis whose central empirical finding is that several fashionable representations add nothing over a well-maintained baseline, the credibility of the negative results is itself a primary result, and it is earned by this documentation rather than asserted.

Provenance documentation ensures that every computational result can be traced to its source data, scripts, parameters and execution environment. For each analysis a provenance manifest records the input files with cryptographic hashes, the software versions and dependencies, the hyperparameters and configuration files, the random seeds used for every stochastic procedure, the computational resources employed, and the output file locations and hashes. Manifests are stored in version-controlled repositories alongside the analysis scripts, so that results can be reproduced from the recorded state rather than from descriptions. An additional rule applies to the activity labels used in the two representation benchmarks: the primary labels are machine-learning predictions from an external model rather than assay measurements, and the external transfer analysis uses a separately sourced label set. Both facts are recorded at every point where a performance number is reported, so that a model's ability to predict a model's output is never presented as biological activity prediction. Failed experiments, exploratory analyses and superseded results are retained and labelled rather than deleted.

For every computational study reported in this thesis, provenance manifests are maintained in the results directories of the underlying project folders. Each computational experiment is assigned a unique job identifier, and results are stored in versioned output directories named by job identifier and timestamp. The quantum molecular structure encoding programme of Section 2.7.4 adopts the same provenance scheme: its encoding stage is frozen with recorded input hashes, its pipeline-confirmation execution is logged, and until its canonical benchmark completes, its comparative status remains not computed and no comparative quantum result appears in this thesis.

2.8.10 External data and reference-set versioning

Several inputs to the analyses described in this chapter are external resources that change on their own release cycles, and each one is therefore versioned as part of the frozen experimental setup rather than treated as a stable background. The docking benchmarks are fixed panel releases, so the panel identifier and download date are recorded in the manifests; the protein structures are specific PDB entries cited by accession, with the retrieval date recorded because remediation of

deposited entries can alter coordinates without changing the accession. The reference sets behind the chemical-space descriptors are pinned with the same discipline: the approved-drug reference, the African natural-product reference set and the network database used for the polypharmacology score are each recorded with their release versions, and the network confidence threshold is part of the manifest rather than a default inherited from a tool. The activity-label model is pinned by model identifier and version, since a silent update of the underlying model would change every label in the panel and invalidate comparisons across runs.

The versioning rule is that no reported number may depend on an unversioned external resource. Where a resource cannot be pinned, for example a web service whose historical states are not archived, the dependence is declared and the affected analysis is labelled as non-reproducible at the level of that input, with the rest of its chain still frozen. This rule costs little at analysis time and prevents the most common irreproducibility in computational drug discovery: results that cannot be regenerated because a database quietly changed underneath them. The same discipline extends to in-house artefacts passed between analyses: a descriptor file consumed by a downstream model is hashed and referenced by hash, so that a stale or partially rewritten file is detectable rather than silently consumed.

2.8.11 Reproducibility infrastructure and reporting boundaries

The reproducibility chain runs from frozen inputs to reported numbers without manual intervention. Each analysis is a versioned script whose execution reads only the hashed inputs declared in its manifest, writes only to its versioned output directory, and emits the job identifier, software versions, random seeds and wall-clock resource usage into the output record. Re-running the script against the frozen inputs must reproduce the reported numbers bit-for-bit for deterministic stages and within the declared seed distribution for stochastic ones; any deviation is a defect to be investigated, not a value to be silently repaired. Where a stage fails, the failure propagates to the output record as an explicit status rather than as a gap, and downstream analyses either exclude the affected records with the exclusion documented or halt.

Reporting boundaries are attached to estimands rather than to chapters, so that a reader meeting a number out of context still meets its scope. Docking scores carry the boundary that they are score-retention statistics on anchored poses; trajectory endpoints carry the boundary that they are within-protocol contrasts at single-replicate scope; activity metrics carry the boundary that their labels are model predictions rather than assay measurements; and every correlation carries its cohort size and the power boundary derived from it. The boundaries are not caveats appended after the fact: each one is fixed when the corresponding quantity is defined, and the results chapters state them at the point where the quantity first appears. This convention costs space, but it removes the failure mode in which a number circulates with its scope stripped, which is how over-claiming enters a manuscript without any single sentence being false.

The final reporting convention is the claim ledger. Each results chapter closes by restating its central claims in a fixed four-way status vocabulary: *supported*, meaning supported within the declared protocol and sampling limits; *refuted*, meaning the declared protocol produced evidence against the claim; *not supported*, meaning the analysis failed to detect the claimed effect at the declared power, which is not a refutation; and *open*, meaning the claim cannot be decided computationally and requires prospective experiment. The statuses are scope-qualified by construction, so a claim supported in one cohort is not silently promoted to a general claim, and a hypothesis evaluated only descriptively is recorded as open rather than as supported. The ledger is a completeness device as much as a honesty device: it forces every central claim to be revisited after the analyses, so that a claim that quietly lost its evidential support during the work is surfaced rather than forgotten.

2.9 Conclusion

In this chapter, the computational framework of the thesis has been specified end to end. The construction of the hybrid library was detailed, together with its two expansion routes, the declared scoring terms of the prioritisation pipeline, the four-target docking protocol, its three-part validation, and the correction and audit of the transporter receptor. The resistance-resilience scoring framework was established with its estimand architecture, eligibility rule, class definitions, network and chemical-space descriptors, and retrospective genotype–phenotype confrontation. The molecular-dynamics protocol was described across its three cohorts, from force-field assignment through equilibration to quality control, endpoint estimation, and the interaction-fingerprint complement, together with the distance-ratio definition that links the dynamics to the docking layer.

The descriptor and modelling layers were then specified: the classical fingerprint family with its three encoding principles, the persistent-homology, tensor and quantum-inspired descriptors under a common conformer-generation governance, the kernel diagnostic hierarchy that separates geometry from performance, and the random-forest, graph and transformer architectures with their frozen configurations, together with the quantum molecular structure encoding protocol of the hybrid quantum-classical programme, whose encoding substrate is frozen and whose canonical benchmark is declared but pending. The statistical framework closed the chapter with partition families, activity-label provenance, prevalence-bounded metrics, conformal calibration and the distance-aware triage rule, significance testing with its power boundaries, the Wilson interval for small-sample proportions, the uncertainty taxonomy, reproducibility infrastructure, external-data versioning, and the claim-ledger convention.

In summary, the protocols presented in this chapter furnish the frozen methodological substrate on which every result of the next chapter rests. Each quantity that appears in the results is defined here with its estimand, its uncertainty type and its reporting boundary, so that the results can be read as tests of the declared protocol rather than as a collection of numbers, and the comparison standard that the quantum machine learning programme must satisfy follows directly from this specification.

Chapter 3

RESULTS AND DISCUSSION

3.1 Introduction

This chapter reports the empirical progression of the thesis: four completed studies that diagnose a representational ceiling, and the structure-direct quantum encoding programme, in progress, that tests whether that ceiling can be overcome. The argument is not four independent reports; it is a single investigative arc with an honest-negative empirical foundation. The chemical-space study constructs an African-natural-product-inspired hybrid library and characterizes it by target-anchored docking against four mechanistically distinct antimalarial proteins. The resistance-scoring study converts the docking campaign into a resistance-aware scoring framework, stress-tests that framework with explicit-solvent molecular dynamics, and confronts the static estimand with published genotype-resolved inhibition data. The representation study asks whether the topological and quantum-inspired descriptors that motivate the thesis add predictive value over an extended-connectivity fingerprint baseline. The learned-representation study extends the same comparison to graph networks and transformers under chemical distribution shift. The structure-direct encoding study then reports the quantum machine learning programme that those four results define: its archived exploratory phase, its frozen encoding substrate, its confirmed pipeline and its predeclared benchmark.

The studies share one library, one label source, and one evaluation philosophy. That shared substrate makes the chapter a single argument rather than a sequence of independent reports. The argument runs as follows: the chemical space is genuinely novel while remaining scaffold-anchored — the scaffold paradox that every later section returns to; docking-derived resistance metrics separate candidate chemotypes but are estimand-bound; and every representation tested here, whether topological, tensor-compressed, kernel-based, or learned, is diagnostically informative while failing to improve on ECFP4 for activity prediction. These four results converge on a shared boundary: no representation examined encodes three-dimensional geometry directly. The honest-negative results of the two representation studies are therefore not side observations of this thesis; they are the empirical baseline against which true quantum machine learning, the subject of the quantum machine learning programme, must be judged.

Throughout, computational predictions are kept explicitly separate from biological activity. Docking scores are empirical scoring-function outputs, activity labels are machine-learning predictions rather than assay measurements, and resistance classes are operational categories computed from score ratios. Where a result cannot support the interpretation a reader might expect, that boundary is stated at the point of the result rather than in a closing caveat.

The chapter is organised as follows. Section 3.2 reports the construction of the hybrid library and the target-anchored docking campaign: the scaffold-guided expansion and its novelty statistics, the prioritisation diagnostics of the expansion routes, the latent-space encoder and the predicted ADMET profiles, the four-target docking profile with protocol validation, the mutation-resilience analysis, and the joint prioritization with the cross-metric information content; and a closing discussion subsection that reads the library statistics as predictions for the studies that follow and presents the resulting shortlist. Section 3.3 presents the resistance-aware scoring results: the framework scope and descriptor characterization, the class counts and protocol reproducibility, the static-to-dynamic estimand divergence that forms the central result, the trajectory-derived endpoints, the distinct information carried by the three ranking dimensions, and the retrospective ChEMBL confrontation that bounds the static estimand from experimental data; followed by a discussion of what the divergence implies about the role of dynamics in a prioritization pipeline. Section 3.4 evaluates the quantum-inspired and topological representations: the topological decomposition of the scaffold paradox, the tensor-compression behaviour, the activity benchmark with its hybrid ablation, the quantum kernel against its tuned classical comparator, the fusion and transfer analyses, the quantum-inspired descriptors' surviving diagnostic contributions, the scaffold-grouped re-evaluation with the docking cross-link, and a discussion of the three mechanisms that make the ceiling structural. Section 3.5 extends the comparison to learned representations under chemical distribution shift: the three partition families,

the localization of the deficit in the structural extrapolation tail, the topological salience that separates what a descriptor uses from what it contributes, and a discussion of what these results mean for model choice in practice. Section 3.6 reports the structure-direct quantum encoding programme: the archived exploratory phase and its garbage-in-garbage-out diagnosis, the frozen encoding substrate and confirmed pipeline, the reformulated benchmark question against classical three-dimensional descriptors, a status discussion separating what the programme has established from what it remains, and the comparison standard any hardware-based claim must satisfy. Section 3.7 closes the chapter with the synthesis that states, on the basis of the results reported here, the comparison standard the quantum machine learning programme must meet, collects the estimand register of every principal quantity reported, and states the falsification conditions under which the chapter's own conclusions would need revision.

3.2 Chemical space construction and target-anchored docking

3.2.1 Scaffold-guided expansion produced a novel but scaffold-anchored library

The library was assembled from 396 African natural products and 454 synthetic antimalarial compounds and expanded by two complementary routes. A similarity search through the Cheese API retrieved 100 analogues per seed from ZINC15/ENAMINE-REAL using Morgan fingerprints (radius 2, 2048 bits) at a Tanimoto threshold of 0.7, and STONED-SELFIES generated a further 100 variants per seed through single-character SELFIES mutations over an alphabet of 50. Generated structures were PAINS-filtered, validated with RDKit and Open Babel, and deduplicated on canonical SMILES, yielding a hybrid library of 65 856 unique molecules, of which 94.1 % remain Lipinski-compliant.

The library is not merely large; it occupies space that the seeds do not. In a 5000-molecule novelty sample, 92.6 % of the expanded molecules had a maximum whole-molecule Tanimoto similarity below 0.4 to the African natural-product seed set, while 69.3 % of the annotated seed scaffolds (70 of 101) were recovered in the expanded library. Maximum scaffold similarity consistently exceeded maximum whole-molecule similarity (mean 0.379 against 0.206), which locates the novelty in the peripheral substituents rather than in the ring systems. Figure 3.2.1 shows the resulting coverage and the scaffold retention that accompanies it. This asymmetry is the structural motif that recurs in the representation study, where persistent homology is used to decompose it.

Multi-parameter optimization combined docking-derived scores, DiffDock confidence, QED, an ADMET composite, and a Lipinski penalty. The 484 centroids obtained by k-means on a 64-dimensional SMILES-VAE latent representation were reduced to 76 centroids above the optimization threshold, mapped to 53 clusters and expanded to 40 481 molecules, of which 19 913 survived synthetic-accessibility and MPO filtering. Centroid-based sampling cut the docking campaign from 263 424 to 1936 calculations, a 99.3 % reduction that brings an estimated 2600 to 5300 CPU-hours of exhaustive docking below 20 CPU-hours while retaining the 69.3 % scaffold recovery. That figure is a computational-cost statement, not a claim of active-compound retention; a retention experiment against known actives would be required to make the stronger claim.

3.2.2 Prioritisation diagnostics: route composition, latent-space quality and predicted ADMET

The two expansion routes contributed measurably different chemistry, and the difference is the complementarity the design intended. Counted at the expansion stage, before PAINS filtering, standardisation and deduplication produced the canonical 65 856-molecule library, the similarity-search

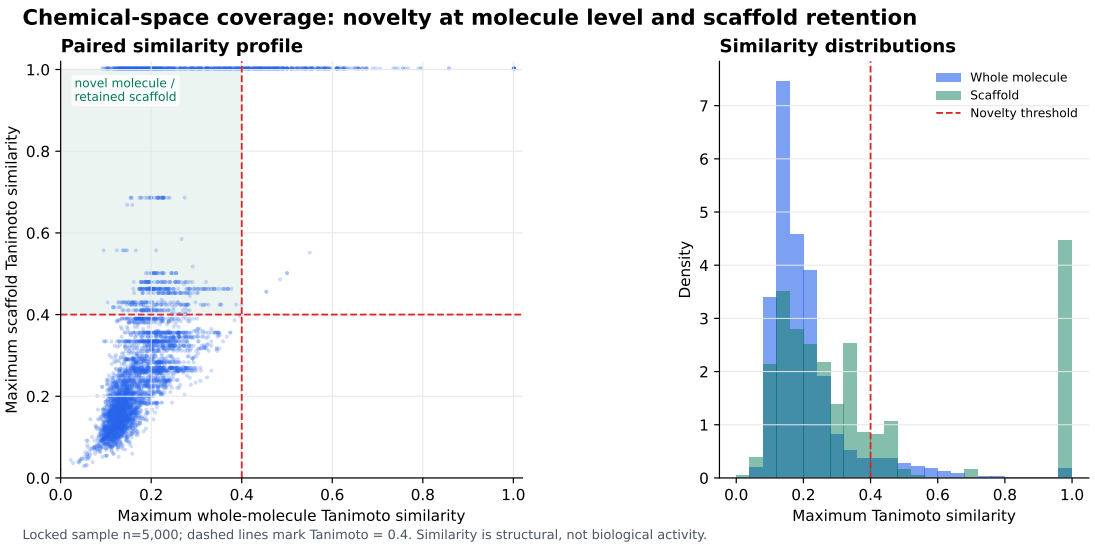


Figure 3.2.1: *Coverage of the hybrid library relative to the African natural-product seed set. Whole-molecule novelty (92.6 % of a 5000-molecule sample below a Tanimoto similarity of 0.4) coexists with scaffold retention (69.3 %; 70 of 101 annotated seed scaffolds). The gap between whole-molecule and scaffold similarity distributions defines the scaffold paradox analysed topologically in the representation study.*

route retrieved 29 805 database analogues whose scaffold inventory is narrow (394 unique Bemis–Murcko scaffolds, a scaffold diversity of 0.013) but whose drug-likeness is high (mean QED 0.697), while the SELFIES-mutation route generated 35 198 variants spanning 13 378 unique scaffolds (diversity 0.380) at a lower mean QED of 0.497. The seed sets bracket this behaviour: the 396 African natural products carry 245 scaffolds (diversity 0.619) at a mean QED of 0.486, and the 454 synthetic antimalarials carry 331 scaffolds (diversity 0.729) at 0.600. Valid-structure fractions were 0.992 for the natural products, 1.000 for the synthetic antimalarials, 0.985 for the similarity search and 0.915 for the SELFIES mutation, the lowest of the four, which quantifies the gap between valence validity by construction and chemical plausibility. Figure 3.2.2 displays the composition: the similarity search contributes database-validated, drug-like analogues with little scaffold novelty, the SELFIES route contributes scaffold-diverse structures at a drug-likeness cost, and the seeds occupy the two extremes of the diversity axis.

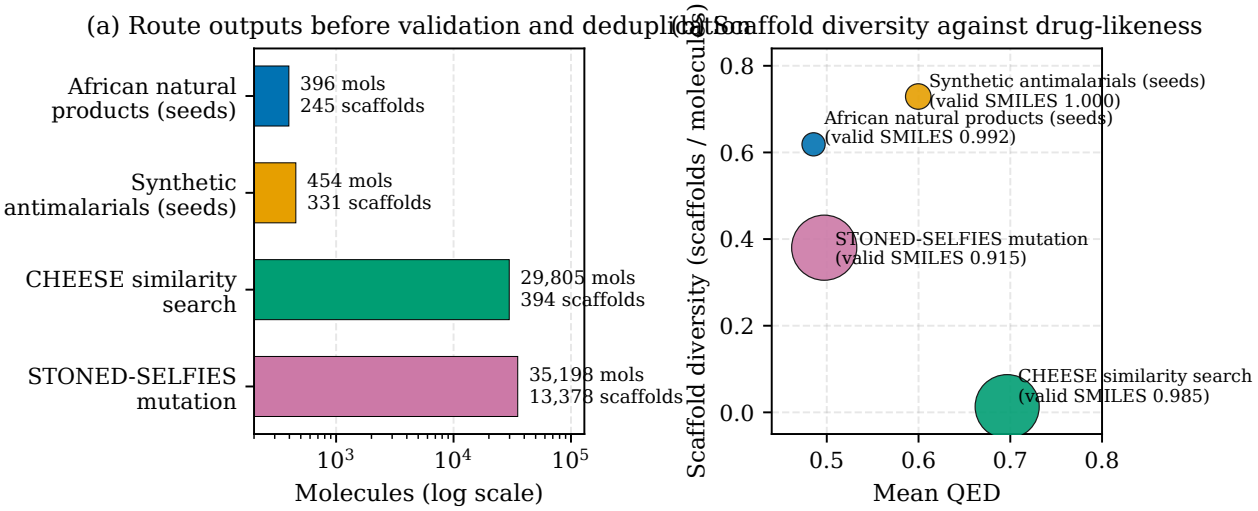


Figure 3.2.2: *Composition of the two expansion routes against their seed sets, counted before validation and deduplication. (a) Route outputs on a logarithmic scale with unique Bemis–Murcko scaffold counts. (b) Scaffold diversity against mean QED, with point area proportional to library size and valid-structure fractions annotated. The similarity-search route is drug-like but scaffold-narrow (29 805 molecules, 394 scaffolds); the SELFIES route is scaffold-diverse at a QED cost (35 198 molecules, 13 378 scaffolds).*

The encoder that supports centroid selection was validated before the clustering was trusted, and the validation is a trade-off report rather than a benchmark victory. Training converged for both latent dimensionalities, with validation-loss standard deviations of 0.022 (32-dimensional) and 0.036 (64-dimensional) over the final ten epochs. The 32-dimensional model reached a final validation loss of 47.85 with 94.18 % SMILES reconstruction accuracy; the 64-dimensional model reached 50.62 with 93.23 % (Figure 3.2.3). On the clustering side (Table 3.2.1, Figure 3.2.4), the 32-dimensional space is better on silhouette (0.297 against 0.274), on the Davies–Bouldin index (0.302 against 0.376, lower being better) and on reconstruction, while the 64-dimensional space is better on the Calinski–Harabasz index (565.6 against 306.6), which rewards separation across many clusters. The 64-dimensional representation was retained for the production clustering on that separation criterion with comparable reconstruction accuracy, and the choice is recorded as a selection under a mixed evidence profile, not as a dominance claim. Both silhouette values sit in the moderate band (0.25 to 0.50) expected for a continuous chemical space. That is acceptable for the task the clustering performs—selecting structurally diverse representatives—and would not be acceptable if the clusters were read as discrete chemical classes; at the production $k = 484$ partition the latent-space silhouette is 0.229, as reported with the representation comparison in Section 3.4.

Metric	32D	64D
Validation loss	47.85	50.62
Reconstruction loss	32.27	37.76
KL divergence	15.58	12.86
Reconstruction accuracy (%)	94.18	93.23
Silhouette score	0.297	0.274
Calinski–Harabasz index	306.6	565.6
Davies–Bouldin index (lower is better)	0.302	0.376

Table 3.2.1: *Model-selection comparison of the 32- and 64-dimensional SMILES-VAE latent spaces. Training metrics are final-epoch values; clustering metrics are computed on the k-means partition of each space at model-selection scope. The two dimensionalities are a trade-off, not a ranking: 32D leads on silhouette, Davies–Bouldin and reconstruction; 64D on Calinski–Harabasz.*

The internal structure of the partition supports the claim that a centroid is a structurally coherent sampling unit rather than an arbitrary point. Mean intra-cluster Tanimoto similarity, averaged over clusters, is 0.227 against 0.101 for cross-cluster pairs, a factor of about 2.2 (Figure 3.2.5); the mean pairwise similarity between the 484 centroids themselves is 0.32, as reported with the chapter synthesis in Section 3.7. Per-cluster physicochemical profiles (Figure 3.2.6) show the ten leading clusters of the 484-cluster partition occupying compact, strongly overlapping bands in molecular weight, lipophilicity and polar surface area, so screening one representative per cluster does not stratify the property space by construction.

Predicted ADMET endpoints were characterised library-wide with the same machine-learning platform that feeds the prioritisation composite[99], and the endpoint distributions stratify by compound source (Figure 3.2.7): molecular weight, lipophilicity, solubility, blood–brain-barrier permeability, bioavailability, clearance, intestinal absorption, cytochrome P450 inhibition, hERG blockade and toxicity endpoints, with the natural products predicted to be on average the more bioavailable class and the two generated subclasses filling the intermediate region, as expected from their dual parentage. These are in-silico endpoints used for ranking within the multi-parameter score; they are not experimental pharmacokinetics, and no candidate advances on them alone.

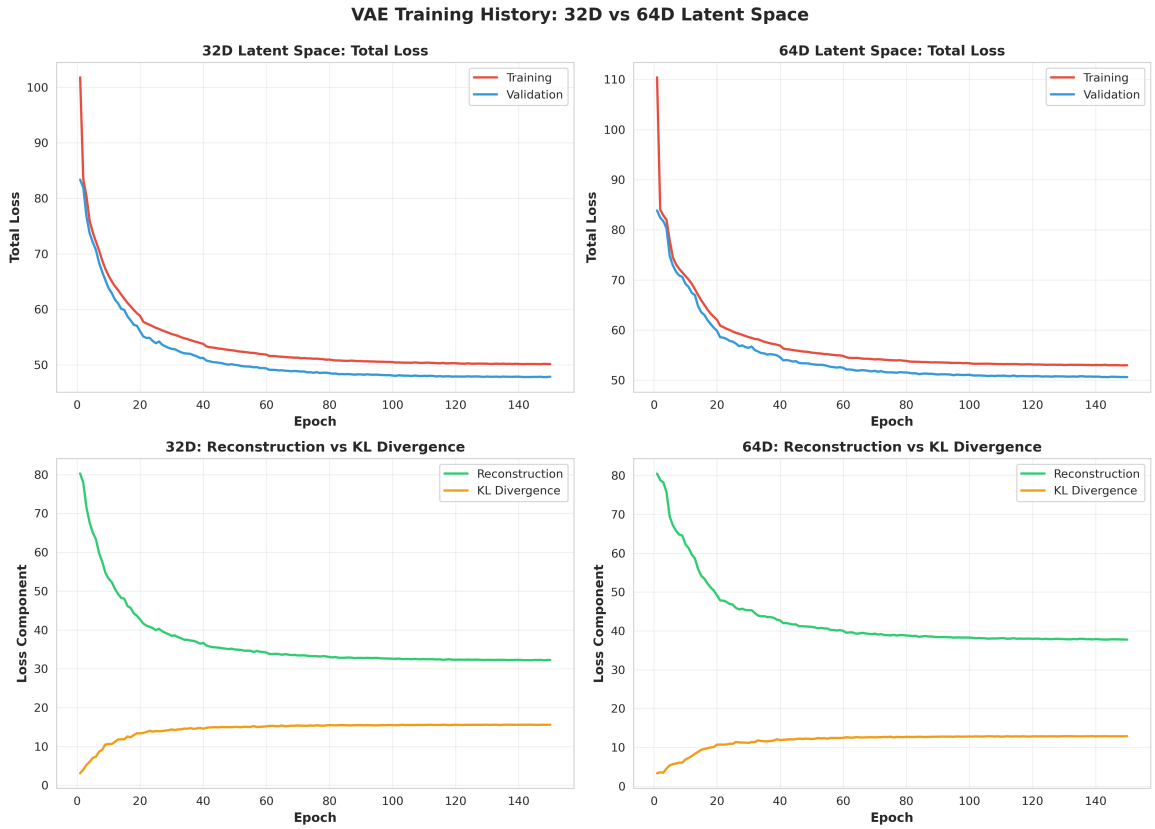


Figure 3.2.3: *Training histories of the 32- and 64-dimensional SMILES-VAE models over 100 epochs. Both runs converge, with final-epoch validation-loss standard deviations of 0.022 and 0.036 respectively and reconstruction accuracies of 94.18 % and 93.23 %.*

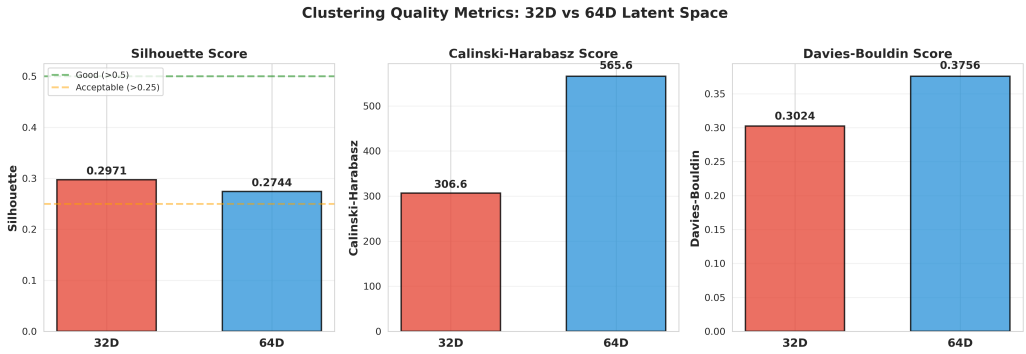


Figure 3.2.4: *Clustering-quality diagnostics of the k-means partitions of the 32- and 64-dimensional latent spaces: silhouette score, Calinski–Harabasz index and Davies–Bouldin index. The comparison is mixed—32D leads on silhouette and Davies–Bouldin, 64D on Calinski–Harabasz—which motivated the selection of the 64-dimensional space on the separation criterion.*

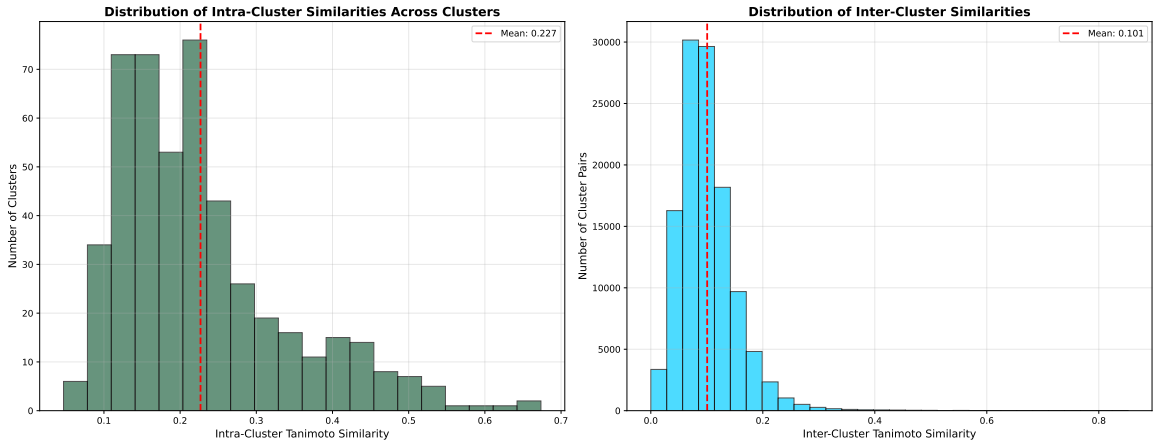


Figure 3.2.5: *Tanimoto similarity distributions within clusters (left, mean 0.227, averaged over clusters) and across cluster pairs (right, mean 0.101) of the 484-cluster partition. The roughly 2.2-fold separation supports reading each cluster representative as a structurally coherent sampling unit.*

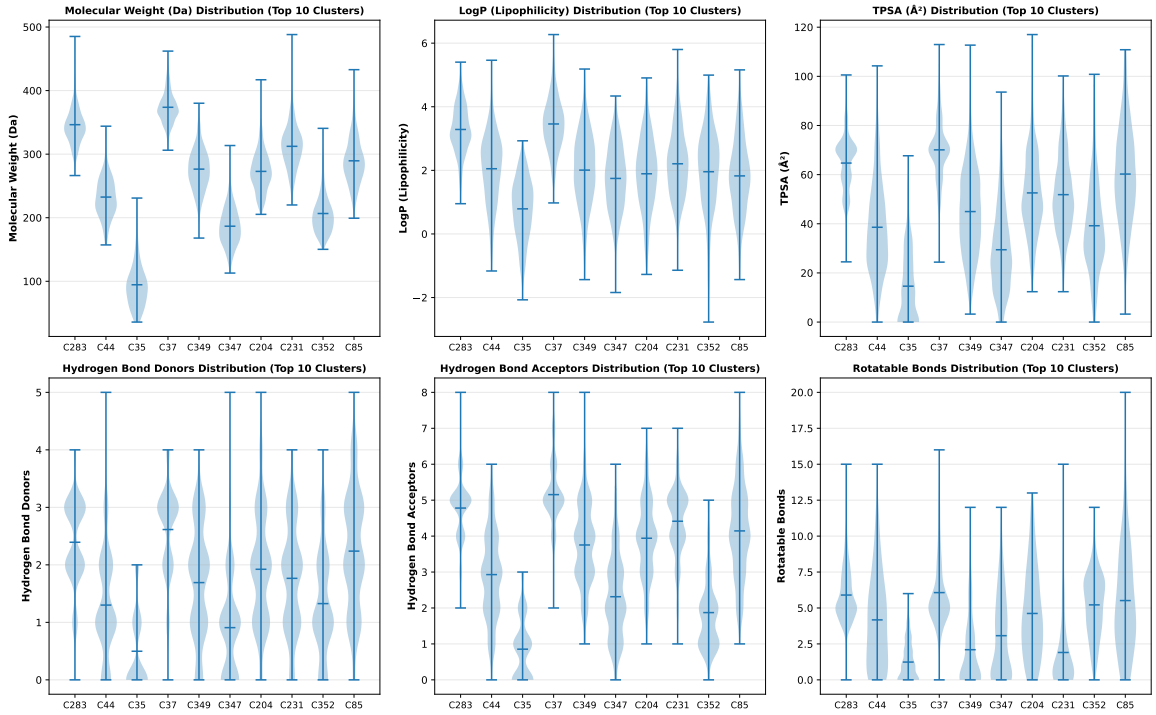


Figure 3.2.6: *Physicochemical distributions (lipophilicity, molecular weight and topological polar surface area) across the ten leading clusters of the 484-cluster k-means partition. The property bands overlap strongly between clusters, so centroid-based sampling does not stratify the property space by construction.*

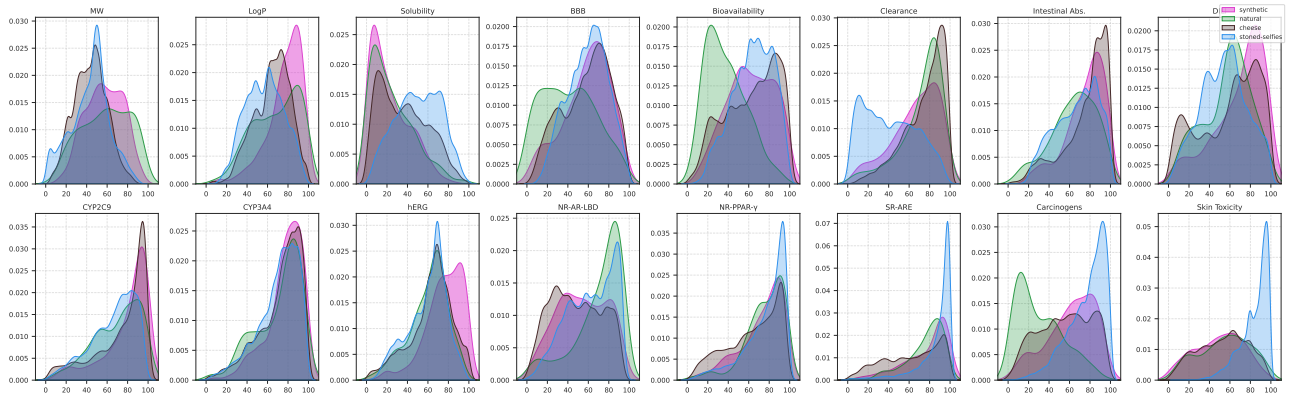


Figure 3.2.7: *Predicted ADMET and toxicity endpoint distributions stratified by compound source (synthetic antimalarials, African natural products, similarity-search analogues and SELFIES variants), from the machine-learning platform used for the prioritisation composite[99]. The endpoints feed the ADMET term of the multi-parameter score and are used for ranking only.*

3.2.3 Target-anchored docking across four mechanistically distinct proteins

Four protein targets were chosen to span independent resistance-relevant vulnerabilities rather than to provide four equivalent binding sites. *Pf*DHFR (PDB 7F3Y, methotrexate copy A702) anchors the antifolate axis; *Pf*CRT is derived from the 6UKJ cryo-EM structure of the 7G8 isoform and was corrected to a 3D7-like wild-type receptor by restoring lysine at position 76, where the deposited structure carries the K76T resistance substitution; the corrected receptor was audited by P2Rank, whose top pocket scored 162.7 at probability 0.999 with its centre 5.7 Å from the docking grid centre. *Pf*ClpP (PDB 2F6I, chain-A Ser252/His223/Asp219 catalytic triad) represents proteostasis, and *Pf*ATP4 (PDB 9N10, D451 phosphorylation-loop region and DPPR hinge) represents ion homeostasis. The *Pf*CRT pocket is a proxy cavity and the *Pf*ATP4 region is mechanism-motivated: neither carries a co-crystallized small-molecule inhibitor, so the panel is a gradient of structural confidence, not a set of four validated binding sites.

Ligands were standardized with Open Babel and converted to the PDBQT format with Meeko. Docking used target-specific cubic boxes: 25 Å edge length for *Pf*DHFR and *Pf*CRT, 20 Å for *Pf*ClpP, and 24 Å for *Pf*ATP4. All 68 candidate–target pairs formed by the 17-member polypharmacology-oriented Set-C cohort passed the geometric pose-quality gate, which requires a rank-1 centroid inside the box, at least one atom within 10 Å of the target anchor, and at least 90 % of ligand atoms inside the padded box. Score ranges were −6.35 to −4.86 kcal mol^{-1} for *Pf*DHFR, −12.01 to −6.37 kcal mol^{-1} for the corrected *Pf*CRT receptor, −7.03 to −5.05 kcal mol^{-1} for *Pf*ClpP and −7.57 to −4.63 kcal mol^{-1} for *Pf*ATP4. The most favourable estimate in the cohort was PP-03 in the *Pf*CRT cavity at −12.01 kcal mol^{-1}; PP-13 led the *Pf*ATP4 region at −7.57 kcal mol^{-1}. Because Vina scoring functions are not calibrated across heterogeneous pockets, no cross-target arithmetic mean was computed; target breadth is read from the pattern of per-target evidence. The cohort was 100 % Lipinski- and Veber-compliant with a mean QED of 0.703.

The docking protocol itself was assessed before any cohort-level interpretation, and the assessment is deliberately mixed. Redocking of co-crystallized reference ligands reproduced the crystal pose for four of five alignable ligands (heavy-atom RMSD below 2.0 Å; 80 % success), with at least one successful reproduction for each of the three targets used in the primary analysis. The single failure was the *Pf*DHFR reference ligand methotrexate, at an RMSD of approximately 30 Å, because the *Pf*DHFR grid is centred on the NADPH-adjacent site rather than the catalytic methotrexate pocket. Enrichment against the MMV Malaria Box under the dual-filter consensus gave ROC-AUC values of 0.924 (*Pf*DHFR), 0.971 (*Pf*CRT) and 1.000 (*Pf*ATP4). Against the external DEKOIS 2.0 panel for *Pf*DHFR, however, Vina alone reached only 0.450 (95% CI 0.37 to 0.53) with $\mathrm{EF}_{5\%}$ of 0.00. A

controlled two-arm comparison on the identical DEKOIS panel gave overlapping results whether the retained methotrexate was present (0.502) or stripped (0.563), so cofactor steric blockade does not explain the near-chance enrichment. Table 3.2.2 summarizes these checks.

Check	Target	Result	Interpretation
Redocking	*Pf*DHFR	1 of 2 (MTX ≈ 30 Å)	Grid centred on NADPH-adjacent site
Redocking	*Pf*CRT	1 of 1	Crystal pose reproduced
Redocking	*Pf*ClpP	RMSD 1.65 Å	Peptide-fragment pose
Redocking	*Pf*ATP4	RMSD 1.23 Å	ADP cofactor pose
MMV (dual filter)	*Pf*DHFR	ROC-AUC 0.924	399 actives / 399 negatives
MMV (dual filter)	*Pf*CRT	ROC-AUC 0.971	Score-stratified labels
MMV (dual filter)	*Pf*ATP4	ROC-AUC 1.000	Score-stratified labels
DEKOIS 2.0 (Vina only)	*Pf*DHFR	ROC-AUC 0.450 [0.37, 0.53]	40 actives / 1200 property-matched decoys
DEKOIS two-arm control	*Pf*DHFR	0.502 vs 0.563 (MTX retained vs stripped)	Cofactor blockade not the cause

Table 3.2.2: *Validation of the target-anchored docking protocol. Redocking assesses pose reproduction; MMV Malaria Box and DEKOIS 2.0 assess enrichment. The DEKOIS result constrains absolute Vina scores for* Pf*DHFR but does not invalidate within-target rankings or the mutant-to-wild-type ratios used by the resistance-resilience score, which are computed on the same receptor.*

3.2.4 Why the internal and external validation results are not in conflict

The three validation assessments ask different questions, and separating them is what turns a mixed-looking table into a coherent protocol statement. Redocking asks whether the protocol reproduces a known binding geometry, and it does for four of five alignable ligands, with the single failure explained by grid placement rather than by scoring behaviour. The MMV enrichment asks whether ranked scores recover known actives in a curated setting; for *Pf*CRT and *Pf*ATP4 its labels are themselves score-stratified, so it measures retrodictive consistency with the scoring function rather than independent discrimination, as Section 2.3 declares. The DEKOIS 2.0 benchmark asks the hardest of the three questions — whether Vina alone separates actives from decoys matched on bulk physical properties so that ranking success cannot follow from property differences alone — and the answer for *Pf*DHFR is near chance, unchanged by the removal of the retained cofactor. A protocol that passes the first two questions and fails the third is not self-contradictory; it is a protocol whose valid uses have to be stated, which is what the rest of this thesis does.

The consequence is mechanical. Systematic scoring bias enters both terms of any ratio computed on the same receptor, so mutant-to-wild-type retention values are the kind of quantity the protocol can support, while absolute affinity claims and cross-target score arithmetic are the kind it cannot — which is why no cross-target mean was computed for the cohort. Table 3.2.3 states the per-target

position of the campaign in one place, and the DEKOIS value is retained in it unchanged, because the honest bound belongs in the summary rather than in a footnote.

Read against the later sections, the validation pattern anticipates the chapter's central distinction: the same scores, used as static estimands, fail both retrospective checks of Section 3.3, and the representation benchmarks of Section 3.4 and Section 3.5 show the analogous failure at the level of model evaluation, where an area under the curve measured under one partition family does not transfer to another. Declaring the estimand at the point of the result is the single methodological habit that keeps all three failures interpretable.

3.2.5 Mutation resilience is target-specific and independent of target breadth

The resistance-resilience score (RRS) is defined per target and per mutation as the magnitude of the mutant Vina score divided by the magnitude of the wild-type Vina score for the same candidate and receptor, expressed as a percentage and written here as $\mathrm{RRS}_{i,m,t}$ (Equation (2.4.1)). Candidates whose wild-type score magnitude falls below 5.0 kcal mol^{-1} are excluded from that target's denominator rather than assigned a surrogate value, which prevents near-zero denominators from inflating the ratio. Only targets satisfying the wild-type binding criterion contribute to a candidate's mean. Classes are assigned from retention thresholds, with Class A* additionally requiring a wild-type anchor of at least 7.0 kcal mol^{-1}: the cohort resolved into seven A*, four B and six C profiles, spanning 73.2 to 115.6% of wild-type score magnitude (Table 3.2.4).

Two features of Table 3.2.4 deserve emphasis because they constrain how the ranking may be used. First, five candidates received their classification from the *Pf* CRT mutant panel alone because their *Pf* DHFR wild-type scores in the separate mutation panel fell below the binding threshold. PP-11 is the clearest case: it is Class A* on the corrected *Pf* CRT channel at a mean RRS of 100.0 % while remaining a *Pf* DHFR non-binder in the mutation panel, which shows that breadth and resilience are separable properties and that pooling them would conflate binding and non-binding states. Second, several mutant ratios exceed 100 %, most prominently PP-15 on *Pf* DHFR (112.7 to 131.8% across four mutants). Such values mean that the mutant docking estimate is more favourable than the wild-type estimate on the same receptor; they are scoring-function ratios and carry no implication of biological gain of function.

The corrected *Pf* CRT channel requires its own caution. Under the 3D7-like LYS-76 receptor and the cavity-anchored grid, the K76T and K76A substitutions produced no detectable change in Vina scores for any candidate (RRS 99.1 to 100.6%), and a pipeline-null control in which the wild-type receptor was processed through the mutation pipeline without a mutation gave the same distribution (99.4 to 100.5%). The 80 % and 70 % class boundaries are therefore finer than the null uncertainty on this channel, and *Pf* CRT RRS values near 100 % are neutral within the null rather than evidence of active resistance circumvention.

3.2.6 Joint prioritization and the information content of network descriptors

Reading breadth and resilience together produces a smaller and more informative shortlist than either metric alone. Counting a target as favourable when its Vina estimate falls at or below that target's median across the cohort (per-target medians: *Pf* DHFR −5.92, *Pf* CRT −8.06, *Pf* ClpP −6.02, *Pf* ATP4 −6.38 kcal mol^{-1}), six candidates were favourable on all four targets, and five of those (PP-15, PP-05, PP-06, PP-11 and PP-13) additionally carried A* classifications. PP-15 combines the highest mean RRS in the cohort (115.6 %) with 4/4 target favourability and resilience distributed across both mutation panels. PP-06 combines 4/4 target favourability with

Target	Receptor	Site status	Grid (Å)	Score range (kcal mol⁻¹)	Validation evidence
Pf DHFR	7F3Y	Crystallographic site (grid NADPH-adjacent)	25	−6.35 to −4.86	Redocking 1 of 2 (MTT ≈ 30 Å); MM/GBSA 0.92; DEKOIS 0.45
Pf CRT (corrected)	6UKJ, K76 restored	Proxy cavity, P2Rank-audited	25	−12.08 to −6.37	Redocking 1 of 1; MM/GBSA 0.97 (scaffold labels)
Pf ClpP	2F6I	Catalytic site (Ser252/His223/Asp219)	20	−7.03 to −5.05	Redocking RMSD 1.65 (peptide fragment)
Pf ATP4	9N10	Mechanism-motivated region (D451/DPPR hinge)	24	−7.57 to −4.63	Redocking RMSD 3.23 (AD); MM/GBSA 1.00 (scaffold labels)

Table 3.2.3: *Per-target summary of the target-anchored campaign: receptor, site status, grid size, within-cohort score range and the validation evidence available for each target. Score ranges are within-target quantities and are not comparable across rows; two of the four sites are proxy or mechanism-motivated cavities, so the panel is a gradient of structural confidence.*

Candidate	Class	$\mathrm{RRS}_{Pf\mathrm{DHFR}}$ (%)				$\mathrm{RRS}_{Pf\mathrm{CRT}}$ (%)		$\mathrm{RRS}_{\mathrm{mean}}$ (%)
		N51I	C59R	S108N	I164L	K76T	K76A	
PP-01	A*	86.4	85.5	87.6	83.9	99.8	99.9	90.5
PP-02	A*	—	—	—	—	100.2	100.0	100.1
PP-03	C	68.1	68.3	65.8	69.0	100.0	100.0	78.5
PP-04	C	68.8	66.9	68.8	73.5	99.9	100.4	79.7
PP-05	A*	—	—	—	—	100.0	100.1	100.1
PP-06	A*	—	—	—	—	100.0	100.0	100.0
PP-07	B	71.8	71.8	72.4	72.2	99.7	99.9	81.3
PP-08	C	62.1	62.1	67.2	66.3	99.6	100.1	76.2
PP-09	B	70.6	73.0	75.8	76.5	99.8	100.0	82.6
PP-10	B	76.0	76.4	75.7	75.8	100.1	100.2	84.0
PP-11	A*	—	—	—	—	100.0	100.0	100.0
PP-12	C	75.2	66.1	62.9	72.5	100.0	100.0	79.5
PP-13	A*	—	—	—	—	99.1	99.4	99.3
PP-14	C	62.2	65.5	68.0	64.9	99.9	100.0	76.8
PP-15	A*	123.2	112.7	125.9	131.8	100.1	100.1	115.6
PP-16	B	72.9	74.7	73.7	75.9	100.6	99.6	82.9
PP-17	C	59.3	61.2	59.5	58.9	100.1	100.1	73.2

Table 3.2.4: *Per-target resistance-resilience scores for the 17-member Set-C cohort, expressed as percentages of the corresponding wild-type Vina score magnitude. Em dashes mark wild-type non-binders ($|S_{\mathrm{Vina},i,\mathrm{WT},t}| <$ 5 kcal mol^{-1}), excluded from the denominator. $\mathrm{RRS}_{\mathrm{mean}}$ is the candidate mean over all eligible target–mutation ratios. Classes are operational categories (A*: wild-type anchor $\geq$ 7 kcal mol^{-1} and all available RRS $\geq 80\,\%$; B: all $\geq 70\,\%$ but not all $\geq 80\,\%$; C: at least one $\geq 80\,\%$ without meeting A/A* or B; D: none $\geq 80\,\%$), not biological resistance phenotypes.*

the most favourable *Pf* ClpP cavity score in the cohort ($-7.03\,\text{kcal mol}^{-1}$). These are computational hypotheses for assay follow-up, not hits.

The cross-metric analysis is important for the thesis because it establishes what chemical-space and network descriptors can and cannot stand in for. On the 17-member cohort, the polypharmacology network similarity correlated negatively with mean RRS ($\rho = -0.714$, $p = 0.0013$), RRS correlated positively with mean wild-type score magnitude ($\rho = +0.691$, $p = 0.0021$), and the number of favourable targets correlated positively with mean RRS ($\rho = +0.714$, $p = 0.0013$); all three survive a Bonferroni threshold of $\alpha = 0.017$ for three tests. The African-chemotype structural index showed no such association ($\rho = -0.190$, $p = 0.465$), so natural-product character does not predict mutation resilience; the index itself places the cohort at a mean of 0.543, with only 2 of 17 candidates above the 0.70 reference level, so the cohort is natural-product-inspired in its sources without being uniformly extreme on the index. Figure 3.2.8 displays these relationships. The power boundary must be reported with them: at $n = 17$ a two-sided Spearman test reaches 80 % power only for $|\rho| \geq 0.62$, and at the Bonferroni-corrected threshold the power to detect $\rho = 0.5$ is approximately 0.37. The surviving associations are consistent with moderate effects; the absence of an ACSI–RRS association is not evidence of equivalence.

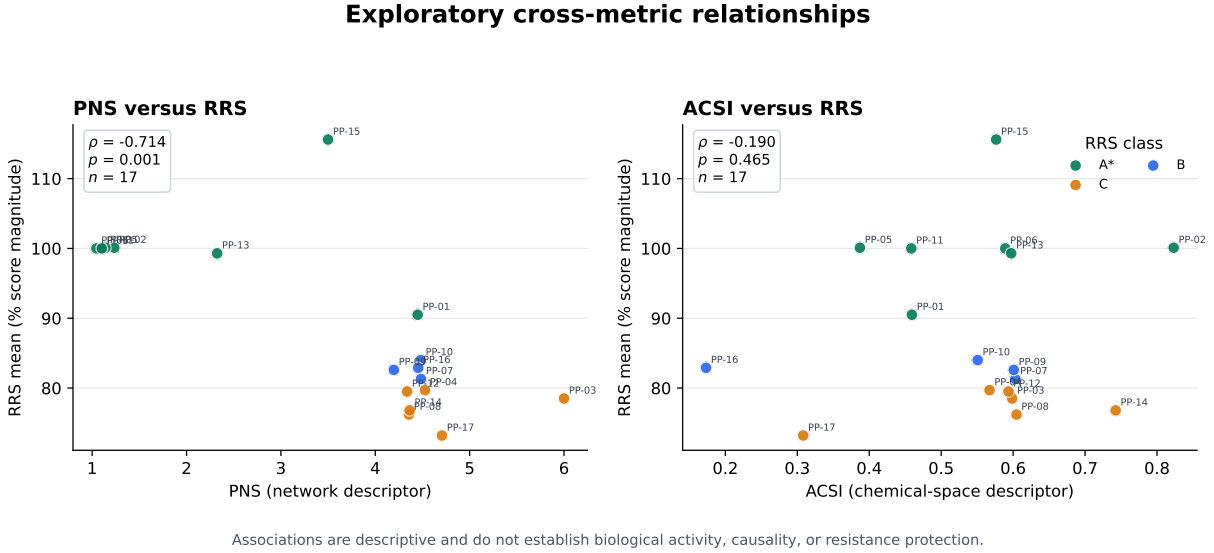


Figure 3.2.8: *Exploratory relationships between resistance resilience and the secondary ranking descriptors for the 17-member cohort, coloured by operational RRS class. The polypharmacology network similarity and the mean wild-type score magnitude both track mean RRS after Bonferroni correction, whereas the African-chemotype structural index does not. These are descriptive source-panel associations and do not establish causality, biological activity, or resistance protection.*

3.2.7 Discussion: what the chemical-space results predict for the studies that follow

The asymmetry that defines the library — 92.6 % whole-molecule novelty on 69.3 % scaffold retention — is not a coverage statistic to be reported and set aside; it is a prediction about every model built on this chemical space. Each representation examined in this thesis describes molecules through local atom environments, predefined dictionaries, bounded topological distances, or message passing over the same connectivity graphs. If the novelty of a chemical space lies in peripheral substitution on conserved cores, then those encodings encounter familiar features in almost every molecule and represent the novel part of it weakly. The scaffold paradox is therefore the reason the representation benchmark of Section 3.4 and the learned-representation benchmark of Section 3.5 enforce scaffold-separated partitions: the partitions are the evaluation-side counterpart of the library's

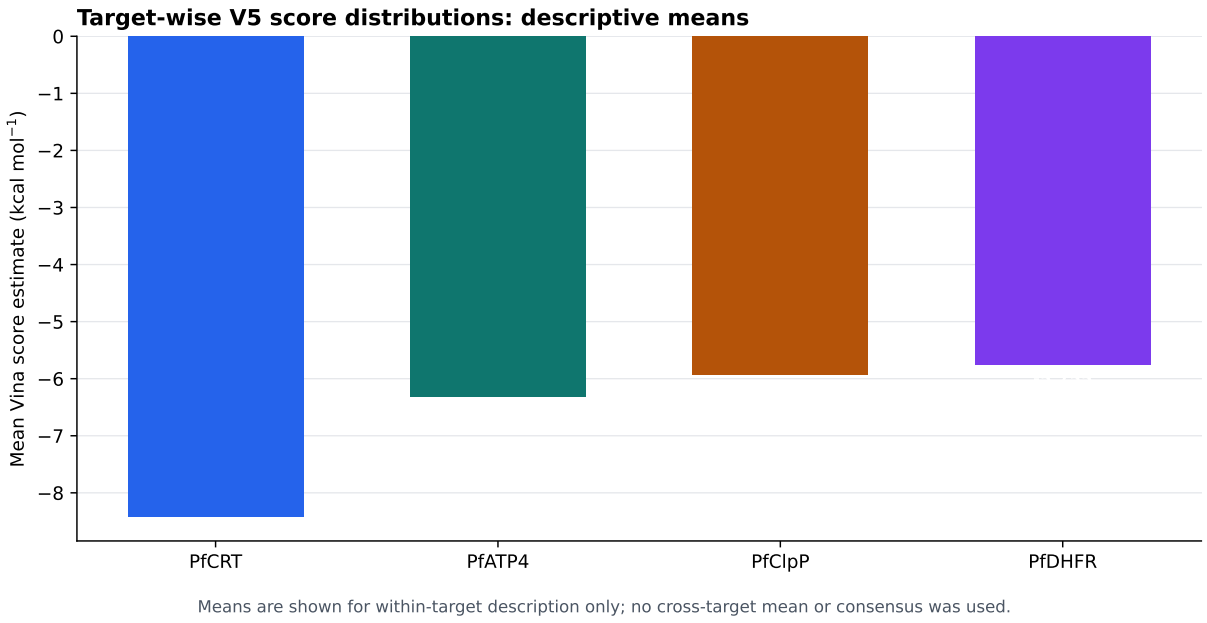


Figure 3.2.9: *Target-wise docking profile of the 17-member Set-C cohort. Because AutoDock Vina scores are not calibrated across dissimilar pockets, breadth is summarized as the count of targets at or below each target's within-cohort median rather than as a cross-target average. The within-target threshold is a prioritization device, not an affinity cutoff.*

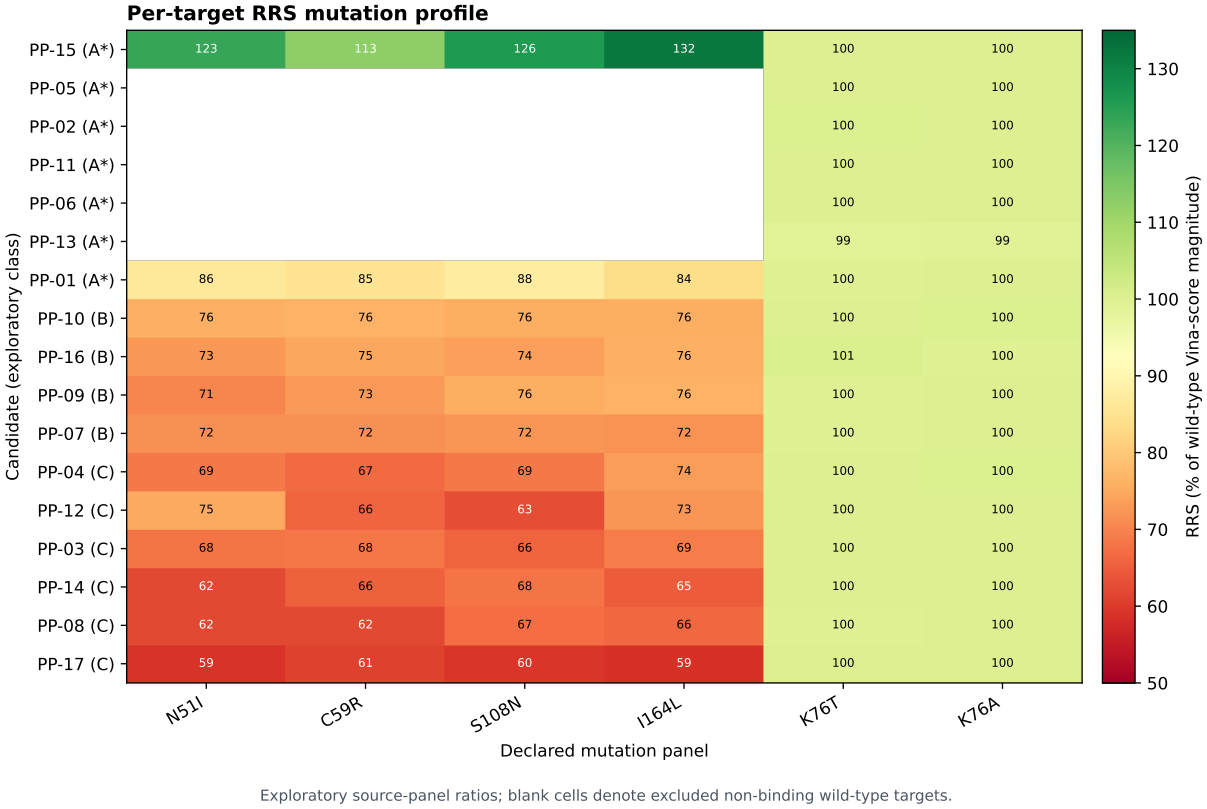


Figure 3.2.10: *Per-target mutation-resilience profiles for the Set-C cohort. Values are ratios of Vina-score magnitudes relative to the declared wild-type denominator; blank cells mark wild-type non-binders excluded from the denominator. The classes are computational operational categories, not biological resistance phenotypes.*

structure, and the chemical-space result is what makes them the relevant test rather than a generic robustness convention.

The two expansion routes each supply what the other lacks — database-validated drug-likeness from the similarity search, scaffold diversity from STONED-SELFIES mutation — and the internal checks reported above, the mixed-evidence latent-space selection and the roughly 2.2-fold separation of intra-cluster from cross-cluster similarity, establish that the sampling units making the campaign affordable are structurally coherent. Two boundaries travel with these results into the rest of the thesis. The clusters separate structurally but not property-wise, so centroid sampling does not stratify the property space and cannot be read as a coverage guarantee in physicochemical terms; and the 99.3 % reduction in docking calculations is a compute statement whose retention of active compounds has not been measured and is not claimed.

The joint prioritization also produces the shortlist that the resistance-scoring study later stress-tests. Table 3.2.5 summarizes its leading entries with the evidence base that set each class.

Candidate	Class	$\mathrm{RRS}_{\mathrm{mean}}$ (%)	Evidence base	Distinctive property
PP-15	A*	115.6	*Pf*DHFR mutants (112.7 to 131.8%) + *Pf*CRT	Highest mean RRS in the cohort; 4/4 favourable targets; resilience on both mutation panels
PP-01	A*	90.5	*Pf*DHFR mutants (83.9 to 87.6%) + *Pf*CRT	Class-A* lead carried into the 16-system molecular-dynamics pilot
PP-06	A*	100.0	*Pf*CRT mutant panel	4/4 favourable targets; most favourable *Pf*ClpP cavity estimate ($-7.03\,\mathrm{kcal\,mol^{-1}}$)
PP-13	A*	99.3	*Pf*CRT mutant panel	Most favourable *Pf*ATP4 estimate ($-7.57\,\mathrm{kcal\,mol^{-1}}$)
PP-11	A*	100.0	*Pf*CRT mutant panel	*Pf*DHFR non-binder in the mutation panel: breadth and resilience separable
PP-05	A*	100.1	*Pf*CRT mutant panel	4/4 favourable targets

Table 3.2.5: *Leading candidates of the joint prioritization, with the evidence base that set each class. Breadth entries count targets at or below their within-cohort median Vina estimate. Classes are operational categories computed from score ratios, not biological resistance phenotypes; every entry is a computational hypothesis for assay follow-up.*

Every entry is a computational hypothesis for assay follow-up. Two structural features of the shortlist matter beyond the candidates themselves: breadth and resilience appear as separable properties, because several candidates carry their classification from a single target panel while remaining non-binders elsewhere; and the strongest mean retention values come from a channel whose own pipeline-null control spans 99.4 to 100.5 %, which is why the class boundaries are read as operational categories anchored to that null rather than as measurements of resistance circumvention.

3.3 Resistance-aware scoring and the static-to-dynamic estimand

3.3.1 Framework scope and descriptor characterization

The resistance-scoring study works on the same upstream library and hardens it into a resistance-aware workflow whose layers are deliberately kept separate. A docking campaign of 136 protein–

ligand systems was constructed from the 17 Set-C candidates docked against five *Pf* DHFR states (wild type plus N51I, C59R, S108N and I164L) and three *Pf* CRT states (wild type plus K76T and K76A). No mutant crystal structure was available, so homology models were generated with SWISS-MODEL from the wild-type templates 7F3Y and 6UKJ and retained only when QMEAN exceeded −4.0, GMQE exceeded 0.7, Ramachandran-favoured residues exceeded 95 % and backbone RMSD was below 1.5 Å; all six models met these thresholds. Figure 3.3.1 shows the resulting two-estimand structure of the study.

The candidate cohort sits in a compact, moderately polar region of chemical space: mean molecular weight 271.9(693) g mol^{-1}, mean LogP 2.84(82), mean QED 0.70(6), mean fraction of sp^3-hybridized carbons 0.22(15) and 0.6(9) stereocentres per scaffold, with a mean MPO score of 0.728(21) and SYBA synthetic accessibility of 36.6(234). These values describe the sources from which the panel was drawn, including prenylated chromones, flavonoid derivatives and indole alkaloids isolated from African medicinal flora such as *Cryptolepis sanguinolenta*, *Enantia chlorantha* and *Artemisia* species; they are not ethnobotanical validation of any candidate.

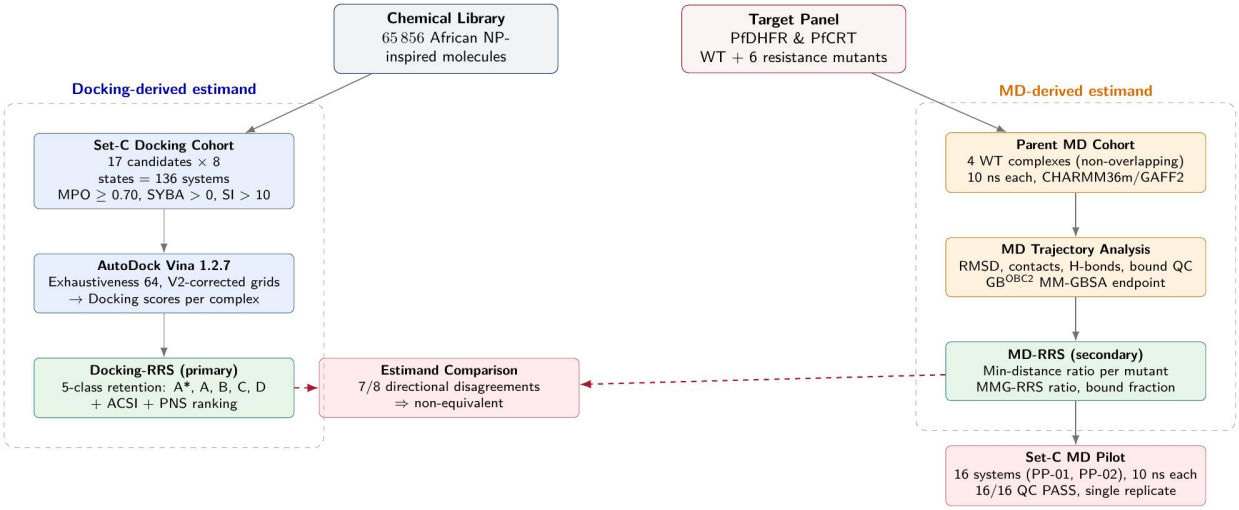


Figure 3.3.1: *Integrated resistance-aware polypharmacology and molecular-dynamics triage framework of the resistance-scoring study. Static multi-target docking feeds the resistance-resilience score, the African-chemotype structural index and the polypharmacology network similarity; 10 ns explicit-solvent molecular dynamics then runs as a secondary structural check. Docking-derived and trajectory-derived quantities are distinct estimands and are never combined into a single score.*

3.3.2 Resistance classes depend on the cohort definition, and the protocol is reproducible

Restricting the primary analysis to the 12 candidates with eligible *Pf* DHFR and *Pf* CRT wild-type scores yields class counts of A*:1, A:1, B:4, C:5 and D:1, with mean target-specific RRS of 75.1 for *Pf* DHFR (12 candidates) and 86.2 for *Pf* CRT (17 candidates). Classifying all 17 candidates on their eligible targets raises the counts to A*:5, A:1, B:5, C:5 and D:1, but the five *Pf* CRT-only compounds are not comparable with the two-target set. Reporting both is the honest option; collapsing them into a single headline count would hide the coverage difference. The class assignment is robust to the declared thresholds but not to stricter ones: moving the wild-type eligibility threshold from 4.0 kcal mol^{-1} to 5.0 kcal mol^{-1} leaves the counts unchanged, whereas raising the retention thresholds from 70 %/80 % to 80 %/90 % increases Class D from 1 to 9.

Protocol reproducibility was quantified rather than asserted. Multi-seed redocking dispersion was ±0.05 kcal mol^{-1} for PP-01 on *Pf* DHFR, ±0.04 kcal mol^{-1} for PP-01 on *Pf* CRT and ±0.03 kcal mol^{-1} for PP-15, and 77.9 % of class assignments remained unchanged under a bounded ±1 kcal mol^{-1} score perturbation. An independent 39-ligand sensitivity panel produced 312 finite docking scores

with a bootstrap mean RRS of 100.45 (95% CI 99.59 to 101.32) and a Class-A fraction of 0.974 (CI 0.923 to 1.000); GNINA convolutional-neural-network cross-scoring agreed with Vina on 100 % of class assignments, with no per-mutant rank transfer claimed (per-mutant rank correlations are near zero for C59R and I164L). Because almost the whole sensitivity panel falls in Class A (bootstrap Class-A fraction 0.974), this agreement is close to degenerate: it constrains gross class assignment only, not ranking quality. Together these checks establish protocol-level consistency, not pose accuracy or binding fidelity.

3.3.3 Static docking and short molecular dynamics disagree in a consistent direction

The central P2 result is that two quantities one might treat as interchangeable are not. Static grid docking evaluates unrelaxed steric complementarity, while a 10 ns explicit-solvent trajectory samples local side-chain reorganization, hydrogen-bond rearrangement and solvent shielding. Trajectory-based retention is summarised by a distance ratio, $\mathrm{MD\text{-}RRS}_d = 100 \times \langle d_{\min} \rangle_{\mathrm{mut}} / \langle d_{\min} \rangle_{\mathrm{WT}}$, computed on the mean minimum heavy-atom protein–ligand distance; values at or below 100 % indicate pose retention at least as tight as the wild-type endpoint, not increased affinity. Across the 16-system Set-C pilot (PP-01 and PP-02 across the *Pf*DHFR and *Pf*CRT wild-type and mutant states), all 16 trajectories passed the geometric contact-stability threshold of a minimum protein–ligand distance below 3.5 Å. Among the eight mutant states with an eligible wild-type docking baseline, static docking predicted reduced score retention (RRS below 100 %) while explicit-solvent relaxation retained poses in 87.5 % of matched comparisons (7 of 8; 95% Wilson CI 52.9 to 97.8%; Figure 3.3.2). At single-replicate scope this is a protocol-local calibration signal, not a demonstrated mechanism, and it is reported as such.

The Class-A* lead PP-01 illustrates the pattern across its eight-system panel (Table 3.3.1). Static docking penalised *Pf*DHFR C59R (docking RRS 85.5 %) and *Pf*CRT K76T (docking RRS 92.5 %), yet both trajectories retained their poses with bound fraction 1.000, mean minimum heavy-atom distances of 2.91 Å ($\mathrm{MD\text{-}RRS}_d = 98.9\,\%$) and 3.02 Å ($\mathrm{MD\text{-}RRS}_d = 102.8\,\%$), and MM-GBSA endpoints of −26.56(191) kcal mol^{-1} and −29.51(265) kcal mol^{-1} respectively. This is geometric pose retention, not thermodynamic proof: the K76A endpoint carries a 7.75 kcal mol^{-1} inter-replicate sensitivity, so no MM-GBSA-derived resilience claim is made for K76A.

A single-ligand feasibility probe outside the Set-C estimand, PP-15 on wild-type *Pf*DHFR and *Pf*CRT, completed both 10 ns trajectories with 100 % quality-control pass, bound fraction 1.000, mean minimum heavy-atom distances of 2.77 Å and 3.20 Å, and MM-GBSA endpoints of −29.52(33) kcal mol^{-1} and −27.79(49) kcal mol^{-1}. No mutant or RRS estimate is derived from this probe and no dual-target resilience claim is made from it.

3.3.4 Trajectory-derived endpoints are informative only where the complex stays bound

Endpoint free-energy estimation depends on the ligand remaining in the pocket, which is not guaranteed. Of the four non-overlapping parent-study wild-type complexes simulated for 10 ns, two dissociated and two remained bound (Table 3.3.2). Only the *Pf*CRT–214 complex produced an interpretable MM-GBSA endpoint (−18.25(40) kcal mol^{-1} over 101 stored snapshots, minimum distance 3.19 Å, 78 persistent contacts). The *Pf*ClpR-labelled 164 system and the *Pf*DHFR 201 system separated beyond 60 Å with zero contact persistence and were excluded, and the *Pf*ATP4 438 system was excluded because multi-chain membrane-transporter parameterization introduced force-field conversion artifacts. Reporting the two dissociations is essential to interpreting the estimand-divergence result: a prediction of retained binding can only be tested where a bound state exists to test.

Candidate	Target	State	ΔG_{bind} (kcal mol^{-1})	$\mathrm{SD}_{\mathrm{prop}}$	MMG-RRS (%)	Docking RRS (%)
PP-01	*Pf*CRT	K76A	−35.29	1.17	115.3	90.86
PP-01	*Pf*CRT	K76T	−29.51	2.65	96.4	92.47
PP-01	*Pf*CRT	WT	−30.61	1.65	100.0	–
PP-01	*Pf*DHFR	C59R	−26.56	1.91	105.0	85.47
PP-01	*Pf*DHFR	I164L	−26.13	2.34	103.3	83.87
PP-01	*Pf*DHFR	N51I	−29.83	1.50	118.0	86.40
PP-01	*Pf*DHFR	S108N	−30.21	3.62	119.5	87.60
PP-01	*Pf*DHFR	WT	−25.29	2.94	100.0	–
PP-02	*Pf*CRT	K76A	−24.53	1.27	97.8	94.62
PP-02	*Pf*CRT	K76T	−30.45	1.38	121.4	87.91
PP-02	*Pf*CRT	WT	−25.08	2.56	100.0	–
PP-02	*Pf*DHFR	C59R	−28.85	1.88	95.7	n/a
PP-02	*Pf*DHFR	I164L	−31.06	1.52	103.0	n/a
PP-02	*Pf*DHFR	N51I	−34.02	2.68	112.8	n/a
PP-02	*Pf*DHFR	S108N	−27.09	1.55	89.8	n/a
PP-02	*Pf*DHFR	WT	−30.16	2.06	100.0	–

Table 3.3.1: *Set-C pilot endpoint summaries for the 16 matched systems of the resistance-scoring study (exploratory; not used to assign RRS classes). ΔG_{bind} is the mean over 100 snapshots $\pm$ the propagated within-trajectory standard deviation, which is not a replicate-level uncertainty estimate. MMG-RRS is the mutant-over-wild-type endpoint ratio ($\times$100; wild type = 100) and docking RRS is shown for comparison (n/a: no eligible wild-type docking reference for PP-02 on* Pf*DHFR). The PP-01* Pf*CRT K76A endpoint (−35.29 kcal mol^{-1}) carries a 7.75 kcal mol^{-1} inter-replicate sensitivity (second trajectory −27.54 kcal mol^{-1}) and supports no resilience claim. Single-replicate summaries are within-protocol contrasts only.*

System	Atoms	BB RMSD (Å)	Lig. RMSD (Å)	Min. dist. (Å)	Contacts	Interpretation
*Pf*ClpR-labelled 164	43 035	1.31	1.11	67.42	0.0	Unbound
*Pf*DHFR 201	134 153	3.05	1.70	78.21	0.0	Unbound
*Pf*CRT 214	76 920	4.47	2.45	3.19	78.1	Bound; $\Delta G_{\mathrm{bind}} = -18.25(40)$ kcal mol^{-1}
*Pf*ATP4 438	420 801	12.27	2.13	2.25	178.4	Bound; endpoint excluded (conversion artifact)

Table 3.3.2: *Production molecular-dynamics stability metrics for the four non-overlapping parent-study wild-type complexes (10 ns, PBC-unwrapped trajectories). These compounds are not among the 17 Set-C candidates and do not validate the Set-C ranking. MM-GBSA endpoints are reported only for complexes that remained bound; the* Pf*ATP4 system was excluded because force-field conversion for the multi-chain transporter introduced artifacts.*

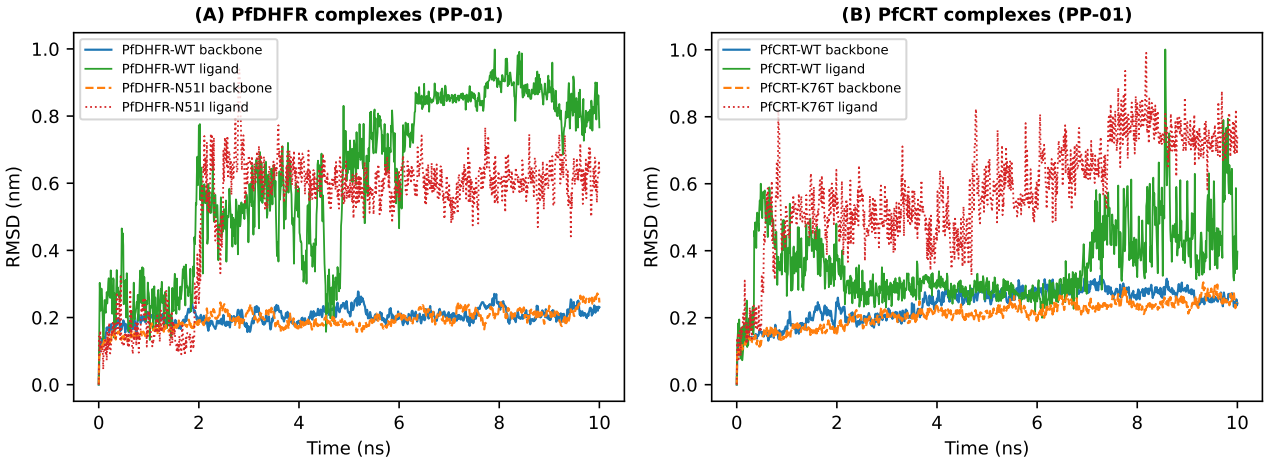


Figure 3.3.2: *Pose retention and thermal equilibration for the PP-01 systems of the resistance-scoring study over 10 ns (canonical R1 trajectories). Explicit-solvent backbone (solid) and ligand-after-backbone-fit (dashed/dotted) RMSD time-series computed with `gmx rms` after PBC-whole reconstruction (fit group: backbone); axes in nm. Panel (A), Pf DHFR: backbone RMSD 0.203(24) nm (WT) and 0.193(29) nm (N51I); ligand 0.613(240) nm and 0.519(190) nm. Panel (B), Pf CRT: backbone 0.241(48) nm (WT) and 0.211(39) nm (K76T); ligand 0.357(113) nm and 0.581(155) nm. All four trajectories pass the geometric quality-control gate (mean minimum heavy-atom distance 2.7 to 3.2 Å, bound fraction 1.000); ligand RMSD reflects in-pocket mobility, not dissociation. No ligand dissociation was observed at single-replicate scope.*

The interaction fingerprint of the 16 quality-control-passed trajectories provides a structural counterpart to the geometric result. Across 100 snapshots per system, *Pf* CRT Tyr16 maintained 100 % contact occupancy in five of six *Pf* CRT systems (PP-01 and PP-02 against wild type, K76T and K76A), and the *Pf* DHFR hotspot contacts Asp54, Leu46, Met55 and Ile14 reached full occupancy across the PP-01 and PP-02 states (Figure 3.3.3). A sparse conserved contact network of this kind is compatible with pose retention, but it does not establish a general mechanism, and the inter-replicate sensitivity of the K76A endpoint (7.75 kcal mol^{-1} between two trajectories, against a wild-type thermal noise floor of 2.01 kcal mol^{-1}) is a reminder that endpoint energies at this scope are not resilience measurements.

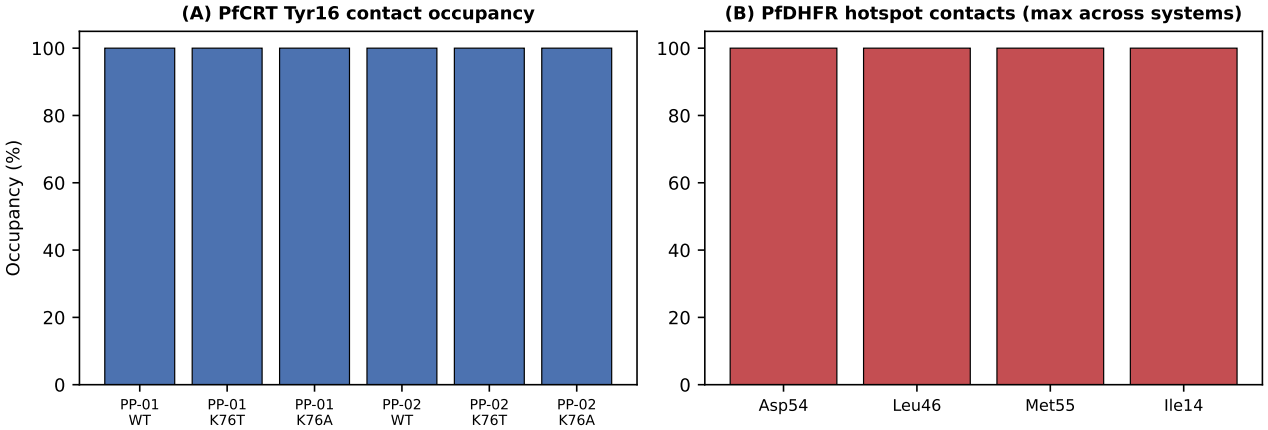


Figure 3.3.3: *Conserved hotspot anchors in the explicit-solvent trajectories (ProLIF interaction-fingerprint occupancy over hydrophobic, van der Waals and hydrogen-bond contact types, 100 snapshots per system). Panel (A): Pf CRT Tyr16 maintains 100 % contact occupancy in five of six Pf CRT systems. Panel (B): Pf DHFR hotspot contacts (Asp54, Leu46, Met55, Ile14) reach maximum occupancy across the PP-01 and PP-02 states. Contact persistence is compatible with pose retention but does not by itself establish a mechanism.*

3.3.5 The three ranking dimensions carry distinct information

Within the two-target set, none of the three pairwise Spearman associations reached significance after Bonferroni correction: polypharmacology network similarity against RRS gave $\rho = -0.2098$ (adjusted $p = 1.0000$), the African-chemotype structural index against RRS gave $\rho = -0.4056$ (adjusted $p = 0.5766$), and RRS against the weakest wild-type score gave $\rho = -0.1661$ (adjusted $p = 1.0000$; Figure 3.3.4). On all 17 candidates the polypharmacology association strengthened to $\rho = -0.5588$ (adjusted $p = 0.0667$), though unequal target coverage limits interpretation; controlling for molecular weight and scaffold frequency gives a moderate partial association of $\rho_{\text{partial}} = -0.6154$. The descriptor constructions are robust to their declared arbitrary choices: recomputing the network ranking at STRING confidence cutoffs of 400, 700 and 900 leaves the polypharmacology ranks nearly unchanged (rank correlations 0.9632 to 0.9975 across cutoff pairs), and varying any African-chemotype weight by $\pm 20\,\%$ preserves the ranking with Spearman correlations of 0.9167 to 0.9804 and top-five Jaccard overlaps of 0.6667 to 1.0000. These checks address ranking stability against a network-data or weighting choice, not biological target importance. The important negative result is that RRS, the African-chemotype structural index and the polypharmacology network similarity are not redundant with one another at this cohort size, so resistance-aware ranking cannot be replaced by a chemical-space descriptor. The failure of that claim to reach significance, however, must not be read as equivalence.

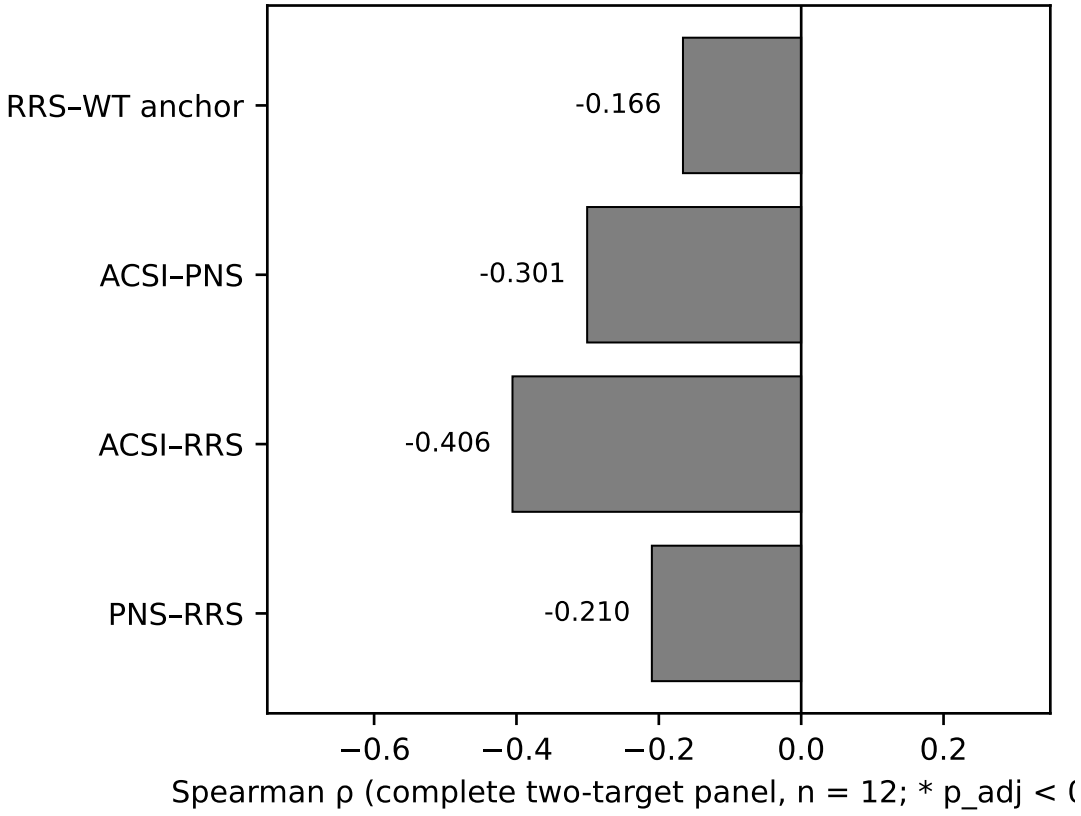


Figure 3.3.4: *Cross-metric Spearman correlations on the complete two-target set ($n = 12$): PNS–RRS $\rho = -0.2098$, ACSI–RRS $\rho = -0.4056$, ACSI–PNS $\rho = -0.3007$ and RRS–weakest-WT-anchor $\rho = -0.1661$; none is significant after Bonferroni correction (adjusted $p \geq 0.51$; 100 000 permutations, seed 42). The absence of significant inter-metric association is consistent with distinct ranking dimensions rather than redundant signals. With 12 observations these intervals are wide and the tests are descriptive.*

3.3.6 Static score differences fail a retrospective phenotype check

As an external, purely retrospective test of the static docking estimand, the resistance-scoring study confronted rigid-receptor score differences with published genotype-resolved inhibition data.

A curated ChEMBL panel for *Pf*DHFR (target CHEMBL1939) contains 28 compounds with paired wild-type and quadruple-mutant (N51I/C59R/S108N/I164L) IC_{50} measurements, 24 of which also carry paired double-mutant (C59R/S108N) values, with triple-mutant (C59R/S108N/I164L) pairs complete for all 28. The compounds were docked against wild-type, double, triple and quadruple *Pf*DHFR receptors under the declared Vina protocol, with mutant receptors generated by in-silico side-chain replacement on unchanged backbone coordinates (112 of 112 docking jobs completed), and the score difference $\Delta\Delta G = s_{\mathrm{mut}} - s_{\mathrm{WT}}$ was compared with the published fold-shift $\log_2(\mathrm{IC}_{50,\mathrm{mut}}/\mathrm{IC}_{50,\mathrm{WT}})$ by Spearman rank correlation, direction agreement and an exact binomial test against chance agreement.

The confrontation is negative at every genotype. Spearman correlation between $\Delta\Delta G$ and the $\log_2$ fold-shift reached 0.19 ($p = 0.375$, $n = 24$) for the double mutant, -0.04 ($p = 0.853$, $n = 28$) for the triple mutant and -0.13 ($p = 0.504$, $n = 28$) for the quadruple mutant, with direction agreement of 10/24 (41.7 %), 9/28 (32.1 %) and 8/28 (28.6 %) respectively. For the triple and quadruple genotypes the agreement rate departs from chance toward systematic disagreement (exact binomial $p = 0.044$ and $p = 0.018$ unadjusted; 0.132 and 0.054 after Bonferroni correction), and the reference antifolates pyrimethamine (919-fold shift) and cycloguanil (145-fold shift) were mispredicted in direction at every genotype. Mutant scores averaged slightly more favourable than wild-type scores (mean $\Delta\Delta G$ -0.19 to -0.12 kcal mol^{-1}, with 68 to 79 % of compounds below zero), a systematic offset smaller than the ± 1 kcal mol^{-1} score sensitivity of the protocol itself.

Two boundaries fix what the result means. The reference measurements are whole-parasite growth IC_{50} values from DHFR-genotyped but non-isogenic strains, so the comparison audits the static $\Delta\Delta G$ estimand against a phenotype-level reference, not enzyme-level binding; and the Set-C candidates are not part of the panel, so the result neither validates nor invalidates their RRS tiers. Its role in this chapter is to bound the docking estimand from experimental data, independently of the chance-level DEKOIS enrichment: a single rigid-grid score difference does not estimate phenotype-level resistance, which is the conclusion the docking–MD divergence reaches from the computational side.

A second retrospective line of evidence uses approved drugs rather than candidate compounds, and it reproduces the same null character on clinically familiar ligands. Five approved antimalarials spanning distinct mechanisms of action (artemisinin, pyrimethamine, chloroquine, lumefantrine and cipargamin) were docked on the same *Pf*DHFR and corrected *Pf*CRT mutation panels and scored under the declared RRS protocol. The control does not recover the clinically documented resistance signatures: pyrimethamine, the antifolate whose clinical fold-shift is the largest in the curated panel above, retains an RRS of approximately 100 % across all four *Pf*DHFR mutants, and chloroquine retains approximately 99 % on *Pf*CRT K76T and K76A, indistinguishable from the pipeline-null spread of the corrected channel (99.4 to 100.5%). Artemisinin and cipargamin show *Pf*DHFR RRS values of approximately 113 to 118%, while chloroquine and lumefantrine are *Pf*DHFR non-binders in this pocket, consistent with their non-antifolate mechanisms. The inability of the protocol to recover known resistance biology is therefore not an artefact of the candidate cohort: it appears on independent, clinically relevant ligands, and it reinforces the standing rule that docking-derived retention values must not be read as resistance-circumvention evidence for any compound.

3.3.7 Operational guidance for resistance-aware triage

The results of the resistance-scoring study condense into four operational rules for any pipeline that ranks compounds against mutation panels, and they are stated here as rules because each one is the direct consequence of a result reported above rather than of a general principle.

The first rule is that rigid-receptor score differences must not be used alone to rank resistance variants. The static $\Delta\Delta G$ estimand failed both external checks reported in this chapter: it recovered known actives at chance level on the enrichment benchmark, and its agreement with published

genotype-resolved fold-shifts ranged from below chance to indistinguishable from chance at every genotype. A score with these two boundaries may still be a useful triage input, but a ranking built solely on it inherits both failures at once.

The second rule is that disagreement between static docking and short explicit-solvent dynamics should be treated as a flag for experimental follow-up, not as evidence against either method. The disagreement observed here was consistent in direction across matched mutant states, yet it arises from non-equivalent estimands: the grid score evaluates an unrelaxed receptor, the trajectory samples local relaxation. Neither is wrong, and the states where they disagree are precisely the states where an experiment is most informative.

The third rule is that the three ranking dimensions of the workflow should be reported as separate descriptive columns rather than combined into a single index. None of the pairwise associations reached significance at the available cohort size, so a composite index would impose weighting assumptions that the data cannot support; the weights would be arbitrary, and the composite would inherit the interpretive limits of its weakest component while hiding them from the user.

The fourth rule is that the wild-type eligibility gate must be enforced before any ratio-based score is computed. Excluding weak wild-type baselines prevents the artificial score inflation analysed in the methods, and the gate's declared threshold should be reported with any score that depends on it, because the coverage of the analysis, which candidates are scoreable at all, is a consequence of the gate rather than of the data.

3.3.8 What the divergence implies about the role of dynamics in a prioritization pipeline

The direction of the 87.5 % divergence is informative about how the two tools should be arranged rather than about which one is correct. A rigid-grid score evaluates unrelaxed steric complementarity, so it charges at full magnitude the strains that explicit-solvent relaxation removes; across the matched mutant states examined here, the static penalty is therefore the conservative bound on retention, and relaxation recovers the pose in most comparisons. Dynamics enters the workflow on this reading not as a better scorer but as a specificity filter on the static ranking — which is why the molecular-dynamics cohort is a 16-system pilot on the leading candidates rather than the full 136-system campaign: the trajectories are deployed where the static answer would change a decision, not as a second global ranking.

The cost asymmetry that motivates that arrangement also bounds what the trajectories can be asked. Endpoint free-energy estimation is interpretable only where the complex remains bound, which held for two of the four non-overlapping parent-study wild-type complexes and failed for the two that separated beyond 60 Å; it is further bounded by replicate sensitivity, exemplified by the *Pf* CRT K76A endpoint whose inter-replicate spread of 7.75 kcal mol^{-1} exceeds the wild-type thermal noise floor of 2.01 kcal mol^{-1}. The interaction fingerprints supply structural corroboration of retention — the conserved *Pf* CRT Tyr16 anchor and the *Pf* DHFR hotspot contacts — but a sparse conserved contact network is evidence of a retained pose, not a mechanism of resistance circumvention, and no mechanism is claimed from it.

There is also a forward connection. Both estimands are geometric evaluations of one structure — one unrelaxed, one averaged over a nanosecond-scale ensemble — and neither converts three-dimensional information into a form that can be learned across molecules. The estimand lesson of this section, to state what is measured at the point of the result, and the representation lessons of Section 3.4 and Section 3.5, to state what is encoded, are the two faces of one boundary, and the experimental side of that boundary is fixed by the two retrospective checks above: on the data available here, neither geometric estimand tracks phenotype-level resistance.

3.4 Quantum-inspired and topological representations

3.4.1 The scaffold paradox has a topological decomposition

The representation study evaluates three descriptors that are proposed to capture features classical fingerprints miss: a persistent-homology topological fingerprint (TFP), a Tucker-compressed tensor network embedding (TNE) and a simulated quantum kernel score (QKS). The representation analyses use a primary panel of 19 849 molecules, of which 19 836 have complete TNE embeddings; binary activity labels come from the Ersilia eos80ch model at a threshold of 0.5, giving 74.2 % active and 25.8 % inactive.

The library statistics of the chemical-space study reappear here as a topological question. Persistent homology resolves the apparent conflict between 92.6 % whole-molecule Tanimoto distinction and 69.3 % scaffold recovery by separating the two homology dimensions: H_1 features, which encode ring topology, are conserved between seed and expanded sets (mean H_1 count 3.61), whereas H_0 features, which encode connectivity, diverge. The decomposition is independently corroborated by scaffold-only Tanimoto similarity (0.379) being 1.84 times higher than whole-molecule similarity (0.206). Figure 3.4.1 shows the two persistence distributions. The mechanistic reading is that single-character SELFIES mutations permute side chains and functional groups while preserving cyclic cores; the contribution of the analysis is to measure that property with a topology-aware quantity rather than to discover it.

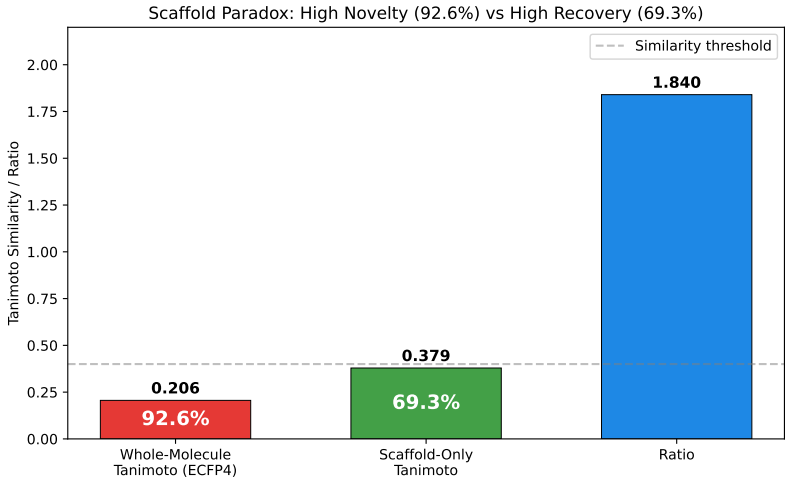


Figure 3.4.1: *Resolution of the scaffold paradox by persistent homology. Panel (A): H_1 persistence distributions for seed natural products and structurally expanded candidates, showing conserved ring topology. Panel (B): H_0 persistence distributions, showing diverging connectivity. Together the two panels account for the 92.6 % whole-molecule Tanimoto distinction alongside 69.3 % scaffold recovery.*

3.4.2 Compression retains docking-relevant information on the derivation panel

Tucker decomposition at bond dimension $d = 8$ compresses each molecular tensor to a 192-element core, which at a mean of 39 atoms corresponds to a 6.1-fold compression of the real-atom representation with a mean relative reconstruction error of 0.113. Compression is not uniform across chemical space: molecules with common ring systems compress better (mean reconstruction error 0.098) than structurally atypical molecules (0.137), and the atom-coordinate factor matrix correlates with molecular volume ($\rho = 0.71$). In a $k = 484$ k-means analysis, TNE produced a higher Silhouette coefficient (0.35) than ECFP4 (0.18) or the VAE latent representation (0.229). Figure 3.4.3 shows the compression behaviour and its regression against docking scores.

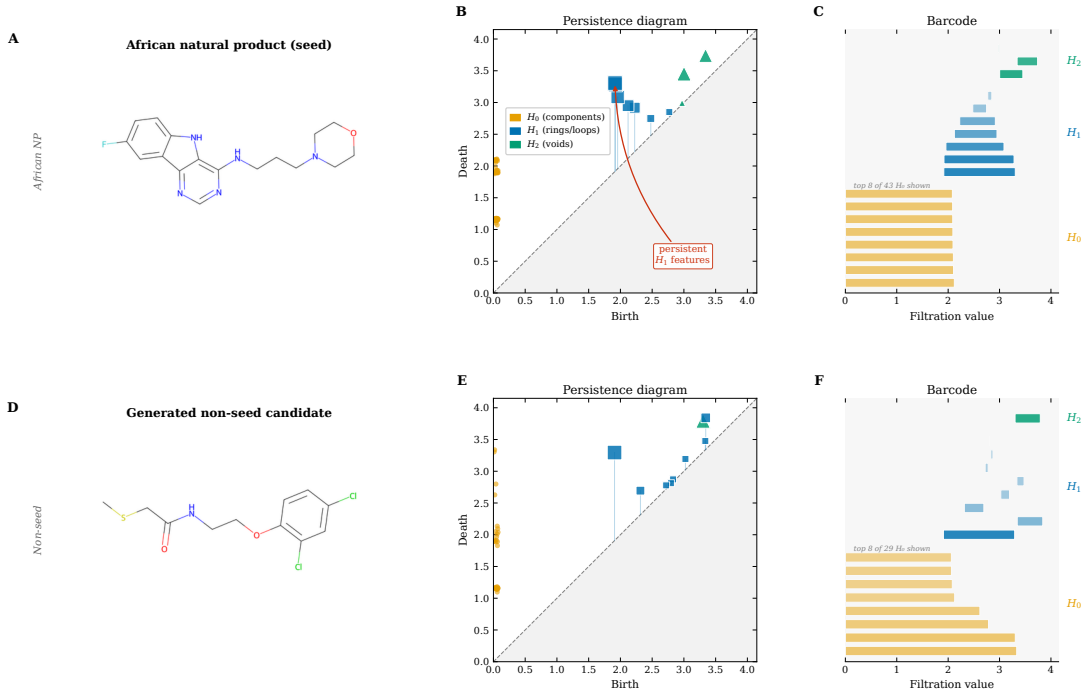


Figure 3.4.2: *Representative persistence diagrams underlying the topological fingerprint. Vietoris–Rips persistence was computed up to homology dimension two on weighted molecular graphs whose edge weights are $w_{ij} = d_{ij}(1 + Z_i Z_j/100)$, so that interactions involving heavier atoms contribute more strongly. The primary activity analysis uses a 78-feature vector combining homology summaries, persistence-image and Betti-curve components.*

Whether that cluster separation translates into predictive information was tested directly. In cross-validated regression against Tartarus docking scores, TNE reached $R^2 = 0.473$ for *Pf*DHFR against 0.461 for ECFP4, while ECFP4 was stronger for *Pf*ATP4 and *Pf*CRT. This is a genuine retention-of-information result, and it is bounded in the same sentence: the regression uses the same molecular set from which the embeddings were derived, so it demonstrates information retention on the derivation panel rather than generalisation to novel compounds.

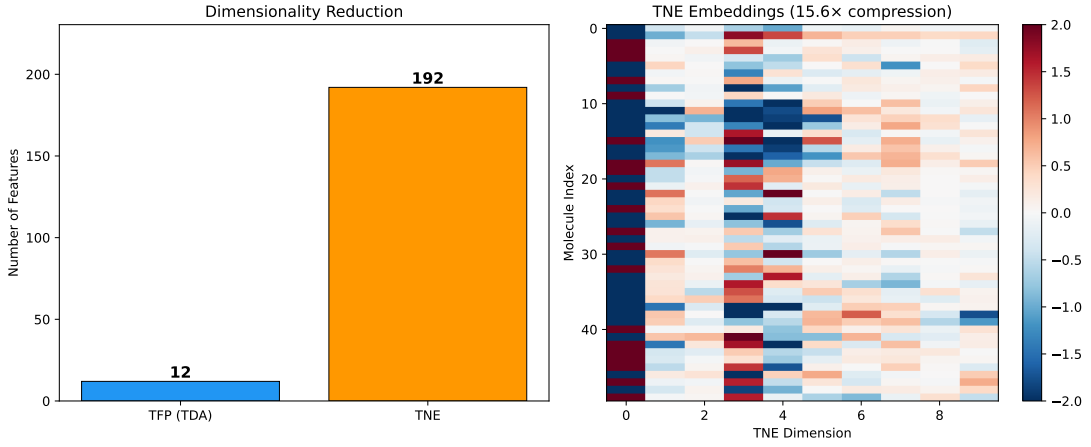


Figure 3.4.3: *Compression behaviour of the Tucker tensor network embedding. The 192-element core retains docking-score-predictive information on the derivation panel, reaching $R^2 = 0.473$ for Pf*DHFR *against 0.461 for ECFP4, while ECFP4 remains stronger for Pf*ATP4 *and Pf*CRT*. Reconstruction error increases for structurally atypical molecules, so compression quality is chemical-space dependent.*

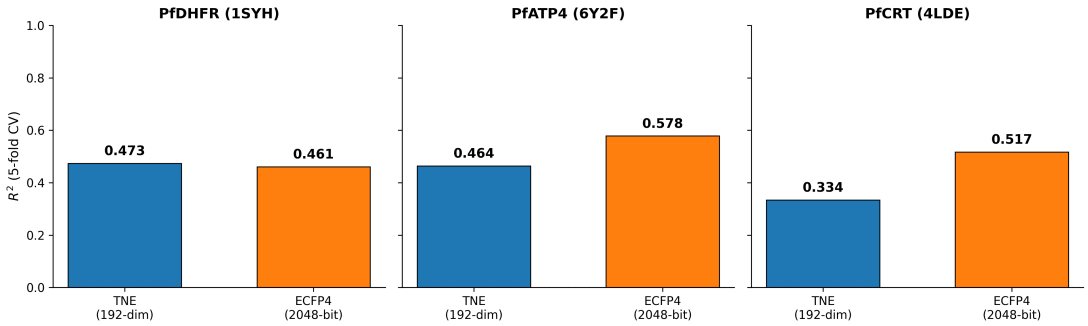


Figure 3.4.4: *Cross-validated regression of docking scores on molecular representations. The tensor network embedding is competitive with ECFP4 on PfDHFR but not on the remaining targets. Because embeddings and regression endpoint come from the same molecular set, this quantifies information retention rather than prospective transfer.*

3.4.3 ECFP4 remains the strongest descriptor and the hybrid does not close the gap

The activity-prediction benchmark is the decisive test, and its result is negative for every quantum-inspired representation (Table 3.4.1). On the 19 836-molecule complete-embedding set under 5-fold stratified cross-validation, ECFP4 reached the highest area under the receiver operating characteristic curve (0.948), followed by the atom-pair (0.940), binding-pattern (0.939), functional-class (0.918), MACCS (0.905) and 2D-pharmacophore (0.896) fingerprints. The 78-feature topological fingerprint reached 0.876 and the tensor network embedding 0.722. The quantum kernel score, evaluated by support-vector machine on the full representation panel ($n = 19\,849$) with a precomputed six-qubit kernel, reached 0.823; because it uses a different panel and classifier it is reported alongside but not as a like-for-like random-forest comparison.

The hybrid descriptor concatenating TFP, TNE and the quantum kernel principal components reached 0.888 ± 0.007, significantly below ECFP4 ($p < 0.0001$) yet above every standalone quantum-inspired descriptor. The ablation is the informative part: removing the quantum kernel components costs 0.040 AUC, removing TFP costs 0.014, and removing TNE slightly improves the hybrid (0.011), consistent with the low standalone TNE score. The hybrid's measurable contribution is therefore concentrated in the quantum kernel components, and the configuration is not recommended over ECFP4 for primary screening on this panel. One further caveat belongs with the table: the hybrid hyperparameters were selected on a 200-molecule development subset that was not held out from the final evaluation, so 0.888 should be read as an upper bound rather than an unbiased estimate.

3.4.4 The quantum kernel is a well-behaved kernel, not a quantum advantage

The most consequential result for this thesis is the kernel comparison. Against a gamma-tuned radial basis function (RBF) baseline, the six-qubit quantum kernel showed no detectable difference internally: at $n = 5000$ the comparison gave $p = 0.419$, and at $n = 19\,849$ it gave $p = 0.060$ (quantum 0.823(8) against RBF 0.829(7), paired $t = -2.59$). The quantum kernel did significantly outperform a linear kernel ($p \leq 0.0006$), so the kernel is not uninformative; it is simply indistinguishable from a well-tuned classical kernel. The geometry of the two kernels differs sharply: kernel–target alignment was 0.622(13) for the quantum kernel against 0.145(6) for RBF, a fourfold geometric advantage that translated into a ΔAUC of -0.006 (Figure 3.4.5). Kernel alignment is therefore not a proxy for classification performance on this task, and a favourable Hilbert-space geometry does not by itself justify a quantum claim.

Method	AUC	Accuracy	F1	ΔAUC vs. ECFP4
ECFP4 (baseline)	0.948	0.896	0.931	—
Atom pair	0.940	0.888	0.927	−0.008
Binding pattern	0.939	0.889	0.928	−0.009
FCFP4	0.918	0.869	0.914	−0.029
MACCS	0.905	0.863	0.909	−0.043
Pharmacophore	0.896	0.854	0.904	−0.052
TFP (persistent homology)	0.876	0.838	0.897	−0.072
TNE ($d = 8$)	0.722	0.756	0.857	−0.226
Quantum kernel (SVM, $n = 19\,849$)	0.823	0.819	0.886	—
Hybrid (TFP + TNE + quantum kernel)	0.888	0.843	0.900	−0.060
Ablation (hybrid minus one component):				
Hybrid − TFP	0.874	0.837	0.895	−0.014
Hybrid − TNE	0.899	0.850	0.903	0.011
Hybrid − quantum kernel	0.847	0.820	0.887	−0.040

Table 3.4.1: *Activity-prediction performance and hybrid ablation in the descriptor benchmark. Fingerprint, TFP, TNE and hybrid rows use the complete-embedding intersection ($n = 19\,836$) under 5-fold stratified random-forest cross-validation (200 trees). The quantum kernel row comes from the full representation panel ($n = 19\,849$) under 5-fold support-vector-machine cross-validation with a precomputed six-qubit kernel and is not a like-for-like comparison, so no ΔAUC against ECFP4 is assigned to it. All activity labels are model predictions, not experimental measurements.*

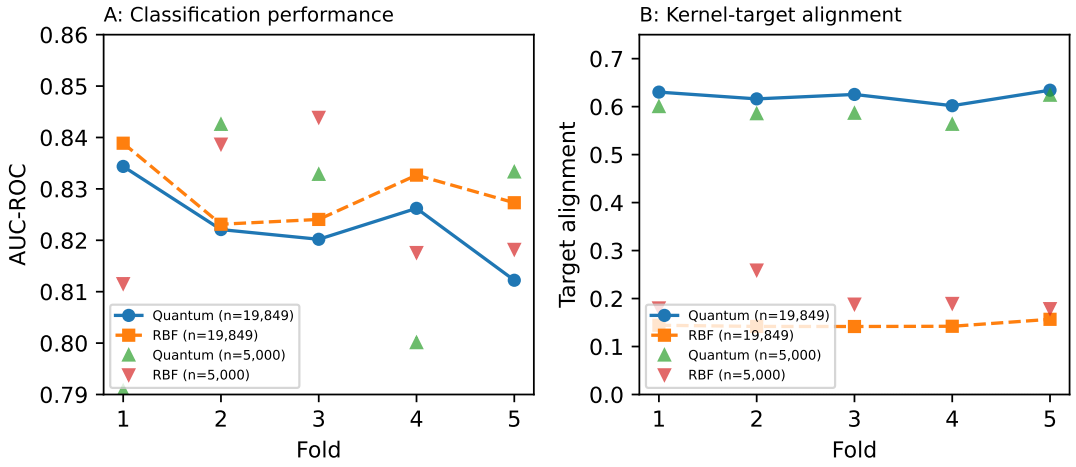


Figure 3.4.5: *Quantum kernel benchmark. Panel (A): per-fold AUC for the quantum kernel and the gamma-tuned RBF baseline at $n = 19\,849$ and $n = 5000$; the two overlap at every fold. Panel (B): kernel–target alignment per fold, where the quantum kernel reaches 0.622(13) against 0.145(6) for RBF, yet this geometric advantage does not translate into classification performance ($\Delta AUC = -0.006$).*

3.4.5 Fusing representations into ECFP4 adds nothing, and transfer exposes fragility

The hybrid of Table 3.4.1 combines the quantum-inspired descriptors with one another and excludes ECFP4, so it cannot show whether they carry information the fingerprint lacks. The ECFP4-inclusive fusion was therefore tested directly. A stacked ensemble combining per-representation random forests through logistic regression over out-of-fold probabilities, the variant least exposed to feature dilution, reached 0.9473(43) against 0.9475(45) for ECFP4 alone (paired $t = -2.29$, $p = 0.084$). Fusions that mixed feature blocks directly were worse: concatenation reached 0.9073 ($\Delta = -0.040$, $p < 0.0001$) and per-block feature selection 0.8958 ($\Delta = -0.052$). Because the deficit grew as the 2048-bit fingerprint was displaced further, the mechanism is dilution of a sparse fingerprint rather than interference between representations. For this endpoint, TFP and TNE add no predictive information to ECFP4.

Transfer to a ChEMBL 34 label-source-shift panel (22 447 antimalarials, $IC_{50} \leq 10$ μM active) separated representations that look similar internally. ECFP4 reached 0.960(4), slightly above its internal value; TFP reached 0.864, the TFP + TNE hybrid 0.800 and TNE 0.645. Under the same shift the quantum kernel fell below RBF (0.817 against 0.847; corrected-resampled $p = 0.021$, BH $q = 0.021$). Retaining the 351 embedding failures (1.6 %) as zero-vector penalties or dropping them for the 22 096 complete cases did not change the ranking (ECFP4 0.960, TFP 0.861, TNE 0.644, TFP + TNE 0.798), so the ordering is not driven by embedding failures (Figure 3.4.6). This panel is independent of the eos80ch label source but molecule-level disjointness from the primary panel was not established, so the analysis is a transferability assessment rather than an independent hold-out validation.

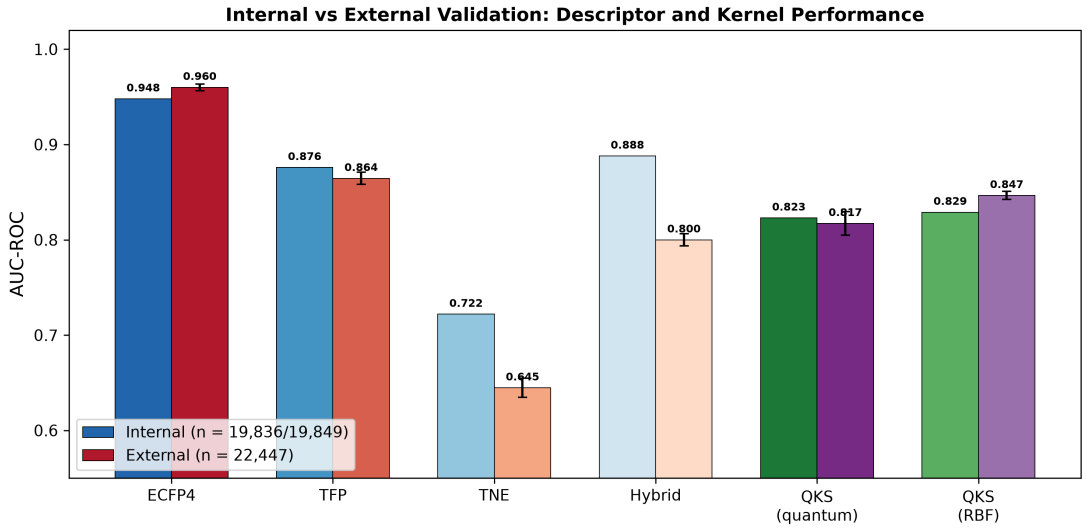


Figure 3.4.6: *Internal versus ChEMBL label-source-shift performance in the descriptor benchmark. Descriptor methods were evaluated with random forests and kernel methods with support-vector machines, so the two families are shown descriptively rather than as a direct model-to-model comparison. The tensor network embedding loses the most under the label-source shift, while the topological fingerprint transfers with a smaller deficit.*

3.4.6 What the quantum-inspired descriptors contribute despite the null result

A null result on the primary endpoint does not leave the descriptors valueless; it delimits what kind of value they have, and the delimitation is itself a result. Three contributions survive the benchmark, each in a different register.

The first is diagnostic. Persistent homology decomposes the scaffold paradox that motivated the whole representation programme: H_1 features, which encode ring topology, are conserved between the seed and expanded sets (mean H_1 count 3.61) while H_0 features, which encode connectivity, diverge, and the decomposition is corroborated independently by the scaffold-only to whole-molecule similarity ratio of 1.84. That analysis converts a library statistic into a structural statement — novelty lies in peripheral substitution — which is then the quantity every later benchmark stratifies by. No fingerprint in the panel produces that statement, because a hashed bit vector has no variable that corresponds to ring-topology conservation as distinct from connectivity divergence.

The second is representational competence without predictive gain. The quantum kernel achieved a kernel–target alignment of 0.622 against 0.145 for the tuned classical kernel and significantly outperformed a linear kernel ($p \leq 0.0006$), so the feature map is not uninformative; it is precisely aligned with the label structure and still does not convert that alignment into classification performance. The tensor embedding retained docking-relevant information on its derivation panel ($R^2 = 0.473$ for *Pf*DHFR against 0.461 for ECFP4) and compressed the atomistic representation 6.1-fold. These are competence results: the representations carry information; they do not carry the right information for this endpoint.

The third is a transfer diagnostic. Under the label-source shift the descriptors separate in a way their internal AUCs do not predict: the topological fingerprint falls to 0.864 and the tensor embedding to 0.645, while the quantum kernel drops below its classical comparator (0.817 against 0.847). A descriptor's behaviour under a label-source change is therefore a property worth measuring in its own right, and the internal benchmark alone would not have revealed it. The honest summary is that the quantum-inspired family earns a role in the workflow as analysis and diagnostic instruments attached to a classical predictor, not as replacements for one — which is exactly the role the synthesis of Section 3.7 assigns it.

3.4.7 The benchmark describes interpolation, and the topological signal cross-links to docking

Two further descriptor-benchmark results bound the interpretation of every AUC reported above. First, the panel is scaffold-poor for its size: 19 836 molecules resolve to 632 Bemis–Murcko scaffolds, or 324 once atom identity is stripped, a mean of 31 molecules per scaffold, with benzene alone forming the core of 1661 of them (8.4 %). Random cross-validation therefore places analogues of the same scaffold on both sides of most fold boundaries, and the reported AUCs describe interpolation within analogue series. Repeating the benchmark with folds grouped by scaffold so that no scaffold appears in both training and test sets costs every descriptor between 0.086 and 0.196 AUC: ECFP4 falls from 0.9485 to 0.8224 (2.5–97.5 percentile of repeat means 0.8122–0.8298), TFP from 0.8691 to 0.7253 and TNE from 0.7205 to 0.6342. The comparative conclusion survives, with ECFP4 strongest and TNE weakest under either split, but the ordering in between does not fully survive: MACCS overtakes FCFP4 and TFP overtakes the pharmacophore fingerprint, which loses 0.196. Performance within scaffolds therefore does not predict transferability across them.

Second, topological features do connect to the docking evidence of the chemical-space study, weakly. Across 17 011 molecules docked against the four targets, the H_0 count ($\rho = -0.248$, $p < 10^{-137}$), H_0 entropy ($\rho = -0.243$) and H_1 entropy ($\rho = -0.190$) correlate negatively with the number of targets bound at $\Delta G \leq -7.0$ kcal mol^{-1} (Figure 3.4.7). Lower values of these features are associated with more targets meeting the threshold. Because the endpoint is a docking-derived count and the associations are weak in magnitude, the result is hypothesis-generating rather than a mechanistic polypharmacology rule. The related association between H_1 count and resistance-resilience score is size-mediated: after controlling for molecular size and structural covariates it does not support an independent topological predictor (Figure 3.4.8).

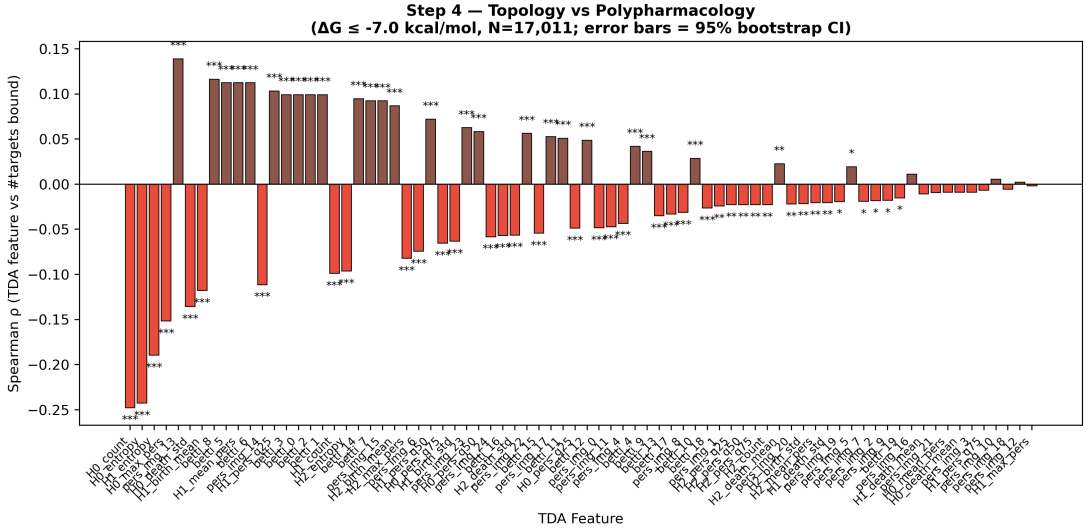


Figure 3.4.7: *Topological features against multi-target docking profiles ($N = 17\,011$ molecules docked against four* P. falciparum *targets). Persistence-derived features correlate negatively but weakly with the number of targets meeting the $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$ threshold. The endpoint is a docking-derived count, so this is a hypothesis-generating association rather than a mechanistic polypharmacology rule.*

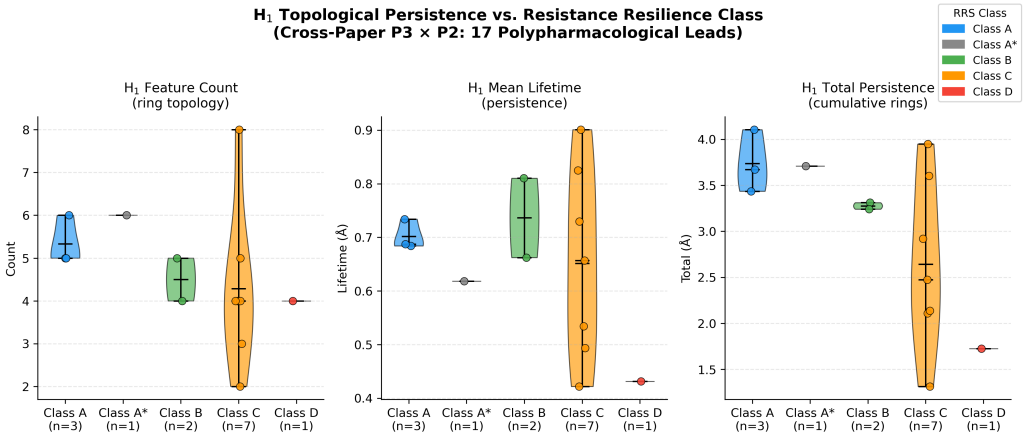


Figure 3.4.8: *Distribution of H_1 cycle persistence across resistance-resilience classes. The apparent association between topological complexity and resistance resilience is size-mediated and does not survive control for molecular size and structural covariates, so H_1 persistence is not an independent topological predictor of the resistance-resilience score.*

3.4.8 Why the ceiling is structural rather than accidental

Three mechanisms, each supported by a result above, explain why no descriptor in the benchmark closes the gap, and together they locate the ceiling in the input information rather than in model capacity or kernel choice.

The first mechanism is what the encodings discard. Hashed circular fingerprints, keyed dictionaries and bounded-topology descriptors all describe two-dimensional connectivity, and the chemical-space study measured where this chemical space's novelty lives: in peripheral substitution on conserved ring systems, with scaffold-only similarity (0.379) nearly twice whole-molecule similarity (0.206) and the topological decomposition assigning conserved ring topology to H_1 and diverging connectivity to H_0. An encoding that flattens exactly that variation cannot be expected to separate the molecules that differ only in it.

The second mechanism is the geometry of the panel and its labels. The panel is scaffold-poor for its size (632 scaffolds for 19 836 molecules, a benzene core in 8.4 % of them, roughly 31 molecules per scaffold) and its labels are model predictions at 74.2 % active prevalence, so most of the available label signal is analogue-series structure that local-environment dictionaries capture well, and the random-split areas under the curve measure that capture. When the partitions enforce extrapolation, every descriptor loses between 0.086 and 0.196 and the internal ordering partially reverses — MACCS overtakes FCFP4, the topological fingerprint overtakes the pharmacophore fingerprint — so performance within scaffolds does not predict transfer across them, and no single descriptor saturates the label signal in the regime that matters prospectively.

The third mechanism is that geometric expressivity does not create information. The quantum kernel reached a fourfold higher kernel–target alignment than the tuned classical kernel and a ΔAUC of -0.006; under the label-source shift it fell below the same comparator (0.817 against 0.847); and on the feature side, the fusion most insulated from dilution changed the baseline by -0.0002 while direct feature mixing lost up to 0.052. A more expressive feature space can re-express the information its input carries, and no further.

The conclusion these mechanisms support is the transition the next section makes: breaking the ceiling requires an encoding that carries the discarded information — three-dimensional geometry and stereochemistry — natively, and the fair comparators for such an encoding are the classical three-dimensional descriptors that already do so, not another fingerprint.

3.5 Learned representations under chemical distribution shift

3.5.1 Under scaffold separation, ECFP4 with a random forest still wins

The learned-representation study extends the representation comparison to graph neural networks and a SMILES transformer on the same frozen 19 836-molecule panel, using 25 fold–seed replicates per model and three partition families (Table 3.5.1). The partition family determines the apparent model ranking. Under stratified random splitting, ECFP4–RF reaches 0.9433 and ChemBERTa 0.9121, and every architecture is within a few points of the fingerprint. Under Bemis–Murcko scaffold separation the ranking is unchanged but the deficits become statistically resolvable: ECFP4–RF reaches 0.8300 while the graph isomorphism network reaches 0.8047 ($\Delta = -0.025$, BH-adjusted $p = 0.0330$), GIN–TFP 0.8138 ($\Delta = -0.016$, adjusted $p = 0.0330$), GIN–TNE 0.8090 ($\Delta = -0.021$, adjusted $p = 0.0378$) and ChemBERTa 0.7867 ($\Delta = -0.043$, adjusted $p = 0.0002$). The Butina cluster split at $C = 0.55$ reproduces the same hierarchy with ECFP4–RF at 0.8331 and

ChemBERTa lowest at 0.7776. Figure 3.5.1 shows the random-versus-scaffold contrast; the gap between the two panels is the quantity that matters, not either AUC alone.

Model	Random	Scaffold	Δ vs. ECFP4 (adjusted p)	Butina
ECFP4–RF (baseline)	0.9433	**0.8300**	—	**0.8331**
GIN	0.9098	0.8047	−0.025 (0.0330)	0.8202
GIN–TFP (topological fusion)	0.9084	0.8138	−0.016 (0.0330)	0.8232
GIN–TNE (tensor fusion)	0.8918	0.8090	−0.021 (0.0378)	0.8191
ChemBERTa	0.9121	0.7867	−0.043 (0.0002)	0.7776

Table 3.5.1: *Primary representation benchmark of the learned-representation study across three partition families. Values are mean test ROC-AUC across five per-seed means (each seed averaging five fold evaluations, 25 fold–seed records per model and split). Scaffold-split differences are evaluated against ECFP4–RF with two-tailed paired t-tests ($df = 4$) and Benjamini–Hochberg adjustment. Activity labels are model predictions on a frozen panel.*

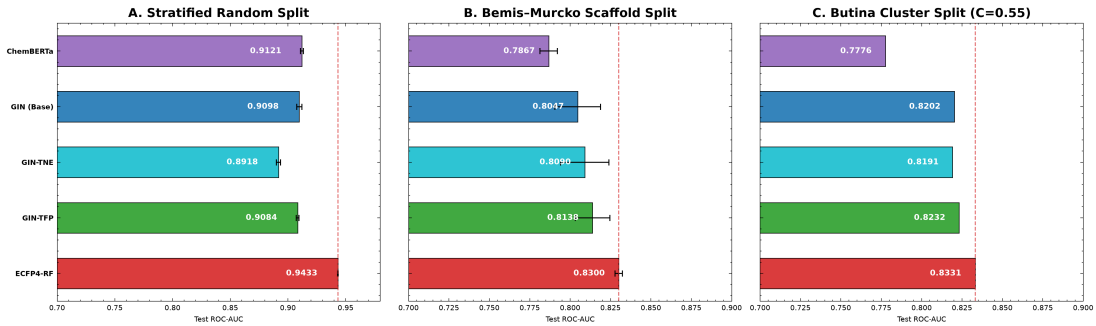


Figure 3.5.1: *Molecular representation performance under random and Bemis–Murcko scaffold splits in the learned-representation study (25 fold–seed replicates per model). Classical ECFP4–RF maintains the highest discrimination under scaffold separation. The collapse of every model between the two panels, rather than the ordering within either panel, is the result that matters for prospective screening.*

3.5.2 The gap is localized in the structural extrapolation tail

Aggregate AUCs hide where the deficit occurs. Stratifying test compounds by the Tanimoto similarity of their nearest training neighbour shows that the ECFP4–RF advantage is largest in the structural extrapolation tail (nearest-neighbour similarity below 0.40, deciles D1–D4), reaching up to 0.053 AUC for the weakest model and 0.027 for GIN–TFP, and narrows sharply, without reversing, to 0.001–0.007 in mid-range interpolation (D5–D8). Model choice is therefore a spatial decision rather than a global one on this panel, which is the observation that motivates the distance-aware routing hypothesis discussed below.

Stratifying by structural complexity sharpens the picture and also disciplines it (Table 3.5.2). Fusing multi-scale persistent-homology images into the graph network (GIN–TFP) yields a positive gain over the base GIN in every cohort: 0.0143 on flat aromatics ($F_{sp^3} < 0.25$, $n = 7797$), 0.0125 on complex three-dimensional scaffolds ($F_{sp^3} \geq 0.45$, $n = 2235$) and 0.0102 on polycyclics (Rings ≥ 4, $n = 363$). The gain is directionally consistent but does not increase monotonically with structural complexity, and on the small polycyclic cohort its bootstrap interval includes zero. Base GIN shows its largest deficit against ECFP4–RF precisely on the stereochemically rich cohorts (0.0494 on complex three-dimensional scaffolds and 0.0606 on polycyclics, against 0.0136 on flat aromatics), and topological fusion partially recovers that deficit, closing 17 % of the cohort-level gap on polycyclics and 36 % at the aggregate scaffold split. The honest reading is that global persistent-image invariants supply information complementary to one-Weisfeiler–Lehman-bounded message passing, without the benchmark isolating that as the unique cause (Figure 3.5.2).

Cohort	ECFP4–RF	GIN	GIN–TFP	Gain vs. GIN
Flat aromatic ($F_{sp^3} < 0.25$; $n = 7797$)	0.7624	0.7488	0.7631	0.0143 ($d = 1.96$; CI 0.0088 to 0.0201)
Complex 3D ($F_{sp^3} \geq 0.45$; $n = 2235$)	0.8457	0.7963	0.8088	0.0125 ($d = 0.46$; CI 0.0040 to 0.0206)
Polycyclic (Rings ≥ 4; $n = 363$)	0.7768	0.7162	0.7264	0.0102 ($d = 0.17$; CI −0.0112 to 0.0306)

Table 3.5.2: *Learned-representation benchmark performance stratified by structural complexity on the frozen 19 836-molecule panel (0 parse failures). Cohort values were recomputed from archived canonical scaffold-split predictions under a predeclared per-seed estimand (mean ± SD over five seeds). The final column gives the paired seed-level Cohen's d and the 95% molecule-level bootstrap interval of the GIN–TFP gain over GIN; the polycyclic interval includes zero. Cohort-level gains are descriptive statistics of this panel.*

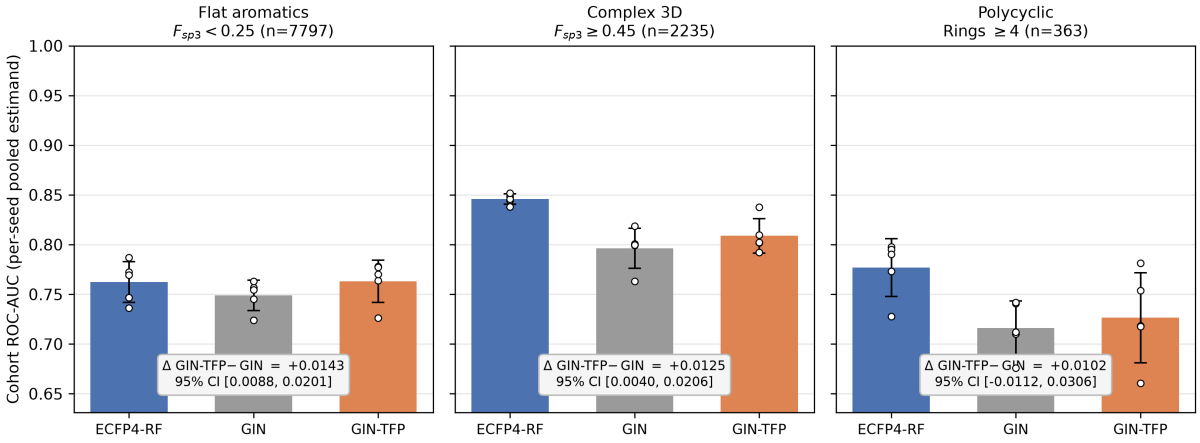


Figure 3.5.2: *Performance by structural-complexity cohort in the learned-representation study under the recomputed per-seed estimand (per-seed points, mean ± SD over five seeds). The GIN–TFP gain over GIN is positive in every cohort, with bootstrap 95% intervals annotated per panel, but it does not increase with structural complexity and the polycyclic interval includes zero. Base GIN shows its largest deficit against ECFP4–RF on the stereochemically rich cohorts.*

A hyperparameter capacity sweep bounds the alternative explanation that the graph deficit is a tuning artifact: across hidden widths of 64, 128 and 256 and dropout probabilities of 0.1 and 0.2, scaffold-split AUC remained within 0.8000 to 0.8081 over 25 fold–seed replicates. Adding parameters does not close the gap to ECFP4–RF, which is consistent with an architectural rather than a tuning explanation.

3.5.3 Topological salience and the limits of what a benchmark can claim

The fused network does use the topological block. Across the 25 replicates the persistent-image block (dimensions 33 to 57) carries the largest mean projection-weight magnitude in the fusion head, 0.0790 against 0.0485 for the H_0 block and 0.0547 for the Betti block (Figure 3.5.3). This salience is reported descriptively: the aggregate scaffold-split gain from topological fusion remains 0.0091 AUC points, so high projection weight is not evidence of importance. Salience and contribution must be measured separately, and here they diverge.

Two further boundaries belong to the learned-representation result itself. First, the early-recognition metrics that a virtual-screening audience would expect are degenerate on this panel: at approximately 74 % active prevalence, $\mathrm{EF}_{1\%}$ saturates at 1.32 to 1.35 against a prevalence-imposed ceiling of approximately 1.35, and precision@1 % lies between 0.978 and 0.999. No prospective precision claim is therefore supported, including for the distance-aware triage rule, which is proposed as a testable operational hypothesis: on this panel split-conformal coverage across the five routed arms meets the nominal 90 % target (0.902 to 0.906, mean set size 1.26 to 1.31) and post-hoc isotonic recalibration reduces residual scaffold-split expected calibration error by a factor of 3 to 5 (from 0.08 to 0.13 to approximately 0.025), but prospective evaluation on an independent low-prevalence library is reserved for future work. Second, the centroid-based screening reduction that makes the pipeline deployable (from 263 000 to 1936 representative evaluations, a 99.3 % overhead reduction, with mean pairwise Tanimoto similarity between centroids of 0.32) is a compute statement, not a performance statement.

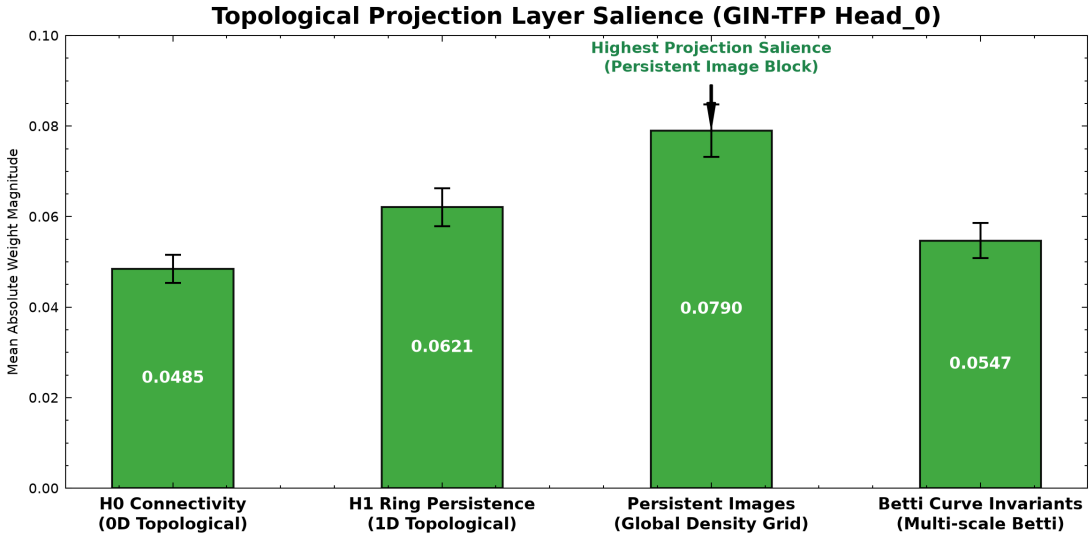


Figure 3.5.3: *Mean projection salience of descriptor blocks in the topological and tensor fusion heads of the learned-representation study. The persistent-image dimensions carry the largest mean weight magnitude, yet the corresponding aggregate scaffold-split gain remains 0.0091 AUC points, so salience is not evidence of predictive contribution.*

3.5.4 What these results mean for model choice in practice

For a library whose prospective compounds resemble its training series, the benchmarks give no reason to displace the fingerprint baseline: ECFP4 with a random forest led every learned architecture under random splitting, its bits remain individually interpretable, and its training cost is negligible beside any neural architecture. In the interpolation regime the differences between models shrink to between 0.001 and 0.007 AUC, which is below the resolution at which a model choice can be justified on performance grounds alone.

For a campaign whose value depends on unseen scaffolds, the same benchmarks say that none of the tested architectures is reliable. The deficit against the baseline reaches 0.053 AUC in the structural extrapolation tail, it persists across a capacity sweep (0.8000 to 0.8081 across widths and dropout settings, against 0.8300 for the baseline) so it is architectural rather than a tuning artefact, and the molecular transformer is the weakest model under both scaffold-separated and cluster partitions, so learning representations from strings does not substitute for the geometric information that no architecture in the panel receives. The defensible reading is spatial: model choice is made per molecule according to where that molecule sits relative to the training set. The distance-aware triage rule operationalizes exactly that reading, with split-conformal coverage across its routed arms at 0.902 to 0.906 against a nominal 90 % target; it remains, by its own declaration, a testable operational hypothesis until it is evaluated on an independent low-prevalence library.

One further practice-level observation concerns the fusion result. Topological features improve the graph network in every structural cohort yet do not scale with complexity, carry the largest projection weights while contributing 0.0091 AUC points in aggregate, and dilute the baseline when mixed into it at feature level. Complementary structural information is present and small, and the gap it fails to close is not closed by architecture; that is the point at which the diagnosis of this chapter hands its question to the encoding level.

The two representation sections therefore close the diagnosis that the chemical-space and resistance sections opened: every representation tested — hashed, topological, tensor-compressed, kernel-based, graph-based or sequence-based — encodes two-dimensional information about a three-dimensional problem, and the deficit concentrates where structural novelty is greatest. What remains is the test itself: whether an encoding that places molecular structure directly into quantum feature space, without passing through a fingerprint, escapes that ceiling.

3.6 Structure-direct quantum molecular encoding

This section reports the quantum machine learning programme of the thesis, which is the only part of the work that uses genuine quantum machine learning and which is reported here as work in progress with its status declared rather than as a source of comparative results. The programme exists because of the four preceding sections: the descriptor benchmark showed that a quantum kernel fed by fingerprint-derived inputs inherits the fingerprint's information ceiling, and the learned-representation benchmark localized every architecture's deficit in the low-similarity extrapolation tail. Structure-direct encoding, in which molecular structure is placed into the circuit parameters rather than into an intermediate classical fingerprint, is the hypothesis those failures generate. The encoding architecture and its four declared phases are specified in Section 2.7.4; this section reports what has been computed, what has been archived, and what remains pending, in that order.

3.6.1 The archived exploratory phase: quantum features fed by a fingerprint

The first implementation of the programme compressed extended-connectivity fingerprint vectors into shallow variational quantum circuits scored by a support-vector oracle. On the 1000-molecule

subsample of the descriptor panel under five-fold cross-validation it reached an area under the ROC curve of 0.8474(129) in its canonical four-qubit, depth-one configuration, a result confirmed by a fifty-trial hyperparameter search; on the same subsample the matched classical comparator, a radial basis function kernel on the same fingerprint, reached 0.8693(265), so the quantum feature extractor sat 2.5 % below its own classical baseline, and under scaffold separation it fell to 0.7831. The diagnosis is the one this thesis predicts from the representation studies: the quantum component received no structural information beyond what the fingerprint had already captured, so it could not recover the stereochemistry the fingerprint discarded — garbage in, garbage out. On 6 October 2026 we archived the entire phase as *exploratory* under a different estimand: it answers the narrow question of whether quantum feature extraction adds value to a fingerprint-fed pipeline on this panel, it does not answer the structure-direct question, and it informs none of the claims of this thesis. It is reported here because an archived result with its estimand named is part of the record, not because it supports the programme.

3.6.2 The structure-direct pipeline and its confirmed phases

The canonical pipeline replaces the fingerprint input with quantum molecular structure encoding: bond-order and Coulomb-adjacency matrices derived from the molecular graph are flattened into one- and two-qubit rotation angles, and the resulting circuit is trained end to end behind a classical TensorFlow head (Section 2.7.4). Four phases are declared with frozen inputs at each boundary, and their status is as follows.

The encoding phase is complete: all 19 849 molecules of the representation panel encoded successfully with zero failures, and the resulting angle datasets are frozen with recorded hashes, so the substrate on which any later comparison rests is fixed before training. The pipeline-confirmation phase is complete: the full hybrid stack ran end to end on a leave-one-out smoke test of the 17-molecule Set-A panel, completing 17 of 17 folds with zero encoding failures and returning an area under the ROC curve of 0.733 at thirty untuned epochs in approximately 200 seconds. That number is a pipeline confirmation and not a benchmark: seventeen molecules at untuned settings support no performance claim, and none is made. The hyperparameter-search phase, a joint search over encoding type, circuit depth, entangling gate and classical-head configuration, is in progress. The canonical benchmark phase, defined on the frozen panel under the partition families, label provenance and multiplicity machinery of Section 2.8, is pending. The comparative status of the programme is therefore NOT COMPUTED, and no comparative quantum result appears in this thesis.

3.6.3 The reformulated question and its predeclared benchmark

The failures of the descriptor and learned-representation benchmarks reformulate the programme's central question. A quantum kernel on fingerprint inputs cannot beat the fingerprint's ceiling, so the operative question is no longer whether quantum processing of classical features helps, but whether encoding molecular structure directly into quantum feature space surpasses the classical alternatives that already encode three-dimensional information. The predeclared question is therefore: does structure-direct quantum encoding surpass classical three-dimensional descriptors — extended three-dimensional fingerprints (E3FP), ultrafast shape recognition with pharmacophoric atom types (USRCAT) and three-dimensional autocorrelation functions (Autocorr3D) — on the high-complexity quartiles of the African-chemotype structural index, under scaffold separation?

Three design choices make that question answerable rather than rhetorical. First, the classical comparators are three-dimensional, not fingerprint-based: if a classical three-dimensional descriptor resolves the scaffold-split deficit that ECFP4 and message passing could not, then the blind spots documented above are a fingerprint problem rather than a classical-computation problem, and the honest-negative outcome — that quantum encoding adds computational cost without predictive gain

on this estimand — is published as a methodological boundary rather than suppressed. Second, the evaluation is stratified by the African-chemotype structural index so that any effect is localized with respect to chemotype complexity rather than averaged away across the panel. Third, the decision machinery is predeclared: effect sizes (Cohen's d) with paired tests under the multiplicity corrections of Section 2.8, the declared power boundaries, and three outcome scenarios — advantage of the structure-direct encoding under scaffold separation, equivalence within the sampling limits, and a deficit attributable to simulator depth — each interpretable under the claim-ledger convention of this thesis. The benchmark has not yet run; this section specifies it so that whichever way it lands, the result is admissible.

3.6.4 What the programme has established, and what it remains

Two things are established at this point, and both are procedural in kind. The first is a frozen substrate: all 19 849 molecules of the representation panel encoded with zero failures and their angle datasets fixed with recorded hashes before any model was trained, so that no encoding choice can be revised after results are seen. The second is an engineering result: the hybrid quantum–classical stack trains end to end, with gradients flowing through the circuit simulation and the classical head, on 17 of 17 leave-one-out folds without encoding failure. Together these answer the question of whether the benchmark can be run at all and whether it will be run on inputs fixed in advance — the procedural counterpart, at the level of one programme, to the frozen-partition discipline that Section 2.8 imposes on the completed studies.

What is not established is any performance statement, and the distinction is kept deliberately. The smoke-test area under the ROC curve of 0.733 was obtained on seventeen molecules at untuned settings and supports no claim about comparative performance; the archived fingerprint-fed phase answers a different question under a different estimand and informs no claim here; and until the canonical benchmark completes, the fifth research question of the introduction remains *open* in the claim-ledger sense — the blind spots it targets are documented, and the test that would decide whether structure-direct encoding overcomes them is specified and frozen.

The predeclaration is what makes even the unwelcome outcomes usable. Advantage under scaffold separation, equivalence within the sampling limits, and a deficit attributable to simulator depth are each interpretable under the claim ledger before any of them is observed, so the programme cannot lose its question by the result. That is the same currency as the descriptor and learned-representation benchmarks before it: a well-specified negative is a boundary the field can build on, and an unspecific positive would have been worth less.

3.6.5 Comparison standard for hardware-based claims

The completed studies earn the right to state what any future quantum claim on this library must satisfy, and the standard has six parts. Any evaluation, on simulator or hardware, must (i) use the frozen panel with its declared African-chemotype structural index quartiles; (ii) enforce scaffold-separated partitions so that the measurement is extrapolation rather than interpolation; (iii) benchmark against the classical three-dimensional descriptors first, because a quantum model that only beats a two-dimensional fingerprint has beaten the wrong baseline; (iv) report effect sizes and uncertainty under the same statistical machinery as every result in this chapter, with multiplicity correction and declared power; (v) support any advantage claim with a theoretical account — reproducing-kernel Hilbert-space dimension, or non-local three-dimensional correlations mediated by entanglement — rather than an area-under-the-curve difference alone, because the descriptor benchmark showed exactly how a favourable Hilbert-space geometry can fail to translate into performance; and (vi) replicate across folds and random seeds with the uncertainty quantified. These requirements are deliberately harder than the ones the completed benchmarks had to meet:

they exist so that the next claim on this chemical space exceeds a measured baseline rather than an aspirational one.

3.7 Synthesis: what the representations do and do not provide

Read as one argument, the completed studies converge on a single answer to the question that titles this thesis. The chemical space is genuinely novel and scaffold-anchored; docking-derived resistance metrics separate candidate chemotypes and identify a shortlist of testable hypotheses; and every representation proposed as an alternative or complement to classical fingerprints, whether persistent-homology, tensor-compressed, kernel-based, graph-based or sequence-based, is diagnostically informative while adding no detectable predictive performance over ECFP4.

The four results that carry that conclusion are quantitative and mutually reinforcing. Adding topological and tensor representations to ECFP4 as a stacked ensemble changed AUC by −0.0002 under a test whose power is limited ($p = 0.084$), and feature-level fusions were worse than ECFP4 alone. The quantum kernel was indistinguishable from a gamma-tuned RBF kernel internally ($p = 0.060$) and lower than it under a ChEMBL label-source shift (0.817 against 0.847), despite a fourfold higher kernel–target alignment. Graph networks and a molecular transformer remained below ECFP4–RF under scaffold, Butina and decile-stratified partitions, with the deficit localized in the low-similarity extrapolation tail and not attributable to model capacity. Persistent homology did not predict mutation resilience once molecular size was controlled. The complementary role the topological descriptors do play is diagnostic rather than predictive: they decompose the scaffold paradox, expose compression behaviour and expose where a learned model's advantage disappears.

That conclusion is the standing baseline for the quantum machine learning programme, which is the only part of this thesis that uses genuine quantum machine learning and which Section 3.6 reports as work in progress. The programme was refocused on quantum molecular structure encoding[104]: bond-order and Coulomb-adjacency matrices are encoded directly as one- and two-qubit rotation angles, with stereochemistry carried by signed rotation parameters, and trained end to end with a classical TensorFlow head, so that the circuit itself, not a fingerprint, is the molecular representation. The refocus follows directly from the results above. The descriptor benchmark showed that a quantum kernel can match a tuned classical comparator yet fail to convert a fourfold kernel–target alignment into classification performance, because its input had already passed through the fingerprint whose information ceiling it inherited; and the learned-representation benchmark localized the deficit of every architecture in the low-similarity extrapolation tail, where structural distinctions are least represented. The two failure modes that this chapter documents, stereochemical collapse in hashed fingerprints and scaffold-extrapolation loss in learned models, are precisely what a structure-direct encoding is designed to address. Its encoding substrate is frozen on the full 19 849-molecule panel, its training pipeline is confirmed end to end, and its canonical benchmark is pending; three outcome scenarios are predeclared, and no comparative result from it is reported here. What the completed studies supply is the comparison standard. A quantum model on this library must be measured against the same frozen panel, the same scaffold-grouped partitions, the same label source and the same multiplicity correction as ECFP4–RF, and must be compared against a tuned classical kernel rather than an untuned one, because the one clear lesson of the descriptor benchmark is that kernel—target alignment and Hilbert-space expressivity do not translate into classification performance. The representational bottlenecks this thesis documents are real: dictionaries that describe local atom environments saturate the available label signal on a scaffold-poor panel, hybrid descriptors dilute a sparse fingerprint, and topological invariants are informative about structure without being predictive about activity. Whether structure-direct quantum encoding escapes those bottlenecks is an empirical question, and the honest-negative record assembled in this chapter is the reason the question can be asked without assuming the answer.

3.8 Conclusion

This chapter established the empirical foundation of the thesis through four linked studies. A hybrid library of 65 856 African-natural-product-inspired molecules is 92.6 % Tanimoto-distinct from its seeds while retaining 69.3 % of their scaffolds, and a centroid-reduced docking campaign against four mechanistically distinct targets shows that mutation resilience and target breadth are separable properties whose ratio-based classes are neutral within a pipeline-null control on the corrected *Pf* CRT channel. Explicit-solvent molecular dynamics retained poses where static docking predicted penalties in 7 of 8 matched mutant states, establishing that docking-derived and trajectory-derived quantities are non-equivalent estimands at single-replicate scope, and trajectory endpoint estimation succeeded only where the complex remained bound. A retrospective confrontation with published genotype-resolved *Pf* DHFR fold-shifts bounded the same docking estimand from the experimental side, with direction agreement of only 28.6 to 41.7 % between rigid-receptor $\Delta\Delta G$ and observed resistance shifts, so the static score is treated throughout as a triage input rather than a resistance measurement. Across topological, tensor-compressed, kernel-based, graph-based and sequence-based representations, ECFP4 with a random forest remained the strongest descriptor, no fusion improved on it, and the quantum kernel matched but did not beat a tuned classical kernel. These boundaries, together with the diagnostic value of persistent homology for generative-model assessment, define the comparison standard that the quantum machine learning programme, whose protocol, archived exploratory phase and predeclared benchmark are reported in Section 3.6, must meet before any hardware-based claim can be made.

3.8.1 The estimand register of the thesis

The chapter reports quantities of at least nine kinds, and the interpretive errors this thesis works to avoid are almost all errors of estimand — reading a quantity as estimating something it does not. Table 3.8.1 collects them in one place, stating for each what it estimates and what it cannot support. The register is the compact form of the estimand discipline that Section 3.3 introduced and that every later section inherits.

3.8.2 What would change the conclusions of this chapter

An honest-negative record is only useful if it states what would overturn it, so the falsification conditions of the chapter's central claims are stated here rather than left implicit. Four would suffice.

The claim that no representation examined beats ECFP4 on this panel would be falsified by any descriptor or architecture, evaluated under the same frozen partitions and label provenance, whose scaffold-split AUC exceeds 0.8300 with a multiplicity-corrected significant difference; the claim is explicitly scoped to the representations tested, so an untested representation — a classical three-dimensional descriptor, for instance — is not contradicted by it, which is why the predeclared benchmark of Section 3.6.3 includes classical three-dimensional comparators and not only the quantum encoding.

The claim that the learned-representation deficit is architectural rather than a tuning artefact would be falsified by a capacity or optimisation sweep that closes the gap to ECFP4–RF under scaffold separation; the sweep reported here (0.8000 to 0.8081 across widths and dropout settings against 0.8300) bounds that possibility at the explored scales, not universally.

The claim that static and dynamic quantities are non-equivalent estimands would be weakened by replicated trajectories showing that the 7 of 8 retention pattern does not survive replicate-level analysis; the single-replicate scope is declared wherever the divergence is used, and the triplicate verification is the experiment that would settle it.

Quantity	What it estimates	What it cannot support
Vina score	Rigid-receptor scoring-function output at an anchored pose	Absolute affinity; cross-target arithmetic means
Resistance-resilience score	Mutant-to-wild-type score retention on the same receptor	Biological resistance phenotype; circumvention evidence
MD-RRS_d	Pose retention relative to wild type over 10 ns	Affinity ranking; microsecond-scale stability
MM-GBSA endpoint	Trajectory-averaged endpoint energy where the complex stays bound	Inter-replicate-comparable free energy; cohort-level resilience
Random-split AUC	Interpolation within analogue series on a frozen panel	Prospective screening performance
Scaffold-split AUC	Extrapolation to unseen ring systems on the same panel	Performance under a changed label source
Kernel–target alignment	Geometric correspondence between kernel and labels	Classification performance
$\mathrm{EF}_{1\%}$ on this panel	Early recognition at 74 % prevalence	Prospective enrichment at deployment prevalence
Fusion-head salience	Learned projection-weight magnitude	Predictive contribution of the corresponding block

Table 3.8.1: *Estimand register: what each principal quantity of this thesis estimates, and the interpretation it cannot support. Every row is enforced at the point of the corresponding result; the register collects the boundaries so that they can be audited together.*

And the diagnosis that the ceiling is a two-dimensional-information ceiling would be falsified, in its strongest form, by a structure-direct quantum encoding that also fails: if the predeclared benchmark of Section 3.6.3 returns equivalence between the quantum encoding and the classical three-dimensional comparators while those comparators themselves fail to close the scaffold-split deficit, then the missing information is not three-dimensionality but something else — label quality, panel size, or endpoint definition — and the four-facet framing of the introduction would require revision. That possibility is not excluded by any result in this thesis; it is one of the three predeclared scenarios, and reporting it as such is what makes the framework falsifiable rather than rhetorical.

General Conclusion

This thesis asked whether quantum, quantum-inspired and learned molecular representations improve antimalarial candidate prioritization over strong classical baselines, and what a genuine quantum machine learning implementation would have to demonstrate on the same chemical space before any claim could be made. The question was addressed through five studies that share one library, one label source and one evaluation protocol: a chemical-space construction and target-anchored docking study, a resistance-aware scoring study stress-tested by explicit-solvent molecular dynamics and bounded by retrospective experimental data, a benchmark of topological, tensor-compressed and quantum-inspired descriptors, a benchmark of graph neural networks and a molecular transformer under chemical distribution shift, and a quantum machine learning programme, refocused on structure-direct encoding, whose protocol is specified and frozen while its canonical benchmark remains pending. The organising result is negative and was stated in advance so that it could not be mistaken for a late qualification: across every representation tested, an extended-connectivity fingerprint with a random forest remained the strongest descriptor for activity prediction on the panels studied, and no fusion, kernel or learned architecture improved on it.

The chemical-space study established the substrate on which every other result rests. An expansion of 396 African natural products and 454 synthetic antimalarials through complementary similarity-search and SELFIES-mutation routes produced a hybrid library of 65 856 unique molecules that is 92.6 % whole-molecule-novel relative to its seeds while retaining 69.3 % of their annotated scaffolds, an asymmetry that locates the novelty in peripheral substitution rather than in ring systems. Centroid-based sampling reduced the docking campaign from 263 424 to 1936 calculations, and target-anchored docking against four mechanistically distinct proteins, with the chloroquine-resistance transporter corrected to a wild-type receptor and audited by an independent pocket-detection method, resolved the 17-member cohort into seven A*, four B and six C resistance-score classes on the corrected channel. Two properties of that result matter beyond the cohort. First, target breadth and mutation resilience proved separable, because five candidates were classified from a single target panel while remaining non-binders elsewhere. Second, the corrected transporter channel behaved as a null under its own mutation pipeline (99.4 to 100.5%), so the class boundaries were anchored to that null rather than to raw thresholds alone. The docking protocol was validated in parts and bounded honestly: redocking reproduced four of five crystal poses, enrichment on known actives was strong under the dual-filter consensus, and the external benchmark returned a near-chance area under the ROC curve that was retained in the record rather than repaired, with a controlled two-arm experiment excluding cofactor blockade as its cause.

The resistance-aware study converted the docking campaign into a triage framework and then subjected it to the stress tests that determine what it may claim. Static docking and short explicit-solvent dynamics disagreed in a consistent direction: across the eight matched mutant states with an eligible wild-type baseline, docking predicted reduced score retention while the trajectories retained the pose in 7 of 8 comparisons (87.5 %; 95% Wilson interval 52.9 to 97.8%). The disagreement is not an error of either method but a divergence of estimands, and it is used as a calibration flag that prioritizes states for experimental follow-up. Two retrospective checks bounded the static estimand from the experimental side. A curated genotype-resolved inhibition panel showed that

rigid-receptor score differences fail to track published fold-shifts at every dihydrofolate-reductase genotype, with direction agreement of only 28.6 to 41.7% and the clinically strongest resistance signature mispredicted in direction; and a control on five approved antimalarials failed to recover the documented resistance phenotypes of pyrimethamine and chloroquine, reproducing the null character of the corrected channels on clinically familiar ligands. Together these checks support one operational conclusion: resistance-resilience classes are score-retention categories, useful as triage inputs and as generators of testable hypotheses, and never as resistance measurements.

The two representation benchmarks produced the thesis's central negative result with its mechanisms attached. The descriptor benchmark showed that the persistent-homology fingerprint and the tensor-compressed embedding carry genuine structural information, the former resolving the novelty-versus-scaffold paradox into conserved ring topology and diverging connectivity, the latter retaining docking-relevant information on its derivation panel, yet neither improved activity prediction: the stacked ensemble that included ECFP4 changed its area under the ROC curve by −0.0002, and the quantum kernel, although it matched a bandwidth-tuned classical kernel (0.823 against 0.829, $p = 0.060$), achieved a fourfold higher kernel–target alignment without converting it into performance, and fell below the same comparator under a label-source shift. The learned-representation benchmark extended the comparison to architectures that learn their own representations. Under random splitting every model sat within a few points of the fingerprint baseline; under scaffold separation the ranking was preserved but the deficits became statistically resolvable, with the graph isomorphism network at 0.8047 and the transformer at 0.7867 against 0.8300 for ECFP4–RF, and a capacity sweep confirmed that the gap is architectural rather than a tuning artifact. The deficit localized in the structural-extrapolation tail and narrowed, without reversing, in the interpolation regime, which converted model choice into a spatial decision and motivated a distance-aware triage rule that remains, by declared design, a testable operational hypothesis rather than a validated procedure. Topological fusion improved the graph network in every structural cohort, but the gain did not scale with structural complexity, its polycyclic interval included zero, and the descriptor salience that accompanied it was not evidence of contribution. Because the activity labels are model predictions on a prevalence-enriched panel, all early-recognition metrics saturate at their prevalence-imposed ceilings, and no prospective screening precision is claimed from any of these results.

These findings are bounded by limitations that are stated rather than smoothed. Docking scores are scoring-function outputs on anchored poses, not affinities; the transporter and ion-pump sites are proxy or mechanism-motivated cavities; the molecular-dynamics evidence is single-replicate at 10 ns and supports pose-retention statements only, with endpoint energies carrying an inter-replicate sensitivity of up to 7.75 kcal mol^{-1}; the activity labels are predictions from an external model on a scaffold-poor panel, so every classification metric measures agreement with one model's opinion of activity; the resistance cohorts are small, and at $n = 17$ a correlation test reaches 80 % power only above a moderate association, so every null association is reported as a failure to detect rather than as evidence of equivalence; and no experimental validation, biochemical or phenotypic, was performed within this work, which is why every candidate shortlist is presented as a hypothesis for assay follow-up rather than as a hit list. The honest-negative framing is itself part of the contribution: each null result is documented with the analysis that would have detected the opposite, so that the boundary is auditable rather than asserted.

The forward direction is the quantum machine learning programme, which answers the fifth research question of the introduction, and whose central question the completed studies made precise. The benchmarks exposed two structural blind spots in the representations examined: hashed fingerprints collapse stereochemistry, and local message passing loses ring geometry across unseen scaffolds; and the descriptor benchmark showed that a quantum component fed by an ECFP4-derived input inherits the fingerprint's information ceiling, which is why its favourable Hilbert-space geometry produced no performance difference. The programme was therefore refocused on quantum molecular structure encoding, which places bond-order and Coulomb-adjacency information directly into one- and two-qubit rotation angles, carries stereocentre configuration and double-bond geometry

as signed parameters, and trains a hybrid quantum-classical model end to end [104]. Its encoding substrate is frozen on the complete 19 849-molecule panel with zero failures, its training pipeline is confirmed end to end, and its canonical benchmark, defined on the same frozen partitions, label provenance and multiplicity machinery as every result above, is pending; three outcome scenarios are predeclared, spanning advantage, equivalence and a depth-limited deficit, and each is interpretable under the claim-ledger convention of this thesis. Until that benchmark completes, the fifth research question is recorded as *open* rather than answered: the blind spots it targets are documented, the test that would decide whether structure-direct encoding overcomes them is specified and frozen, and no result — favourable or otherwise — is claimed in advance of it. Hardware evaluation follows only after the simulator benchmark is complete under that protocol.

Taken together, the thesis delivers three principal contributions to the field.

The first is a frozen, fully declared benchmark of classical, topological, tensor-compressed, quantum-inspired and learned molecular representations on African natural-product chemical space, with scaffold-grouped partitions, declared label provenance, stated power boundaries and a claim ledger, in which honest-negative results are documented with the same machinery as positive ones. Any future quantum implementation on this library inherits a comparison standard rather than a collection of incomparable one-off studies.

The second is a reusable resistance-aware prioritization workflow: a corrected wild-type transporter receptor audited by an independent pocket-detection method, a ratio-based retention score with a declared eligibility rule, a static-to-dynamic divergence diagnostic with its sampling limits quantified, and a retrospective genotype–phenotype confrontation that bounds what the static score can claim from published inhibition data. The workflow is deliberately protocol-local, but its components — the wild-type eligibility gate and the estimand-divergence framing — transfer to other resistance-relevant target panels, a transfer whose timeliness grows with the documented expansion of artemisinin-resistance markers across Africa.

The third is an empirical diagnosis of the representational ceiling that limits antimalarial virtual screening: classical fingerprints, quantum-inspired kernels and learned representations all flatten stereochemistry into two-dimensional topology, and the failure appears as scaffold-extrapolation loss (the graph isomorphism network at 0.8047 against 0.8300 for ECFP4–RF under Bemis–Murcko separation) wherever structural novelty is greatest. Structure-direct quantum encoding is the testable hypothesis that answers this diagnosis; if it surpasses the classical three-dimensional descriptors under the predeclared standard, the question shifts from whether to why — Hilbert-space expressivity, non-local three-dimensional correlations mediated by entanglement, or reproducing-kernel dimension — and if it does not, the honest-negative result bounds what quantum encoding can claim on this chemical space.

The quantum machine learning programme that continues this work must satisfy three requirements established here. First, benchmark against classical three-dimensional descriptors — extended three-dimensional fingerprints, ultrafast shape recognition and three-dimensional autocorrelation functions — before benchmarking against the two-dimensional ECFP4 baseline, because a quantum model that only beats a fingerprint has beaten the wrong comparison. Second, enforce scaffold-separated evaluation, so that the measurement is extrapolation across unseen ring systems rather than interpolation within analogue series. Third, formalize the theoretical advantage — reproducing-kernel Hilbert-space dimension, or entanglement-mediated non-local correlations — before any hardware-based claim, because this thesis showed empirically that a favourable kernel geometry and a fourfold alignment advantage can coexist with no performance difference at all. The next step is the one the thesis was built to enable: the structure-direct quantum benchmark on this library, judged against the comparison standard the completed studies define.

Bibliography

[1] World Health Organization. *World Malaria Report 2025*. Report. World Health Organization, 2025.

[2] J. M. Alghamdi, A. Al-Qahtani, F. Alhamlan, and A. Al-Qahtani. "Recent Advances in the Treatment of Malaria". In: *Pharmaceutics* 16 (2024). DOI: 10.3390/pharmaceutics16111416.

[3] D. J. A. Schedler. "Drug Discovery: A History (Sneader, Walter)". In: *Journal of Chemical Education* 83.2 (Feb. 2006), p. 215. ISSN: 1938-1328. DOI: 10.1021/ed083p215.1.

[4] S. Alizon and P.-O. Méthot. "Reconciling Pasteur and Darwin to control infectious diseases". In: *PLOS Biology* 16.1 (Jan. 2018). Ed. by A. Read, e2003815. ISSN: 1545-7885. DOI: 10.1371/journal.pbio.2003815.

[5] H. Kubinyi. "Drug research: myths, hype and reality". In: *Nature Reviews Drug Discovery* 2.8 (Aug. 2003), pp. 665–668. ISSN: 1474-1784. DOI: 10.1038/nrd1156.

[6] W. R. Galloway, A. Isidro-Llobet, and D. R. Spring. "Diversity-oriented synthesis as a tool for the discovery of novel biologically active small molecules". In: *Nature Communications* 1.1 (Sept. 2010). ISSN: 2041-1723. DOI: 10.1038/ncomms1081.

[7] S. M. Paul, D. S. Mytelka, C. T. Dunwiddie, C. C. Persinger, B. H. Munos, S. R. Lindborg, and A. L. Schacht. "How to improve R&D productivity: the pharmaceutical industry's grand challenge". In: *Nature Reviews Drug Discovery* 9.3 (Feb. 2010), pp. 203–214. ISSN: 1474-1784. DOI: 10.1038/nrd3078.

[8] J. Hughes, S. Rees, S. Kalindjian, and K. Philpott. "Principles of early drug discovery". In: *British Journal of Pharmacology* 162.6 (Feb. 2011), pp. 1239–1249. ISSN: 1476-5381. DOI: 10.1111/j.1476-5381.2010.01127.x.

[9] N. Schaduangrat, S. Lampa, S. Simeon, M. P. Gleeson, O. Spjuth, and C. Nantasenamat. "Towards reproducible computational drug discovery". In: *Journal of Cheminformatics* 12.1 (Jan. 2020). ISSN: 1758-2946. DOI: 10.1186/s13321-020-0408-x.

[10] G. Sliwoski, S. Kothiwale, J. Meiler, and E. W. Lowe. "Computational Methods in Drug Discovery". In: *Pharmacological Reviews* 66.1 (Jan. 2014), pp. 334–395. ISSN: 0031-6997. DOI: 10.1124/pr.112.007336.

[11] J. Lin, D. Sahakian, S. de Morais, J. Xu, R. Polzer, and S. Winter. "The Role of Absorption, Distribution, Metabolism, Excretion and Toxicity in Drug Discovery". In: *Current Topics in Medicinal Chemistry* 3.10 (May 2003), pp. 1125–1154. ISSN: 1568-0266. DOI: 10.2174/1568026033452096.

[12] U. Naithani and V. Guleria. "Integrative computational approaches for discovery and evaluation of lead compound for drug design". In: *Frontiers in Drug Discovery* 4 (Apr. 2024). ISSN: 2674-0338. DOI: 10.3389/fddsv.2024.1362456.

[13] J. Li, M. Alam, C. M. Sha, J. Wang, N. V. Dokholyan, and S. Ghosh. *Drug Discovery Approaches using Quantum Machine Learning*. Apr. 2021. arXiv: 2104.00746 [cs.ET].

[14] B. J. Neves, R. C. Braga, V. M. Alves, M. N. N. Lima, G. C. Cassiano, E. N. Muratov, F. T. M. Costa, and C. H. Andrade. "Deep Learning-driven research for drug discovery: Tackling Malaria". In: *PLOS Computational Biology* 16.2 (Feb. 2020). Ed. by Q. Morris, e1007025. ISSN: 1553-7358. DOI: 10.1371/journal.pcbi.1007025.

[15] R. Raphemot, D. Posfai, and E. R. Derbyshire. "Current therapies and future possibilities for drug development against liver-stage malaria". In: *Journal of Clinical Investigation* 126.6 (June 2016), pp. 2013–2020. ISSN: 1558-8238. DOI: 10.1172/jci82981.

[16] World Health Organization. *Malaria: Artemisinin Partial Resistance — Questions and Answers*. WHO Newsroom Q&A. 2025.

[17] N. W. Young et al. "Mapping the Prevalence of Molecular Markers of *Plasmodium falciparum* Artemisinin Partial Resistance in Africa: A Spatial-Temporal Modelling Study". In: *The Lancet Infectious Diseases* (2026). Advance online publication. DOI: 10.1016/S1473-3099(26)00237-9.

[18] A. Letebo et al. "Genomic Surveillance Reveals Co-Occurrence of *Plasmodium falciparum* Drug Resistance Variants Across Diverse Transmission Settings in Ethiopia". In: *Nature Microbiology* (2026). Advance online publication. DOI: 10.1038/s41564-026-02420-5.

[19] F. Habarugira, J. Batamuriza, R. Ndahimana, J. N. Minega, M. M. Dieng, M. J. Mutsaka-Makuvaza, T. M. Daba, Y. Idaghdour, and L. Mutesa. "The Global Landscape of *Plasmodium falciparum* Drug Resistance Markers, 2005–2025: A Systematic Review and Meta-Analysis". In: *Pathogens* 15.2 (2026).

[20] J. Santos-Lima et al. "Molecular Markers of Antimalarial Drug Resistance in pfk13, pfmdr1, pfdhfr, and pfdhps Genes of *Plasmodium falciparum* in a Rural Municipality in Cubal, Benguela Province, Angola (2022–2023)". In: *Malaria Journal* 25 (2026). DOI: 10.1186/s12936-026-05856-6.

[21] E. S. Mathews and A. R. Odom John. "Tackling resistance: emerging antimalarials and new parasite targets in the era of elimination". In: *F1000Research* 7 (Aug. 2018), p. 1170. ISSN: 2046-1402. DOI: 10.12688/f1000research.14874.1.

[22] C. Kraeker and C. Chandler. ""We Learn From Them, They Learn From Us": Global Health Experiences and Host Perceptions of Visiting Health Care Professionals". In: *Academic Medicine* 88.4 (Apr. 2013), pp. 483–487. ISSN: 1040-2446. DOI: 10.1097/acm.0b013e3182857b8a.

[23] A. G. Atanasov et al. "Natural products in drug discovery: advances and opportunities". In: *Nature Reviews Drug Discovery* 20.3 (Mar. 2021), pp. 200–216. ISSN: 1474-1784. DOI: 10.1038/s41573-020-00114-z.

[24] C. Lipinski and A. Hopkins. "Navigating chemical space for biology and medicine". In: *Nature* 432.7019 (Dec. 2004), p. 861. ISSN: 0028-0836. DOI: 10.1038/nature03193.

[25] C. A. Lipinski, F. Lombardo, B. W. Dominy, and P. J. Feeney. "Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings". In: *Advanced Drug Delivery Reviews* 23.1 (Jan. 1997). In Vitro Models for Selection of Development Candidates, pp. 3–25. ISSN: 0169-409X. DOI: https://doi.org/10.1016/S0169-409X(96)00423-1.

[26] C. A. Lipinski. "Lead- and drug-like compounds: the rule-of-five revolution". In: *Drug Discovery Today: Technologies* 1.4 (Dec. 2004), pp. 337–341. ISSN: 1740-6749. DOI: 10.1016/j.ddtec.2004.11.007.

[27] M. D. Shultz. "Two Decades under the Influence of the Rule of Five and the Changing Properties of Approved Oral Drugs: Miniperspective". In: *Journal of Medicinal Chemistry* 62.4 (Sept. 2018), pp. 1701–1714. ISSN: 1520-4804. DOI: 10.1021/acs.jmedchem.8b00686.

[28] L. Patel, T. Shukla, X. Huang, D. Ussery, and S. Wang. "Machine Learning Methods in Drug Discovery". In: *Molecules* 25 (Nov. 2020), p. 5277. DOI: 10.3390/molecules25225277.

[29] G. W. Ashdown, M. Dimon, M. Fan, F. S.-R. Terán, K. Witmer, D. C. A. Gaboriau, Z. Armstrong, D. M. Ando, and J. Baum. "A machine learning approach to define antimalarial drug action from heterogeneous cell-based screens". In: *Science Advances* 6.39 (2020), eaba9338. DOI: 10.1126/sciadv.aba9338. eprint: https://www.science.org/doi/pdf/10.1126/sciadv.aba9338.

[30] A. Keshavarzi Arshadi, M. Salem, J. Collins, J. S. Yuan, and D. Chakrabarti. "DeepMalaria: Artificial Intelligence Driven Discovery of Potent Antiplasmodials". In: *Frontiers in Pharmacology* 10 (2020), p. 1526. ISSN: 1663-9812. DOI: 10.3389/fphar.2019.01526.

[31] D. Weininger. "SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules". In: *Journal of Chemical Information and Computer Sciences* 28.1 (1988), pp. 31–36. DOI: 10.1021/ci00057a005. eprint: https://pubs.acs.org/doi/pdf/10.1021/ci00057a005.

[32] S. Honda, S. Shi, and H. R. Ueda. "SMILES Transformer: Pre-trained molecular fingerprint for low data drug discovery". In: *arXiv* (Apr. 2019). ISSN: 1557-8666. DOI: 10.1089/cmb.2024.0884.

[33] R. A. Samant Monika, T. A. K. Chadha Nidhi, and H. Yasha. "In Silico Designing and Analysis of Inhibitors against Target Protein Identified through Host-Pathogen Protein Interactions in Malaria". In: *International Journal of Medicinal Chemistry* 2016 (Jan. 2016). ISSN: 2090-2069. DOI: 10.1155/2016/2741038.

[34] Q. A. Arshad, M. Ali, S.-u. Hassan, C. Chen, A. Imran, G. Rasul, and W. Sultani. *A Dataset and Benchmark for Malaria Life-Cycle Classification in Thin Blood Smear Images*. 2021. DOI: 10.48550/ARXIV.2102.08708.

[35] T. Suzuki and M. Katouda. "Predicting toxicity by quantum machine learning". In: *Journal of Physics Communications* 4.12, 125012 (Dec. 2020), p. 125012. DOI: 10.1088/2399-6528/abd3d8. arXiv: 2008.07715 [quant-ph].

[36] S. Mensa, E. Sahin, F. Tacchino, P. Kl Barkoutsos, and I. Tavernelli. "Quantum machine learning framework for virtual screening in drug discovery: a prospective quantum advantage". In: *Machine Learning: Science and Technology* 4.1, 015023 (Mar. 2023), p. 015023. DOI: 10.1088/2632-2153/acb900. arXiv: 2204.04017 [quant-ph].

[37] S. Roy. *Quantum Entanglement Meets AI: A Guide to Ecosystems in Quantum Machine Learning*. Medium. Accessed: 2024-10-15. Oct. 2023. DOI: 10.31237/osf.io/dwhp5.

[38] McKinsey & Company. *Recalculating the Future of Drug Development with Quantum Computing*. McKinsey & Company. Accessed: 2025-04-08. Dec. 2023. DOI: 10.37547/tajet/volume06issue12-03.

[39] H.-Y. Huang, M. Broughton, M. Mohseni, R. Babbush, S. Boixo, H. Neven, and J. R. McClean. "Power of data in quantum machine learning". In: *Nature Communications* 12.1 (May 2021). ISSN: 2041-1723. DOI: 10.1038/s41467-021-22539-9.

[40] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd. "Quantum machine learning". In: *Nature* 549.7671 (Sept. 2017), pp. 195–202. ISSN: 1476-4687. DOI: 10.1038/nature23474.

[41] Y. Cao, J. Romero, J. P. Olson, M. Degroote, P. D. Johnson, M. Kieferová, I. D. Kivlichan, T. Menke, B. Peropadre, N. P. D. Sawaya, S. Sim, L. Veis, and A. Aspuru-Guzik. "Quantum Chemistry in the Age of Quantum Computing". In: *Chemical Reviews* 119.19 (Aug. 2019), pp. 10856–10915. ISSN: 1520-6890. DOI: 10.1021/acs.chemrev.8b00803.

[42] C. Outeiral, M. Strahm, J. Shi, G. M. Morris, S. C. Benjamin, and C. M. Deane. "The prospects of quantum computing in computational molecular biology". In: *WIREs Computational Molecular Science* 11.1 (May 2020). ISSN: 1759-0884. DOI: 10.1002/wcms.1481.

[43] A. M. Smaldone, Y. Shee, G. W. Kyro, C. Xu, N. P. Vu, R. Dutta, M. H. Farag, A. Galda, S. Kumar, E. Kyoseva, and V. S. Batista. *Quantum Machine Learning in Drug Discovery: Applications in Academia and Pharmaceutical Industries*. 2024. DOI: 10.48550/ARXIV.2409.15645.

[44] A. Perdomo-Ortiz, M. Benedetti, J. Realpe-Gómez, and R. Biswas. "Opportunities and challenges for quantum-assisted machine learning in near-term quantum computers". In: *Quantum Science and Technology 3, 030502 (2018)* 3.3 (Aug. 31, 2017), p. 030502. ISSN: 2058-9565. DOI: 10.1088/2058-9565/aab859. arXiv: 1708.09757 [quant-ph].

[45] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles. "Variational Quantum Algorithms". In: *Nature Reviews Physics 3, 625-644 (2021)* 3.9 (Dec. 16, 2020), pp. 625–644. ISSN: 2522-5820. DOI: 10.1038/s42254-021-00348-9. arXiv: 2012.09265 [quant-ph].

[46] F. Bova, A. Goldfarb, and R. G. Melko. "Commercial applications of quantum computing". In: *EPJ Quantum Technology* 8.1 (Jan. 2021). ISSN: 2196-0763. DOI: 10.1140/epjqt/s40507-021-00091-1.

[47] M. Cerezo, G. Verdon, H.-Y. Huang, L. Cincio, and P. J. Coles. "Challenges and Opportunities in Quantum Machine Learning". In: *Nature Computational Science 2, 567-576 (2022)* 2.9 (Mar. 16, 2023), pp. 567–576. ISSN: 2662-8457. DOI: 10.1038/s43588-022-00311-3. arXiv: 2303.09491 [quant-ph].

[48] K. A. Britt and T. S. Humble. "High-Performance Computing with Quantum Processing Units". In: (Nov. 2015). DOI: 10.48550/ARXIV.1511.04386. arXiv: 1511.04386 [cs.ET].

[49] K. Batra, K. M. Zorn, D. H. Foil, E. Minerali, V. O. Gawriljuk, T. R. Lane, and S. Ekins. "Quantum Machine Learning Algorithms for Drug Discovery Applications". In: *Journal of Chemical Information and Modeling* 61.6 (May 2021), pp. 2641–2647. ISSN: 1549-960X. DOI: 10.1021/acs.jcim.1c00166.

[50] M. Avramouli, I. K. Savvas, A. Vasilaki, and G. Garani. "Unlocking the Potential of Quantum Machine Learning to Advance Drug Discovery". In: *Electronics* 12.11 (May 2023), p. 2402. ISSN: 2079-9292. DOI: 10.3390/electronics12112402.

[51] M. Avramouli, I. Savvas, A. Vasilaki, G. Garani, and A. Xenakis. "Quantum Machine Learning in Drug Discovery: Current State and Challenges". In: *Proceedings of the 26th Pan-Hellenic Conference on Informatics*. PCI 2022. ACM, Nov. 2022, pp. 394–401. DOI: 10.1145/3575879.3576024.

[52] K. Batra, K. M. Zorn, D. H. Foil, E. Minerali, V. O. Gawriljuk, T. R. Lane, and s. ekins sean. "Quantum Machine Learning for Drug Discovery". In: (Aug. 2020). DOI: 10.26434/chemrxiv.12781232.v1.

[53] Y. K. Wong, Y. Zhou, Y. S. Liang, H. Qiu, Y. X. Wu, and B. He. "The New Answer to Drug Discovery: Quantum Machine Learning in Preclinical Drug Development". In: *2023 IEEE 4th International Conference on Pattern Recognition and Machine Learning (PRML)*. IEEE, Aug. 2023, pp. 557–564. DOI: 10.1109/prml59573.2023.10348356.

[54] A. Bhatia, M. Saggi, and S. Kais. "Quantum Machine Learning Predicting ADME-Tox Properties in Drug Discovery". In: *Journal of chemical information and modeling* (2023). DOI: 10.1021/acs.jcim.3c01079.

[55] J. Li, M. Alam, C. M. Sha, J. Wang, N. V. Dokholyan, and S. Ghosh. "Invited: Drug Discovery Approaches using Quantum Machine Learning". In: *2021 58th ACM/IEEE Design Automation Conference (DAC)*. IEEE, Dec. 2021, pp. 1356–1359. DOI: 10.1109/dac18074.2021.9586268.

[56] L. F. Chaves and C. J. M. Koenraadt. "Climate Change and Highland Malaria: Fresh Air for a Hot Debate". In: *The Quarterly Review of Biology* 85.1 (Mar. 2010), pp. 27–55. ISSN: 1539-7718. DOI: 10.1086/650284.

[57] A. Chasapi, M. Aivaliotis, L. Angelis, A. Chanalaris, I. Iliopoulos, I. Kappas, C. Karapiperis, N. C. Kyrpides, E. Pafilis, E. Panteris, P. Topalis, G. Tsiamis, I. S. Vizirianakis, M. Vlassi, V. J. Promponas, and C. A. Ouzounis. "Establishment of computational biology in Greece and Cyprus: Past, present, and future". In: *PLOS Computational Biology* 15.12 (Dec. 2019). Ed. by J. A. Papin, e1007532. ISSN: 1553-7358. DOI: 10.1371/journal.pcbi.1007532.

[58] L.-M. Birkholtz, O. Bastien, G. Wells, D. Grando, F. Joubert, V. Kasam, M. Zimmermann, P. Ortet, N. Jacq, N. Saïdani, S. Roy, M. Hofmann-Apitius, V. Breton, A. I. Louw, and E. Maréchal. "Integration and mining of malaria molecular, functional and pharmacological data: how far are we from a chemogenomic knowledge space?" In: *Malaria Journal* 5.1 (Nov. 2006). ISSN: 1475-2875. DOI: 10.1186/1475-2875-5-110.

[59] J. Cox-Singh, T. M. E. Davis, K.-S. Lee, S. S. G. Shamsul, A. Matusop, S. Ratnam, H. A. Rahman, D. J. Conway, and B. Singh. "Plasmodium knowlesi Malaria in Humans Is Widely Distributed and Potentially Life Threatening". In: *Clinical Infectious Diseases* 46.2 (Jan. 2008), pp. 165–171. ISSN: 1537-6591. DOI: 10.1086/524888.

[60] B. Singh, L. K. Sung, A. Matusop, A. Radhakrishnan, S. S. Shamsul, J. Cox-Singh, A. Thomas, and D. J. Conway. "A large focus of naturally acquired Plasmodium knowlesi infections in human beings". In: *The Lancet* 363.9414 (Mar. 2004), pp. 1017–1024. ISSN: 0140-6736. DOI: 10.1016/s0140-6736(04)15836-4.

[61] S. Getachew, S. To, H. Trimarsanto, K. Thriemer, T. G. Clark, B. Petros, A. Aseffa, R. N. Price, and S. Auburn. "Variation in Complexity of Infection and Transmission Stability between Neighbouring Populations of Plasmodium vivax in Southern Ethiopia". In: *PLOS ONE* 10.10 (Oct. 2015). Ed. by I. Mueller, e0140780. ISSN: 1932-6203. DOI: 10.1371/journal.pone.0140780.

[62] C. Barnadas, L. Timinao, S. Javati, J. Iga, E. Malau, C. Koepfli, L. J. Robinson, N. Senn, B. Kiniboro, L. Rare, J. C. Reeder, P. M. Siba, P. A. Zimmerman, H. Karunajeewa, T. M. Davis, and I. Mueller. "Significant geographical differences in prevalence of mutations associated with Plasmodium falciparum and Plasmodium vivax drug resistance in two regions from Papua New Guinea". In: *Malaria Journal* 14.1 (Oct. 2015). ISSN: 1475-2875. DOI: 10.1186/s12936-015-0879-9.

[63] J. Marfurt, F. d. Monbrison, S. Brega, L. Barbollat, I. Müller, A. Sie, M. Goroti, J. C. Reeder, H.-P. Beck, S. Picot, and B. Genton. "Molecular Markers of In VivoPlasmodium vivaxResistance to Amodiaquine Plus Sulfadoxine-Pyrimethamine: Mutations inpvdhfrandpvmdr1". In: *The Journal of Infectious Diseases* 198.3 (Aug. 2008), pp. 409–417. ISSN: 1537-6613. DOI: 10.1086/589882.

[64] E. M. Bunnik, G. Batugedara, A. Saraf, J. Prudhomme, L. Florens, and K. G. Le Roch. "The mRNA-bound proteome of the human malaria parasite Plasmodium falciparum". In: *Genome Biology* 17.1 (July 2016). ISSN: 1474-760X. DOI: 10.1186/s13059-016-1014-0.

[65] M. Linzke, S. L. R. Yan, A. Tárnok, H. Ulrich, M. R. Groves, and C. Wrenger. "Live and Let Dye: Visualizing the Cellular Compartments of the Malaria Parasite Plasmodium falciparum". In: *Cytometry Part A* 97.7 (Nov. 2019), pp. 694–705. ISSN: 1552-4930. DOI: 10.1002/cyto.a.23927.

[66] S. Auburn, U. Böhme, S. Steinbiss, H. Trimarsanto, J. Hostetler, M. Sanders, Q. Gao, F. Nosten, C. I. Newbold, M. Berriman, R. N. Price, and T. D. Otto. "A new Plasmodium vivax reference sequence with improved assembly of the subtelomeres reveals an abundance of pir genes". In: *Wellcome Open Research* 1 (Nov. 2016), p. 4. ISSN: 2398-502X. DOI: 10.12688/wellcomeopenres.9876.1.

[67] M. R. GALINSKI, S. A. LAPP, M. S. PETERSON, F. AY, C. J. JOYNER, K. G. LE ROCH, L. L. FONSECA, and E. O. VOIT. "Plasmodium knowlesi:a superbin vivononhuman primate model of antigenic variation in malaria". In: *Parasitology* 145.1 (July 2017), pp. 85–100. ISSN: 1469-8161. DOI: 10.1017/s0031182017001135.

[68] S. Helleboid, C. Haug, K. Lamottke, Y. Zhou, J. Wei, S. Daix, L. Cambula, G. Rigou, D. W. Hum, and R. Walczak. "The Identification of Naturally Occurring Neoruscogenin as a Bioavailable, Potent, and High-Affinity Agonist of the Nuclear Receptor RORα (NR1F1)". In: *SLAS Discovery* 19.3 (Mar. 2014), pp. 399–406. ISSN: 2472-5552. DOI: 10.1177/1087057113497095.

[69] G. B. Elion. "The Quest for a Cure". In: *Annual Review of Pharmacology and Toxicology* 33.1 (Apr. 1993), pp. 1–25. ISSN: 1545-4304. DOI: 10.1146/annurev.pa.33.040193.000245.

[70] S. J. W. Black. *Sir James W. Black – Nobel Lecture*. https://www.nobelprize.org/prizes/medicine/1988/black/lecture/. Accessed 25 April 2025. Apr. 2025.

[71] U.S. Food and Drug Administration. *The Drug Development Process*. Accessed April 25, 2025. Feb. 2025. DOI: 10.13103/jfhs.2025.40.1.1.

[72] The Lancet. "Phase 0 trials: a platform for drug development?" In: *The Lancet* 374.9685 (July 2009). Publisher: Elsevier, p. 176. ISSN: 0140-6736. DOI: 10.1016/S0140-6736(09)61309-X.

[73] R. M. Gyasi, F. Asante, J. Y. Yeboah, K. Abass, C. M. Mensah, and L. P. Siaw. "Pulled in or pushed out? Understanding the complexities of motivation for alternative therapies use in Ghana". In: *International Journal of Qualitative Studies on Health and Well-being* 11.1 (Jan. 2016), p. 29667. ISSN: 1748-2631. DOI: 10.3402/qhw.v11.29667.

[74] M. Barmpouni, J. P. Gordon, R. L. Miller, C. R. J. Pritchard, J. W. Dennis, V. Grammelis, A. Rousakis, K. Souliotis, G. Poulakou, G. L. Daikos, and A. Al-Taie. "Estimating the Clinical and Economic Impact of Introducing a New Antibacterial into Greek Clinical Practice for the Management of Hospital-Acquired Infections with Limited Treatment Options". In: *Infectious Diseases and Therapy* 12.2 (Dec. 2022), pp. 527–543. ISSN: 2193-6382. DOI: 10.1007/s40121-022-00743-4.

[75] D. J. Newman and G. M. Cragg. "Natural Products as Sources of New Drugs over the Last 25 Years". In: *J. Nat. Prod.* 70.3 (Feb. 2007), p. 477. ISSN: 0163-3864. DOI: 10.1021/np068054v.

[76] N. S. Matowo, J. Moore, S. Mapua, E. P. Madumla, I. R. Moshi, E. W. Kaindoa, S. P. Mwangungulu, D. R. Kavishe, R. D. Sumaye, D. W. Lwetoijera, and F. O. Okumu. "Using a new odour-baited device to explore options for luring and killing outdoor-biting malaria vectors: a report on design and field evaluation of the Mosquito Landing Box". In: *Parasites & Vectors* 6.1 (May 2013). ISSN: 1756-3305. DOI: 10.1186/1756-3305-6-137.

[77] N. Bhagwandin. "Antimicrobial Resistance (AMR) and Multidrug Resistance (MDR) : Overview of current approaches, consortia and intellectual property issues." In: (2023), 24 pages. DOI: https://doi.org/10.34667/tind.29008.

[78] K. Na-Bangchang and J. Karbwang. "Current status of malaria chemotherapy and the role of pharmacology in antimalarial drug research and development". In: *Fundamental & Clinical Pharmacology* 23.4 (July 2009), pp. 387–409. ISSN: 1472-8206. DOI: 10.1111/j.1472-8206.2009.00709.x.

[79] E. G. Tse, M. Korsik, and M. H. Todd. "The past, present and future of anti-malarial medicines". In: *Malaria Journal* 18.1 (Mar. 2019). ISSN: 1475-2875. DOI: 10.1186/s12936-019-2724-z.

[80] A. Kondrachine and P. Trigg. "Global overview of malaria". In: *The Indian journal of medical research* 106 (Aug. 1997), pp. 39–52. ISSN: 0971-5916. DOI: 10.1016/s1383-5769(98)80014-7.

[81] M. K. Dwivedi and P. K. Singh. "Anti-Malarial Drug Resistance: Need for Novel Natural Products". In: *Research Anthology on Recent Advancements in Ethnopharmacology and Nutraceuticals*. IGI Global, 2022, pp. 233–250. ISBN: 9781668435472. DOI: 10.4018/978-1-6684-3546-5.ch013.

[82] O. Folake, I. Doosuur, I. Surayyah, B. Aishatu N, and G. Hope. "An Overview of the Historical Development of Antimalaria Drug Resistance". In: *Journal of Current Chemical and Pharmaceutical Sciences* 08.01 (2018). ISSN: 2277-2871. DOI: 10.21767/2277-2871.1000112.

[83] M. Kakde. "Drug resistance and current challenges to overcome". In: *International journal of therapeutic innovation* 1.1 (Aug. 2023), pp. 31–37. ISSN: 3048-4626. DOI: 10.55522/ijti.v1i1.0011.

[84] O. J. Patrick, A. A. Sokan-Adeaga, and R. O. Emmanuel. "Historical Evolution and Molecular Mechanisms of Antimalarial Drug Resistance with Emerging Challenges and Future Directions". In: *Discover Public Health* 23 (2026), p. 749.

[85] S. Sinha, B. Medhi, and R. Sehgal. "Challenges of drug-resistant malaria". In: *Parasite* 21 (2014), p. 61. ISSN: 1776-1042. DOI: 10.1051/parasite/2014059.

[86] T. N. Wells. "New medicines to improve control and contribute to the eradication of malaria". In: *Malaria Journal* 9.S2 (Oct. 2010). ISSN: 1475-2875. DOI: 10.1186/1475-2875-9-s2-i15.

[87] F. Ntie-Kang, K. D. Nyongbela, G. A. Ayimele, and S. Shekfeh. ""Drug-likeness" properties of natural compounds". In: *Physical Sciences Reviews* 4.11, 169 (Sept. 2019), p. 169. DOI: 10.1515/psr-2018-0169.

[88] D. I. P. "Towards conservation of Nigerian medicinal plants". In: *Journal of Medicinal Plants Research* 6.19 (May 2012). ISSN: 1996-0875. DOI: 10.5897/jmpr10.612.

[89] F. Ntie-Kang, L. L. Lifongo, L. M. Mbaze, N. Ekwelle, L. C. Owono Owono, E. Megnassan, P. N. Judson, W. Sippl, and S. M. N. Efange. "Cameroonian medicinal plants: a bioactivity versus ethnobotanical survey and chemotaxonomic classification". In: *BMC Complementary and Alternative Medicine* 13.1 (June 2013). ISSN: 1472-6882. DOI: 10.1186/1472-6882-13-147.

[90] P. Amoa Onguéné, F. Ntie-Kang, L. L. Lifongo, J. C. Ndom, W. Sippl, and L. M. Mbaze. "The potential of anti-malarial compounds derived from African medicinal plants. Part I: A pharmacological evaluation of alkaloids and terpenoids". In: *Malaria Journal* 12.1 (Dec. 2013). ISSN: 1475-2875. DOI: 10.1186/1475-2875-12-449.

[91] L. L. Lifongo, C. V. Simoben, F. Ntie-Kang, S. B. Babiaka, and P. N. Judson. "A Bioactivity Versus Ethnobotanical Survey of Medicinal Plants from Nigeria, West Africa". In: *Natural Products and Bioprospecting* 4.1 (Feb. 2014), pp. 1–19. ISSN: 2192-2209. DOI: 10.1007/s13659-014-0005-7.

[92] E. Salmerón-Manzano, J. A. Garrido-Cardenas, and F. Manzano-Agugliaro. "Worldwide Research Trends on Medicinal Plants". In: *International Journal of Environmental Research and Public Health* 17.10 (May 2020), p. 3376. ISSN: 1660-4601. DOI: 10.3390/ijerph17103376.

[93] M. A. Ungogo, G. U. Ebiloma, N. Ichoron, J. O. Igoli, H. P. de Koning, and E. O. Balogun. "A Review of the Antimalarial, Antitrypanosomal, and Antileishmanial Activities of Natural Compounds Isolated From Nigerian Flora". In: *Frontiers in Chemistry* 8 (Dec. 2020). ISSN: 2296-2646. DOI: 10.3389/fchem.2020.617448.

[94] E. Adase, P. Ankutse, D. Kumadoh, M.-A. Archer, M. O. Kyene, G. N. Yeboah, and D. O. Asamoah Agyare. ""A Review of Parquetina nigrescens (Afzel.) Bullock, A Plant for Traditional Medicine: Phytochemical and Pharmacological Properties"". In: *Evidence-Based Complementary and Alternative Medicine* 2022 (Nov. 2022). Ed. by D. D. R. Arcanjo, pp. 1–21. ISSN: 1741-427X. DOI: 10.1155/2022/6076707.

[95] O. O. Fajinmi, O. O. Olarewaju, and J. Van Staden. "Propagation of Medicinal Plants for Sustainable Livelihoods, Economic Development, and Biodiversity Conservation in South Africa". In: *Plants* 12.5 (Mar. 2023), p. 1174. ISSN: 2223-7747. DOI: 10.3390/plants12051174.

[96] J. Medina-Franco. "Advances in computational approaches for drug discovery based on natural products". In: *Revista latinoamericana de quimica* 41.5 (Sept. 2013), p. 95. ISSN: 2218-273X. DOI: 10.3390/biom11050630.

[97] L. Chen, Y. Y. Li, Q. Zhao, H. Peng, and T. J. Hou. "ADME Evaluation in Drug Discovery. 10. Predictions of P-Glycoprotein Inhibitors Using Recursive Partitioning and Naive Bayesian Classification Techniques". In: *Mol. Pharmaceutics* 8.3 (Mar. 2011), p. 889. ISSN: 1543-8392. DOI: 10.1021/mp100465q.

[98] S. Wang, Y. Li, J. Wang, L. Chen, L. Zhang, H. Yu, and T. Hou. "ADMET Evaluation in Drug Discovery. 12. Development of Binary Classification Models for Prediction of hERG Potassium Channel Blockage". In: *Mol. Pharmaceutics* 9 (2012), p. 996. DOI: 10.1021/mp300023x.

[99] K. Swanson, P. Walther, J. Leitz, S. Mukherjee, J. C. Wu, R. V. Shivnaraine, and J. Zou. "ADMET-AI: a machine learning ADMET platform for evaluation of large-scale chemical libraries". In: *Bioinformatics* 40.7 (June 2024). Ed. by A. Valencia. ISSN: 1367-4811. DOI: 10.1093/bioinformatics/btae416.

[100] L. Guan, H. Yang, Y. Cai, L. Sun, P. Di, W. Li, G. Liu, and Y. Tang. "ADMET-score – a comprehensive scoring function for evaluation of chemical drug-likeness". In: *MedChemComm* 10.1 (2019), pp. 148–157. ISSN: 2040-2511. DOI: 10.1039/c8md00472b.

[101] Yash Patel and C. Toxtli. *Migrating from Classical Machine Learning to Quantum Machine Learning: An overview and case study on Drug Discovery*. en. 2019. DOI: 10.13140/RG.2.2.19999.38567.

[102] S. Ekins, A. C. Puhl, K. M. Zorn, T. R. Lane, D. P. Russo, J. J. Klein, A. J. Hickey, and A. M. Clark. "Exploiting machine learning for end-to-end drug discovery and development". In: *Nature Materials* 18.5 (Apr. 2019), pp. 435–441. ISSN: 1476-4660. DOI: 10.1038/s41563-019-0338-z.

[103] S. van der Voort, A. Bender, and B. A. Westerman. *The Role of Computer-AidedDrug Design in Drug Discovery*. Jan. 2024. DOI: 10.1002/9783527840748.ch9.

[104] M. Boy, J. Körner, D. Bisaghchi, P. Girvan, and N. Bros. "Encoding Molecular Structures in Quantum Machine Learning". In: *Machine Learning: Science and Technology* 6.4 (2025). Introduces the quantum molecular structure encoding (QMSE) scheme; preprint arXiv:2507.20422, p. 045076.

[105] M. V. S. Temgoua, J.-P. Tchapet Njafa, P. Samafou, W. F. Mbacham, and S. G. Nana Engo. "Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis". In: *ChemRxiv* (2026). Preprint. DOI: 10.26434/chemrxiv.15006437/v2.

[106] M. V. S. Temgoua, J.-P. Tchapet Njafa, S. G. Nana Engo, P. Samafou, and W. F. Mbacham. "Topological and tensor-network representations resolve chemical paradoxes in African antimalarial natural products". In: *ChemRxiv* (2026). Preprint. DOI: 10.26434/chemrxiv.15007167/v1.

[107] M. V. S. Temgoua, J.-P. Tchapet Njafa, P. Samafou, W. F. Mbacham, and S. G. Nana Engo. "Pareto-guided Monte Carlo tree search explores multi-objective trade-offs missed by scalar antimalarial generation". In: *ChemRxiv* (2026). Preprint. DOI: 10.26434/chemrxiv.15007306/v1.

[108] M. Lžičař and H. Gamouh. "CHEESE: 3D Shape and Electrostatic Virtual Screening in a Vector Space". In: *ChemRxiv* (2024). Preprint. DOI: 10.26434/chemrxiv-2024-cswth.

[109] A. Nigam, R. Pollice, M. Krenn, G. d. P. Gomes, and A. Aspuru-Guzik. "Beyond generative models: superfast traversal, optimization, novelty, exploration and discovery (STONED) algorithm for molecules using SELFIES". In: *Chemical Science* 12.20 (2021), pp. 7079–7090. DOI: 10.1039/d1sc00231g.

[110] R. Gómez-Bombarelli, J. N. Wei, D. Duvenaud, J. M. Hernández-Lobato, B. Sánchez-Lengeling, D. Sheberla, J. Aguilera-Iparraguirre, T. D. Hirzel, R. P. Adams, and A. Aspuru-Guzik. "Automatic Chemical Design Using a Data-Driven Continuous Representation of Molecules". In: *ACS Central Science* 4.2 (2018), pp. 268–276. DOI: 10.1021/acscentsci.7b00572.

[111] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. "Optuna: A Next-generation Hyperparameter Optimization Framework. A Next-generation Hyperparameter Optimization Framework". In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. KDD '19. ACM, July 2019, pp. 2623–2631. DOI: 10.1145/3292500.3330701.

[112] G. R. Bickerton, G. V. Paolini, J. Besnard, S. Muresan, and A. L. Hopkins. "Quantifying the chemical beauty of drugs". In: *Nature Chemistry* 4.2 (2012), pp. 90–98. DOI: 10.1038/nchem.1243.

[113] M. Sicho and D. Svozil. "SYBA: Bayesian estimation of synthetic accessibility of organic compounds". In: *Journal of Cheminformatics* 12 (2020), p. 35. DOI: 10.1186/s13321-020-00439-2.

[114] P. Ertl, S. Roggo, and A. Schuffenhauer. "Natural product-likeness score and its application for prioritization of compound libraries". In: *Journal of Chemical Information and Modeling* 48.1 (2008), pp. 68–74. DOI: 10.1021/ci6002386.

[115] G. Corso, H. Stärk, B. Jing, R. Barzilay, and T. Jaakkola. "DiffDock: Diffusion steps, twists, and turns for molecular docking". In: *International Conference on Learning Representations (ICLR)*. 2023.

[116] A. Waterhouse, M. Bertoni, S. Bienert, G. Studer, G. Tauriello, R. Gumienny, F. T. Heer, T. A. P. de Beer, C. Rempfer, L. Bordoli, R. Lepore, and T. Schwede. "SWISS-MODEL: homology modelling of protein structures and complexes". In: *Nucleic Acids Research* 46.W1 (2018), W296–W303. DOI: 10.1093/nar/gky427.

[117] D. Szklarczyk, R. Kirsch, M. Koutrouli, K. Nastou, F. Mehryary, R. Hachilif, A. L. Gable, T. Fang, N. T. Doncheva, S. Pyysalo, P. Bork, L. J. Jensen, and C. von Mering. "The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest". In: *Nucleic Acids Research* 51.D1 (2023), pp. D638–D646. DOI: 10.1093/nar/gkac1000.

[118] M. J. Abraham, T. Murtola, R. Schulz, S. Páll, J. C. Smith, B. Hess, and E. Lindahl. "GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers". In: *SoftwareX* 1–2 (2015), pp. 19–25. DOI: 10.1016/j.softx.2015.06.001.

[119] J. Huang, S. Rauscher, G. Nawrocki, T. Ran, M. Feig, B. L. de Groot, H. Grubmüller, and A. D. MacKerell. "CHARMM36m: an improved force field for folded and intrinsically disordered proteins". In: *Nature Methods* 14.1 (2017), pp. 71–73. DOI: 10.1038/nmeth.4067.

[120] N. Michaud-Agrawal, E. J. Denning, T. B. Woolf, and O. Beckstein. "MDAnalysis: a toolkit for the analysis of molecular dynamics simulations". In: *Journal of Computational Chemistry* 32.10 (2011), pp. 2319–2327. DOI: 10.1002/jcc.21787.

[121] M. S. Valdés-Tresanco, M. E. Valdés-Tresanco, P. A. Valiente, and E. Moreno. "gmx_MMPBSA: a new tool to perform end-state free energy calculations with GROMACS". In: *Journal of Chemical Theory and Computation* 17.10 (2021), pp. 6281–6291. DOI: 10.1021/acs.jctc.1c00645.

[122] C. Bouysset and S. Fiorucci. "ProLIF: a library to encode molecular interactions as fingerprints". In: *Journal of Cheminformatics* 13 (2021), p. 72. DOI: 10.1186/s13321-021-00548-6.

[123] G. Landrum. *RDKit: Open-source cheminformatics*. https://www.rdkit.org. Version 2025.03.6.

[124] S. Riniker and G. A. Landrum. "Better informed distance geometry: using what we know to improve conformation generation". In: *Journal of Chemical Information and Modeling* 55.12 (2015), pp. 2562–2574. DOI: 10.1021/acs.jcim.5b00654.